Introduction to Group Theory

MAT 2143

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These lecture notes were produced using my course notes from Winter 2016 and Winter 2019. They are loosely based on the following texts:

Thomas W. Judson, *Abstract Algebra, Theory and Applications*, Annual Edition 2018.

W. Keith Nicholson, *Introduction to Abstract Algebra*, Third Edition, . John Wiley & Sons Inc., 2007.

INTEGERS

Mathematical induction. A general proof technique in mathematics that is extremely useful when we want to prove a certain statement for all values of $n \in \mathbb{N}$ is proof by induction.

Recall that $\mathbb{N} = \{1, 2, 3, \ldots\}$ is the set of natural numbers.

Example 1.1. We want to prove that $1 + \cdots + n = \frac{n(n+1)}{2}$ for every $n \in \mathbb{N}$. The statement is easily verified for $n = 1$. Now we assume it is true for some value $n = n_0$, and using this assumption, we prove that it holds for $n = n_0 + 1$:

$$\begin{aligned} 1 + \cdots + n_0 + (n_0 + 1) &= (1 + \cdots + n_0) + (n_0 + 1) \\ &= \frac{n_0(n_0 + 1)}{2} + (n_0 + 1) \\ &= (n_0 + 1)\left(\frac{n_0}{2} + 1\right) = \frac{(n_0 + 1)(n_0 + 2)}{2}. \end{aligned}$$

Principle of induction. Let $S(n)$ be a predicate that depends on a variable $n \in \mathbb{N}$. Assume that

(i) $S(n_0)$ is true for some $n_0 \in \mathbb{N}$.

(ii) We can prove that for any $k \in \mathbb{N}$ where $k \geqslant n_0$, if $S(k)$ is true, then $S(k + 1)$ is also true.

Then $S(n)$ is true for every $n \in \mathbb{N}$ satisfying $n \geqslant n_0$.

Before giving another example of application of the induction principle, we recall the definition of divisibility.

Definition 1.2. Given $m, n \in \mathbb{Z}$, we say m *divides* n (or m is a *divisor* of n; or n is *divisible* by m) and we write $m|n$, if $n = md$ for some $d \in \mathbb{Z}$.

We write $m \nmid n$ if m does not divide n (or equivalently, if n is not divisible by m).

Example 1.3. We want to prove that for every $n \in \mathbb{N}$, the integer $4^{n+1} + 4^n + 1$ is divisible by 3. For $n = 1$ we have

$$4^2 + 4^1 + 1 = 21$$

which is divisible by 3. Now assume the statement is true for some $n = k$. Then for $n = k + 1$ we have

$$4^{k+2} + 4^{k+1} + 1 = \left[4(4^{k+1} + 4^k + 1) - 4\right] + 1 = 4(4^{k+1} + 4^k + 1) - 3.$$

By the induction hypothesis, $4^{k+1} + 4^k + 1$ is divisible by 3, i.e.,

$$4^{k+1} + 4^k + 1 = 3q \text{ for some integer } q.$$

It follows that

$$4(4^{k+1} + 4^k + 1) - 3 = 3q - 3 = 3(q - 1),$$

so that $4(4^{k+1} + 4^k + 1) - 3$ is also divisible by 3.

Principle of induction (strong version). Let $S(n)$ be a predicate that depends on a variable $n \in \mathbb{N}$. Assume that

(i) $S(n_0)$ is true for some $n_0 \in \mathbb{N}$.
(ii) We can prove that for every $k \in \mathbb{N}$ where $k \geqslant n_0$, if $S(n_0), \ldots, S(k)$ are true then $S(k+1)$ is also true.

Then $S(n)$ is true for every $n \in \mathbb{N}$ satisfying $n \geqslant n_0$.

Both of the induction principles are logically equivalent to the following statement:

Well-ordering principle. Every nonempty subset of $\mathbb{N}$ has a (unique) smallest member.

Theorem 1.4. (Division Algorithm) *Let $n, d \in \mathbb{Z}$ such that $d \geqslant 1$. Then there exists a unique pair of integers $q, r \in \mathbb{Z}$ such that $n = qd + r$ and $0 \leqslant r < d$.*

Proof. Existence:
Step 1. Set $X := \{k \in \mathbb{Z} \,:\, k = n - td \text{ for some } t \in \mathbb{Z} \text{ and } k \geqslant 0\}$. Clearly $X \subseteq \mathbb{N} \cup \{0\}$.

Step 2.: We show that $X \neq \varnothing$:

- If $n \geqslant 0$ then we can set $t = 0$. It follows that $n \in X$.
- If $n < 0$, then we set $t = n$. It follows that $n(1-d) = n - nd \in X$. Note that $n(1-d) \geqslant 0$ because $1 - d \leqslant 0$.

Step 3. By the well-ordering principle, there exists a smallest element $r \in X$. Thus $r = n - t_0 d$ for some $t_0 \in \mathbb{Z}$. Set $q := t_0$.

Step 4. From $r \in X$ it follows that $r \geqslant 0$. Next we show $r < d$. Assume on the contrary, that $r \geqslant d$, so that $r - d \geqslant 0$. Then

$$r - d = (n - t_0 d) - d = n - (t_0 + 1)d \in X,$$

which contradicts the fact that $r = \min X$.

Uniqueness: Assume that q', r' is another pair such that $n = q'd + r'$ and $0 \leqslant r < d$. We want to show $q = q'$ and $r = r'$. We have

$$n = qd + r = q'd + r' \;\Rightarrow\; d(q' - q) = r - r'.$$

If $q = q'$, then it follows that $r - r' = 0$, hence $r = r'$, and the proof is complete. Now assume that $q \neq q'$. Without loss of generality we can assume $q < q'$ (the case $q > q'$ is similar). Thus

$$r - r' = d(q' - q) \geqslant d, \text{ so that } r \geqslant r' + d \geqslant d.$$

This contradicts the assumption $r < d$. □

Example 1.5. Here are some explicit examples of the division algorithm:

(i) Let us assume $n = 19$ and $d = 4$. We want to find q, r. We have

$$X = \{\ldots, 19, 15, 11, 7, 3, -1, -5, \ldots\},$$

where the gray entries are not in X. It follows that $r = \min X = 3$ and $q = 4$.

(ii) $n = -13$ and $d = 7$. Then $X = \{\ldots, -13, -6, 1, 8, \ldots\}$, where the gray entries are not in X. It follows that $r = \min X = 1$ and $q = -2$.

Exercises.

Let m, n, d be integers. Prove the following statements.

(i) $n|n$.
(ii) If $m \neq 0$, $d \in \mathbb{N}$ and $d|m$ then $d \leqslant |m|$.
(iii) If $d|m$ and $m|n$ then $d|n$.
(iv) If $d|n$ and $n|d$ then $d = \pm n$.
(v) If $d|m$ and $d|n$ then $d|xm + yn$ for every $x, y \in \mathbb{Z}$.

Definition 1.6. Let m and n be integers. Assume that at least one of m and n is nonzero. An integer $d \geqslant 1$ is called the *greatest common divisor* (or the gcd) of m and n, if it satisfies the following properties:

(i) d is a common divisor of m and n, i.e., $d|m$ and $d|m$.
(ii) For every common divisor d' of d, we have $d'|d$.

From the above definition, it is <u>not</u> immediately obvious that the gcd of m and n exists. If it does, then it is also the largest common divisor of m and n; because for every d' such that $d'|m$ and $d'|n$, we have $d'|d$, hence $d' \leqslant |d| = d$.

Remark 1.7. (i) If $\gcd(m, n)$ exists, then it is unique. This is because if $d \geqslant 1$ and $d' \geqslant 1$ are two possible $\gcd$'s for m and n, then in particular we have

$$d|d' \text{ and } d'|d,$$

so that $d = \pm d'$. Since $d, d' \geqslant 1$, it follows that $d = d'$.

(ii) By convention, $\gcd(0, 0)$ is not defined.

(iii) If either $m = 0$ or $n = 0$, then $\gcd(m, n) = \max\{|m|, |n|\}$.

Theorem 1.8. (Bezout's Theorem) *Let m and n be integers such that either $m \neq 0$ or $n \neq 0$. Then there exist $x, y \in \mathbb{Z}$ such that*

$$\gcd(m, n) = xm + yn.$$

In particular, $\gcd(m, n)$ exists.

Proof. Step 1. Consider the set

$$S := \{k \in \mathbb{N} \ : \ k = xm + yn \text{ for some } x, y \in \mathbb{Z}\}.$$

Clearly $S \neq \varnothing$. Thus, by the well-ordering principle, S has a smallest member. Set $d := \min S$, so that

$$d = x_0 m + y_0 n \text{ where } x_0, y_0 \in \mathbb{Z}.$$

Step 2. We prove that $d|m$ and $d|n$. We use the method of proof by contradiction. Assume, on the contrary, that $d \nmid m$. By Theorem 1.4 we have

$$m = dq + r \text{ for } 0 < r < d.$$

Thus

$$r = m - dq = m - (x_0 m + y_0 n) = (1 - x_0)m + (-y_0)n \in S \text{ and } r < d,$$

which contradicts the assumption that $d = \min S$. The argument for $d|n$ is the same.

Step 3. Suppose that $d' \in \mathbb{Z}$ is a common divisor of m and n (i.e., $d'|m$ and $d'|n$). We prove that $d'|d$. We have

$$d'|m \;\Rightarrow\; d'|x_0 m \quad \text{and} \quad d'|n \;\Rightarrow\; d'|y_0 n.$$

Finally, we have

$$d'|x_0 m \text{ and } d'|y_0 n \;\Rightarrow\; d'|x_0 m + y_0 n \;\Rightarrow\; d'|d. \qquad \square$$

Definition 1.9. Two integers m and n are called *co-prime* if $\gcd(m, n) = 1$.

Bezout's Theorem implies that for co-prime integers m and n, there exist $x, y \in \mathbb{Z}$ such that $xm + yn = 1$.

Remark 1.10. A digression about definitions: it is customary in mathematical writing to use "if" (rather than "if and only if"). This is because a definition is not a logical implication. It is just a sentence that describes the meaning of a new concept.

Corollary 1.11. *Let $d, m, n \in \mathbb{Z}$ be integers, such that either $m \neq 0$ or $n \neq 0$. Assume that $d = \gcd(m, n)$. Then $\gcd(\frac{m}{d}, \frac{n}{d}) = 1$.*

Proof. It is enough to show that if $d' \in \mathbb{Z}$ satisfies $d'|\frac{m}{d}$ and $d'|\frac{n}{d}$, then $d' = \pm 1$.

By Bezout's Theorem, there exist $x, y \in \mathbb{Z}$ such that $mx + ny = d$. Thus we have

$$\frac{m}{d}x + \frac{n}{d}y = 1.$$

Note that $\frac{m}{d}, \frac{n}{d} \in \mathbb{Z}$. Now

$$d'|\frac{m}{d} \text{ and } d'|\frac{n}{d} \text{ imply that } d'|\left(\frac{m}{d}x + \frac{n}{d}y\right),$$

hence $d'|1$, and therefore $d' = \pm 1$. $\square$

The Euclidean algorithm. Theorem 1.8 proves the existence of $x, y \in \mathbb{Z}$ such that

$$xm + yn = \gcd(m, n).$$

But does there exist a method for computing the integers x, y? The *Euclidean algorithm* provides such a method. Here is how the Euclidean algorithm works:

Start with integers $m \geqslant n > 0$. Set $r_0 = m, r_1 = n$, and define r_i for $i \geqslant 2$ by

$$r_{i-2} := q_i r_{i-1} + r_i \;\text{ where } 0 \leqslant r_i < r_{i-1},$$

where q_i, r_i are obtained by the division algorithm (Theorem 1.4). We have

$$r_1 > r_2 > r_3 > \cdots \geqslant 0.$$

Set $k := \min\{i \;:\; r_{i+1} = 0\}$.

Proposition 1.12. $r_k = \gcd(m, n)$.

Proof. Step 1. Set $d := \gcd(m, n)$. We have $d|r_0$ and $d|r_1$. Next we note that:

$$\text{For every } i \geqslant 2, \text{ if } d|r_{i-2} \text{ and } d|r_{i-1} \text{ then } d|(r_{i-2} - q_i r_{i-1}) = r_i.$$

It follows by induction that $d|r_i$ for every $i \geqslant 1$. In particular, $d|r_k$.

Step 2. To complete the proof, it is enough to show that $r_k|m$ and $r_k|n$. Indeed when these relations are proved if follows that r_k is a common divisor of m and n, hence $r_k|d$. Now

$$d|r_k \text{ and } r_k|d \;\Rightarrow\; d = \pm r_k.$$

Since $d, r_k > 0$, we obtain $d = r_k$.

Step 3. We now prove the claim of Step 2. To this end, we show that $r_k|r_i$ for every $1 \leqslant i \leqslant k$. This is done by "reverse" induction:

(i) We have $r_k|r_k$ (trivial) and $r_k|r_{k-1}$ because $r_{k-1} = q_{k+1}r_k + r_{k+1} = q_{k+1}r_k$.

(ii) For $i \geqslant 1$, if $r_k|r_i$ and $r_k|r_{i-1}$ then

$$r_k|r_i \text{ and } r_k|r_{i-1}q_i \;\Rightarrow\; r_k|(q_i r_{i-1} + r_i) = r_{i-2}.$$

From (i) and (ii) above it follows that $r_k|r_i$ for all $1 \leqslant i \leqslant k$. In particular, $r_k|r_0 = m$ and $r_k|r_1 = n$. □

To compute $x_0, y_0 \in \mathbb{Z}$ such that $\gcd(m, n) = x_0 m + y_0 n$, we use the following reverse calculation:

$$\begin{aligned}
\gcd(m, n) = r_k &= r_{k-2} - q_k r_{k-1} \\
&= r_{k-2} - q_k(r_{k-3} - q_{k-1}r_{k-2}) \\
&= (1 + q_k q_{k-1})r_{k-2} - q_k r_{k-3} \\
&= \cdots
\end{aligned}$$

Each time we replace the remaining r_i with the largest index by $r_{i-2} - q_i r_{i-1}$, so that we obtain a linear combination of r_{i-1} and r_{i-2}.

Example 1.13. Let us compute $\gcd(108, 63)$. We set $r_0 = m = 108$ and $r_1 = n = 63$. Then

$$\begin{array}{ll}
108 = 1 \times 63 + 45 & r_2 = 45 \\
63 = 1 \times 45 + 18 & r_3 = 18 \\
45 = 2 \times 18 + 9 & r_4 = 9 \\
18 = 2 \times 9 + 0 & r_5 = 0.
\end{array}$$

Therefore $\gcd(108, 63) = r_4 = 9$. We also have

$$\begin{aligned}
9 &= 45 - 2 \times 18 \\
&= 45 - 2 \times (63 - 1 \times 45) \\
&= 3 \times 45 - 2 \times 63 \\
&= 3 \times (108 - 1 \times 63) - 2 \times 63 \\
&= 3 \times 108 - 5 \times 63.
\end{aligned}$$

Therefore for $x_0 = 3$ and $y_0 = -5$ we have $9 = x_0 m + y_0 n$.

Definition 1.14. An integer p is called *prime* if it satisfies the following two properties:

(i) $p > 1$.
(ii) The only divisors of p are ± 1 and $\pm p$.

An integer $n > 1$ which is not prime is called a *composite* number. An important property of prime numbers is the following:

Lemma 1.15. *(i) Let $a, b \in \mathbb{Z}$, and let p be a prime number. Assume that $p|ab$. Then either $p|a$ or $p|b$.*
(ii) Let $a_1, \ldots, a_k \in \mathbb{Z}$ for some $k \in \mathbb{N}$. Let p be a prime number such that $p|a_1 \cdots a_k$. Then there exists some $1 \leqslant i \leqslant k$ such that $p|a_i$.
(iii) Let $d, m, n \in \mathbb{Z}$ such that $d|mn$ and $\gcd(d, m) = 1$. Then $d|n$.

Proof. (i) Assume that $p \nmid a$. We will show that $p|b$. Indeed, set $d := \gcd(a, p)$. Since the only positive divisors of p are 1 and p, we should have $d \in \{1, p\}$. But if $d = p$ then we have $p|a$. Thus we can assume that $d = 1$.

Now by Bezout's theorem there exist $x, y \in \mathbb{Z}$ such that $xa + yp = 1$. We have

$$b = b(xa + yp) = abx + pby.$$

Since $p|ab$, we have $p|abx$. Also, $p|pby$. Thus

$$p|abx + pby \text{ and } abx + pby = b, \text{ and consequently } p|b.$$

(ii) For $k = 1$ this is trivial and for $k = 2$ this is part (i). Now we prove the statement by induction on k. Assume that it holds for some $k \in \mathbb{N}$. We need to show that for every $m_1, \ldots, m_{k+1} \in \mathbb{Z}$, if p is a prime number such that $p|m_1 \cdots m_{k+1}$, then $p|m_i$ for some $1 \leqslant i \leqslant k + 1$. There are two cases to consider:

- Case I: $p|m_1 \cdots m_k$. Then by induction hypothesis there exists $1 \leqslant i \leqslant k$ such that $p|m_i$.
- Case II: $p \nmid m_1 \cdots m_k$. Setting $a = m_1 \cdots m_k$ and $p = m_{k+1}$, from part (i) it follows that $p|m_{k+1}$.

(iii) The argument is similar to (i). We leave it as an exercise. □

Exercises.

(1) Using induction, prove that if $m_1, \ldots, m_k$ are integers and p is a prime number such that $p|m_1 \cdots m_k$, then there exists an index $1 \leqslant i \leqslant k$ such that $p|m_i$.
(2) Compute $\gcd(56, 24)$. Then compute integers $x, y \in \mathbb{Z}$ such that

$$\gcd(56, 24) = 24x + 56y.$$

(3) Complete the proof of Lemma 1.15(ii).
(4) Write down the proof of Lemma 1.15(iii).
(5) Let $m, n, d \in \mathbb{Z}$ such that $\gcd(d, n) = 1$ and $d|mn$. Prove that $d|m$.

Hint. First choose $x, y \in \mathbb{Z}$ such that $nx + dy = 1$. Then write $m = m(nx + dy)$, and continue as in the proof of Lemma 1.15(i).

An important theorem in elementary arithmetic, whose proof we skip (but it appears in the textbook) is the following.

Theorem 1.16. (Fundamental Theorem of Arithmetic) *Let $n > 1$ be an integer. Then n can be written as a product of prime numbers. Furthermore, the factorization of n as a product of primes is unique: if*

$$n = p_1 \cdots p_k = q_1 \cdots q_\ell$$

where the p_i's and the q_j's are prime numbers, then $k = \ell$ and the q_j's are just a rearrangement of the p_i's.

Theorem 1.17. *There exist infinitely many prime numbers.*

Proof. We give Euclid's original proof. Assume, on the contrary, that there are only finitely many primes. Label them as $p_1, \ldots, p_k$. Now consider

$$N := 1 + p_1 \cdots p_k.$$

Now let p be the smallest divisor of N such that $p > 1$. Then p is a prime number (because every divisor of p will also be a divisor of N). Thus $p = p_i$ for some $1 \leqslant i \leqslant k$. This implies that $p | p_1 \cdots p_k$, and therefore

$$p | (N - p_1 \cdots p_k) = 1,$$

which is a contradiction. □

Lemma 1.18. *Let $N > 1$ be an integer. Assume that the prime factorization of N is*

$$N = p_1^{a_1} \cdots p_k^{a_k}$$

where $p_1, \ldots, p_k$ are distinct prime numbers and $a_1, \ldots, a_k \in \mathbb{N}$. Let p be a prime number such that $p^k | N$ for some $k \in \mathbb{N}$. Then $p = p_i$ for a unique $1 \leqslant i \leqslant k$. Furthermore $k \leqslant a_i$.

Proof. Step 1. From $p^k | N$ it follows that $p | N$, hence $p | p_1^{a_1} \cdots p_k^{a_k}$. Now Lemma 1.15(ii) implies that $p | p_i$ for some $1 \leqslant i \leqslant k$. Since p and p_i are prime numbers, from $p | p_i$ it follows that $p = p_i$.

Step 2. We prove $k \leqslant a_i$. Becasue of the symmetry of the indices we can assume that $i = 1$. Assume that, on the contrary, $k > a_1$; thus $k \geqslant a_1 + 1$. From $p_1^k | N$ we have

$$N = p_1^k d \text{ for some } d \in \mathbb{Z},$$

hence $N = p_1^k d = p_1^{a_1+1}(p_1^{k-(a_1+1)} d)$ and $d' := p_1^{k-(a_1+1)} d \in \mathbb{Z}$. Hence by cancellation of $p_1^{a_1}$ from both sides we obtain

$$p_1^{a_1} \cdots p_k^{a_k} = p_1^{a_1+1} d' \;\Rightarrow\; p_2^{a_2} \cdots p_k^{a_k} = p_1 d'.$$

The latter implies that $p_1 | p_2^{a_2} \cdots p_k^{a_k}$. Thus Lemma 1.15(ii) implies that $p | p_i$ for some $2 \leqslant i \leqslant k$. This is a contradiction because the p_i's are distinct primes. □

The prime factorization provides another method to compute the $\gcd$ of integers:

Proposition 1.19. *Let $a, b \in \mathbb{N}$, with prime factorizations*

$$a = p_1^{a_1} \cdots p_k^{a_k} \text{ and } b = p_1^{b_1} \cdots p_k^{b_k},$$

where $p_1, \ldots, p_k$ are distinct primes and $a_i, b_i \in \mathbb{Z}^{\geqslant 0} := \{x \in \mathbb{Z} \,:\, x \geqslant 0\}$ for all $1 \leqslant i \leqslant k$. Then

$$\gcd(a, b) = p_1^{\min\{a_1, b_1\}} \cdots p_k^{\min\{a_k, b_k\}}.$$

Proof. Step 1. First we show that $M := p_1^{\min\{a_1,b_1\}} \cdots p_k^{\min\{a_k,b_k\}}$ is a common divisor of a and b. To do so, we prove that $M|a$, and the proof for $M|b$ is similar.

Step 2. To prove $M|a$, note that for every $1 \leqslant i \leqslant k$ we have $\min\{a_i, b_i\} \leqslant a_i$, hence $a_i = \min\{a_i, b_i\} + c_i$ for some $c_i \in \mathbb{Z}^{\geqslant 0}$. Thus we have

$$a = M p_1^{c_1} \cdots p_k^{c_k},$$

and $p_1^{c_1} \cdots p_k^{c_k} \in \mathbb{Z}$ because it is a product of primes with non-negative integer exponents. This completes the proof of $M|a$.

Step 3. To show that M is indeed equal to $\gcd(a, b)$, we need to show the following: if $d|a$ and $d|b$ for some $d \in \mathbb{Z}$, then $d|M$. This completes the proof of the fact that $M = gcd(a, b)$.

Step 4. We now proceed towards the proof of the claim of Step 3. Assume that $d \in \mathbb{Z}$ satisfies $d|a$ and $d|b$. If p is a prime such that $p^k|d$ for some $k \in \mathbb{N}$, then

$$p^k|d \text{ and } d|a \text{ implies } p^k|a,$$

and therefore by Lemma 1.18 we have $p = p_i$ for some $1 \leqslant i \leqslant k$, and $k \leqslant a_i$. Similarly, from $d|b$ we obtain $p^k|b$, hence by Lemma 1.18 we have $p = p_j$ for some $1 \leqslant j \leqslant k$, and $k \leqslant b_j$. Since the primes $p_1, \ldots, p_k$ are distinct, it follows that $i = j$, and consequently, $k \leqslant \min\{a_i, b_i\}$.

Step 4. From what is shown in Step 3 it follows that in the prime factorization of d, the only primes that can appear are $p_1, \ldots, p_k$; morevoer, this prime factorization is of the form

$$d = p_1^{c_1} \cdots d_k^{e_k}, \text{ where } e_i \leqslant \min\{a_i, b_i\} \text{ for all } 1 \leqslant i \leqslant k.$$

Therefore $d|M$, because

$$M = d p_1^{\min\{a_1,b_1\}-e_1} \cdots p_k^{\min\{a_k,b_k\}-e_k},$$

and the exponents $\min\{a_1, b_1\} - e_1$ are non-negative integers. □

GROUPS

Integers modulo n. Let $n \geqslant 1$ be an integer. The division algorithm modulo n yields remainders r such that $0 \leqslant r < n$. We want to define addition and multiplication operations on the set of these remainders, namely on

$$\mathbb{Z}_n := \{0, 1, \ldots, n-1\}.$$

For $a, b \in \mathbb{Z}_n$ we define

$$a \dot{+} b := (a + b) \text{ modulo } n,$$

and

$$a \dot{\times} b := ab \text{ modulo } n.$$

Here "modulo n" means the remainder of the given quantity when it is divided by n.

Example 2.1. We have $\mathbb{Z}_4 = \{0, 1, 2, 3\}$. Then we can check that

$$1 \dot{+} 2 = 3 \text{ and } 2 \dot{+} 3 = 1.$$

Also,

$$2 \dot{\times} 3 = 2 \text{ and } 2 \dot{\times} 2 = 0.$$

The formal definition of the set $\mathbb{Z}_n$ is as follows. Consider the binary relation $\equiv_n$ on $\mathbb{Z}$ defined as follows: for $a, b \in \mathbb{Z}$,

$$a \equiv_n b \quad \Leftrightarrow \quad n|(a-b).$$

Exercises.

(1) Let $n \in \mathbb{N}$.
   (i) Prove that $\equiv_n$ is an equivalence relation, i.e., it is reflexive, symmetric, and transitive.
   (ii) Prove that the integers $0 \leqslant k \leqslant n-1$ are representatives of distinct equivalence classes of $\equiv_n$. Furthermore, these equivalence classes are indeed *all* of the equivalence classes of $\equiv_n$.
   (iii) Let $a, b, a', b' \in \mathbb{Z}$. Prove that if $a \equiv_n a'$ and $b \equiv_n b'$, then

(2.1) $$a + b \equiv_n a' + b' \text{ and } ab \equiv_n a'b'.$$

(2) Compute the last 2 digits from the right of 7^{111}.

Definition 2.2. Fix an integer $n \geqslant 1$. For each $a \in \mathbb{Z}$, let $[a]$ denote the equivalence class of a, i.e.,

$$[a] := \{b \in \mathbb{Z} \ : \ b \equiv_n a\}.$$

We denote the set of the equivalence classes of $\equiv_n$ by $\mathbb{Z}_n$. Then by the preceding exercise each of these equivalence classes contains a **unique** element $a \in \mathbb{Z}$ such that $0 \leqslant a \leqslant n-1$. Therefore we can naturally identify $\mathbb{Z}_n$ with $\{0, 1, \ldots, n-1\}$.

Remark 2.3. From (2.1) it follows that the addition and multiplication that we defined on $\mathbb{Z}_n$ are indeed induced from the corresponding operations on $\mathbb{Z}$. We will learn more about this phenomenon in the future.

Proposition 2.4. *Let $n \geqslant 1$, and let $a, b, c \in \mathbb{Z}_n$.*

(i) $a \dot{+} b = b \dot{+} a$.
(ii) $a \dot{\times} b = b \dot{\times} a$.
(iii) $a \dot{+} (b \dot{+} c) = (a \dot{+} b) \dot{+} c$.
(iv) $a \dot{\times} (b \dot{\times} c) = (a \dot{\times} b) \dot{\times} c$.
(v) $a \dot{+} 0 = 0 \dot{+} a = a$.
(vi) $a \dot{\times} 1 = 1 \dot{\times} a = a$.
(vii) $a \dot{\times} (b \dot{+} c) = a \dot{\times} b \dot{+} a \dot{\times} c$.
(viii) $a \dot{+} (-a) = 0$, *where by $-a$ we denote the equivalence class of $n - a$.*
(ix) $\gcd(a, n) = 1$ *if and only if there exists an element $a' \in \mathbb{Z}_n$ such that $a \dot{\times} a' = 1$.*

Proof. We only prove (i) and (ix). The remaining parts are left as an exercise.

(i) Set $r := a \dot{+} b$. Then $a+b = qn+r$, hence $b+a = qn+r$, from which it follows that $b \dot{+} a = r$.

(ix) Assume that $\gcd(a, n) = 1$. By Theorem 1.8 there exist $x, y \in \mathbb{Z}$ such that $ax + ny = 1$. Thus $ax \equiv_n 1$. Now set $a' := x$ modulo n. Then $a' = x + tn$ for some $t \in \mathbb{Z}$, hence

$$aa' = ax + atn = 1 - ny + atn = 1 + n(-y + at),$$

so that $a \dot{\times} a' = 1$.

Conversely, assume that there exists $a' \in \mathbb{Z}_n$ such that $a \dot{\times} a' = 1$. Then $aa' = nq + 1$, so that we have $aa' - nq = 1$. Now if $d|a$ and $d|n$ then we have $d|aa' - nq = 1$, hence $d = \pm 1$. Therefore $\gcd(a, n) = 1$. □

Solving equations in $\mathbb{Z}_n$. From now on we always equip $\mathbb{Z}_n$ with the defined addition and multiplication operations. Moreover, we will stop using the fancy notation $\dot{+}$, $\dot{\times}$, and $\equiv_n$ for addition, multiplication, and equality of $\mathbb{Z}_n$. Instead, we will use the usual symbols in both $\mathbb{Z}$ and $\mathbb{Z}_n$.

Definition 2.5. Let $n \in \mathbb{N}$. An element $a \in \mathbb{Z}_n$ is called *invertible* if there exists some $b \in \mathbb{Z}_n$ such that $ab = 1$. If a is invertible, we denote the inverse of a by a^{-1}.

Proposition 2.6. *Let $n \in \mathbb{N}$, and $a, b \in \mathbb{Z}_n$. Assume that $\gcd(a, n) = 1$. Then the equation $ax = b$ has a unique solution in $\mathbb{Z}_n$, given by $x = a^{-1}b$.*

Proof. Assume that $ax = b$ for some $x \in \mathbb{Z}_n$. By multiplying both sides by a^{-1} (where multiplication takes place in $\mathbb{Z}_n$), we obtain $x = a^{-1}b$. This proves uniqueness and existence simultaneously. □

Computation of a^{-1}. Let $n \in \mathbb{N}$ and let $a \in \mathbb{Z}_n$. Assume that $\gcd(a, n) = 1$. Then a is invertible in $\mathbb{Z}_n$. But how can we compute a^{-1}? This can be done using the proof of Proposition 2.4(ix), as follows. Choose $x, y \in \mathbb{Z}$ such that $ax + by = 1$. Then set $a^{-1} := x$ mod n.

Exercises.

(1) Verify, using the proof of Proposition 2.4(ix), that the above method indeed yields a^{-1}.

Example 2.7. Consider $3 \in \mathbb{Z}_7$. Note that $\gcd(3,7) = 1$, and therefore 3 is invertible in $\mathbb{Z}_7$. To compute 3^{-1}, we note that $3(-2) + 7(1) = 1$, so that

$$3^{-1} = -2 \bmod 7 = 5.$$

Example 2.8. We want to solve the equation $8x = 3$ in $\mathbb{Z}_{35}$. First we compute $\gcd(35, 8)$. The Euclidean algorithm yields:

$$35 \quad 8 \quad 3 \quad 2 \quad 1 \quad 0 \quad \cdots$$

Therefore $\gcd(35, 8) = 1$. Thus 8 is invertible in $\mathbb{Z}_{35}$. Now to compute a^{-1} we need to write the $\gcd$ as a linear combination of 35 and 8:

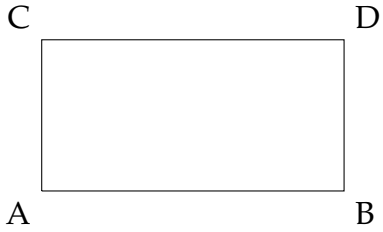
$$\begin{aligned} 1 &= 3 - 2 \\ &= 3 - (8 - 3 \times 2) = (3 \times 3) - 8 \\ &= 3 \times (35 - (4 \times 8)) - 8 = 3 \times 35 - 13 \times 8. \end{aligned}$$

It follows that $8^{-1} = -13 = 35 - 13 = 22$. Now

$$x = (3)(22) = 66 \bmod 35 = 31 \text{ in } \mathbb{Z}_{35}.$$

Symmetries of polygons. A symmetry of a polygon is a map on the set of its vertices which preserves adjacency, distances, and angles. It follows easily that every symmetric is a bijective map on the set of the nodes of the polygon. Note that the composition of any two symmetries is again a symmetry.

Example 2.9. Let us investigate the symmetries of a rectangle:



The rectangle has 4 symmetries:

(i) The identity map (which fixes every vertex), denoted by f_0.

(ii) A horizontal reflection:

$$f_1(A) = C,\ f_1(B) = D,\ f_1(C) = A,\ f_1(D) = B.$$

(iii) A vertical reflection:

$$f_2(A) = B,\ f_2(B) = A,\ f_2(C) = D,\ f_2(D) = C.$$

(iv) The composition of the two maps in (ii) and (iii). There are two ways to compose, but they both yield the same map:

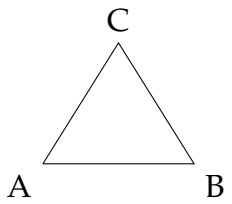
$$f_3(A) = D,\ f_3(B) = C,\ f_3(C) = B,\ f_3(D) = A.$$

The *multiplication table* of the symmetries of the rectangle is given below. The entry on row i and column j is $f_i \circ f_j$.

Example 2.10. Let us investigate symmetries of an equilateral triangle.

	f_0	f_1	f_2	f_3
f_0	f_0	f_1	f_2	f_3
f_1	f_1	f_0	f_3	f_2
f_2	f_2	f_3	f_0	f_1
f_3	f_3	f_2	f_1	f_0

FIGURE 1. The multiplication of symmetries of a rectangle.



There are 6 different symmetries:

(i) The identity map.
(ii) The 60 degree clockwise rotation ρ_1.
(iii) The 60 degree counter-clockwise rotation ρ_2.
(iv) The reflection μ_1 that fixes A.
(v) The reflection μ_2 that fixes B.
(vi) The reflection μ_3 that fixes C.

The *multiplication table* of these rotations is given in the textbook.

Permutations. A permutation is just a bijection on a finite set (usually the set of numbers 1 through n). More formally,

Definition 2.11. Let $n \geqslant 1$. A *permutation* of $\{1, \ldots, n\}$ is a bijection

$$\sigma : \{1, \ldots, n\} \to \{1, \ldots, n\}.$$

Since permutations are functions, we can compose them, as shown the the following example. Clearly the composition of any two permutations is again a permutation.

Example 2.12. Let $n = 4$. The maps $\sigma, \tau : \{1, 2, 3, 4\} \to \{1, 2, 3, 4\}$ defined by

$$\sigma(1) = 3,\ \sigma(2) = 4,\ \sigma(3) = 1,\ \sigma(4) = 2$$

and

$$\tau(1) = 1,\ \tau(2) = 3,\ \tau(3) = 2,\ \tau(4) = 4$$

are permutations of $\{1, 2, 3, 4\}$. Note that $\sigma^2 = \tau^2 = \mathrm{id}$, where id denotes the identity permutation (i.e., $\mathrm{id}(k) = k$ for all $1 \leqslant k \leqslant n$). Furthermore, $\rho := \sigma \circ \tau$ is given by

$$\rho(1) = 3,\ \rho(2) = 1,\ \rho(3) = 4,\ \rho(4) = 2.$$

We sometimes denote the permutation σ by

$$\begin{pmatrix} 1 & 2 & \cdots & n-1 & n \\ \sigma(1) & \sigma(2) & \cdots & \sigma(n-1) & \sigma(n) \end{pmatrix} \tag{2.2}$$

Definition 2.13. The set of all of the permutations of $\{1, \ldots, n\}$ is denoted by S_n.

Example 2.14. The permutation $\sigma \in S_4$ defined by

$$\sigma(1) = 4,\ \sigma(2) = 3,\ \sigma(3) = 1,\ \sigma(4) = 2$$

is denoted by

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix}.$$

Groups: definitions and examples. A *binary operation* on a set X is a map

$$\star : X \times X \to X$$

that assigns to any pair $(a, b) \in X \times X$ an element $a \star b \in X$.

Definition 2.15. A *group* is a set G equipped with a binary operation $\star : G \times G \to G$ which satisfies the following:

(i) The binary operation $\star$ is *associative*, i.e., $(a \star b) \star c = a \star (b \star c)$ for all $a, b, c \in G$.
(ii) G has an *identity element*. That is, there exists an element $e \in G$ such that $a \star e = e \star a = a$ for all $a \in G$. Sometimes e is called the *neutral element* of G.
(iii) Every $a \in G$ has an *inverse*, i.e., for every $a \in G$ there exists an element $a^{-1} \in G$ such that $a \star a^{-1} = a^{-1} \star a = e$.

Example 2.16. Here are some examples of groups:

(i) The set $\mathbb{Z}$ together with the binary operation $a \star b := a + b$ (the usual addition) is a group. The identity element of $\mathbb{Z}$ is 0 and the inverse of every $a \in \mathbb{Z}$ is $-a$.
(ii) The set $\mathbb{R}^+$ consisting of positive real numbers, equipped with the binary operation $a \star b := ab$ (the usual multiplication) is a group. The identity element of $\mathbb{R}^+$ is $1 \in \mathbb{R}^+$, and the inverse of every $a \in \mathbb{R}^+$ is $\frac{1}{a}$.
(iii) The set $\mathbb{Z}_n$ together with the binary operation $a \star b := a + b$ (the usual addition of $\mathbb{Z}_n$) is a group. The identity element of $\mathbb{Z}$ is 0. The inverse of $0 \in \mathbb{Z}_n$ is 0, and the inverse of every $a \in \mathbb{Z}_n \backslash \{0\}$ is $n - a$.

Remark 2.17. The importance of associativity of a group G is that the value of an expression $g_1 \star \cdots \star g_k$ does not depend on how it is parsed using brackets. For example, associativity implies that for all choices of elements $g_1, g_2, g_3, g_4 \in G$,

$$((g_1 \star g_2) \star g_3) \star g_4 = (g_1 \star g_2) \star (g_3 \star g_4) = g_1 \star ((g_2 \star g_3) \star g_4) = \cdots$$

We will prove this statement in Theorem 3.5.

Example 2.18. Let G be the set of symmetries of a rectangle (see Example 2.9), equipped with composition of maps considered as a binary operation. Then G is a group. Associativity of the binary operation of G is a natural property of composition of functions. The identity element of G is f_0, and we have $f_i^{-1} = f_i$ for $1 \leqslant i \leqslant 3$.

Definition 2.19. Let G be a finite group whose elements are $g_1, \ldots, g_n$ in some order. The *Cayley table* of G is the $n \times n$ table whose (i, j)-entry is $g_i \star g_j$. Note that the Cayley table depends on the chosen ordering of the elements of G.

For example, Figure 1 is the Cayley table of the group described in Example 2.18.

Definition 2.20. A group G is called *abelian* if $a \star b = b \star a$ for every $a, b \in G$.

Exercises.

(1) Verify that the set of symmetries of an equilateral triangle, equipped with the composition of maps considered as a binary operation, is a group. Is this group abelian?
(2) Verify that all of the groups in Example 2.16 are abelian.

The symmetric group S_n. Now let $n \in \mathbb{N}$. Consider the binary operation on S_n defined by

$$\sigma \star \tau := \sigma \circ \tau,$$

where the right hand side means composition of the two maps σ and τ.

Remark 2.21. If

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix} \text{ and } \tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$$

then $\sigma \star \tau = \sigma \circ \tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 2 & 4 \end{pmatrix}$

Proposition 2.22. *The set S_n equipped with the above binary operation is a group*

For the proof of this proposition we note that:

(i) Associativity of $\star$ is a natural property of composition of functions.
(ii) The identity element of $\star$ is the identity permutation $\iota \in S_n$, defined by $\iota(k) = k$ for all $1 \leqslant k \leqslant n$.
(iii) The inverse of a permutation σ is simply the inverse map σ^{-1}, defined by

$$\sigma^{-1}(k) = \ell \text{ if and only if } \sigma(\ell) = k.$$

The group of units of $\mathbb{Z}_n$. For $n \geqslant 1$, set $U(n) := \{a \in \mathbb{Z}_n \ : \ \gcd(a, n) = 1\}$. By Proposition 2.4(ix), $U(n)$ is precisely the set of invertible elements of $\mathbb{Z}_n$. Thus,

$$U(n) = \{a \in \mathbb{Z}_n \ : \ \exists_{a' \in \mathbb{Z}_n} aa' = 1\}.\}.$$

Proposition 2.23. *The set $U(n)$, equipped with the multiplication of $\mathbb{Z}_n$, is an abelian group.*

Proof. Step 1. The multiplication of $\mathbb{Z}_n$ induces a binary operation on $U(n)$: If $a, b \in U(n)$ then $ab \in U(n)$ because:

$$a \in U(n) \Rightarrow \exists_{a' \in \mathbb{Z}_n} aa' = 1,$$

and

$$b \in U(n) \Rightarrow \exists_{b' \in \mathbb{Z}_n} bb' = 1,$$

and now we have $(ab)(b'a') = a(bb')a' = a(1)a' = aa' = 1$, hence $ab \in U(n)$. Thus $U(n)$ is closed under multiplication.

Step 2. Associativity of the multiplication on $U(n)$ is inherited from $\mathbb{Z}_n$.

Step 3. The identity element of $U(n)$ is $1 \in \mathbb{Z}_n$. Note that $1 \in U(n)$ because $\gcd(1, n) = 1$.

Step 4. For $a \in U(n)$, the multiplicative inverse a' also belongs to $U(n)$ (because it is also invertible).

Step 5. Finally, $U(n)$ is abelian because of Proposition 2.4(ii). □

Example 2.24. We have $U(12) = \{1, 5, 7, 11\}$. The identity element is 1. We have $5^{-1} = 5$, $7^{-1} = 7$, $11^{-1} = 11$.

Example 2.25. We have $U(11) = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ (note that 11 is prime). We have $2^{-1} = 6$ and $3^{-1} = 4$, etc.

The vector space $\mathbb{R}^n$. The standard addition of vectors is a binary operation on the vector space $\mathbb{R}^n = \{(x_1, \ldots, x_n) \ : \ \forall_{1 \leqslant i \leqslant n} x_i \in \mathbb{R}\}$:

$$(x_1, \ldots, x_n) + (y_1, \ldots, y_n) = (x_1 + y_1, \ldots, x_n + y_n).$$

Proposition 2.26. *$\mathbb{R}^n$ equipped with the standard addition of vectors is an abelian group.*

Proof. We leave the proof as an exercise. We only remark that the identity element of $\mathbb{R}^n$ is the vector $(0, \ldots, 0)$, and the inverse of $(x_1, \ldots, x_n)$ is $(-x_1, \ldots, -x_n)$. □

The matrix group $\mathrm{GL}_2(\mathbb{R})$. Let $\mathrm{GL}_2(\mathbb{R})$ be the set of invertible 2×2 matrices with real entries. Then $\mathrm{GL}_2(\mathbb{R})$ is a group (its binary operation is the usual matrix multiplication). It is *not* an abelian group because for example

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

2.1 Digression on complex numbers. Numbers of the form $a + bi$, where $a, b \in \mathbb{R}$ and $i = \sqrt{-1}$ (so that $i^2 = -1$), are called *complex numbers*. We add complex numbers componentwise:

$$(a + bi) + (c + di) = (a + c) + (b + d)i.$$

We multiply complex numbers just like ordinary numbers, taking into account that $i^2 = -1$:

$$(a + bi)(c + di) = ac + bdi^2 + bci + adi = (ac - bd) + (ad + bc)i.$$

For example,

$$(1 - 3i) + (2 + 4i) = 3 + i \text{ and } (1 - i)(1 + 2i) = 3 + i.$$

The mutiplicative inverse of a nonzero complex number $a + bi$ is

$$\frac{1}{a + bi} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2} i.$$

The quaternion group. Consider the matrices

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{I} := \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad \mathbf{J} := \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}, \quad \mathbf{K} := \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix},$$

where $i = \sqrt{-1}$. Then we have

$$\mathbf{I}^2 = \mathbf{J}^2 = \mathbf{J}^2 = \mathbf{K}^2 = -I,$$

and

$$\mathbf{IJ} = \mathbf{K} = -\mathbf{JI}, \quad \mathbf{JK} = \mathbf{I} = -\mathbf{KJ}, \quad \mathbf{KI} = \mathbf{J} = -\mathbf{IJ}.$$

The following proposition is now starighforward:

Proposition 2.27. *The set of matrices* $Q := \{\pm I, \pm\mathbf{I}, \pm\mathbf{J}, \pm\mathbf{K}\}$ *is a group.*

We remark that the group Q is not abelian.

Exercises.

In the quaternion group Q, What are the inverses of $\mathbf{I}$, $\mathbf{J}$, and $\mathbf{K}$?

ELEMENTARY PROPERTIES OF GROUPS

In this section we study some fundamental properties of general groups.

Proposition 3.1. *Let G be any group.*

(i) *The identity element of G is unique.*

(ii) *For every $a \in G$, the inverse of a is unique.*

Proof. (i) Let $e, e' \in G$ be two identity elements in G. Then

$$ee' = e \text{ because } e' \text{ is an identity element}$$

and

$$ee' = e' \text{ because } e \text{ is an identity element.}$$

From these relations it follows that $e = e'$.

(ii) Let $g \in G$, and assume that $h_1, h_2 \in G$ are two inverses of g. As usual let $e \in G$ be the identity element. Then

$$h_1 = h_1 e = h_1(gh_2) = (h_1 g)h_2 = eh_2 = h_2.$$

□

Proposition 3.2. *Let G be any group.*

(i) *For all $a, b \in G$ we have $(ab)^{-1} = b^{-1}a^{-1}$.*

(ii) *For every $a \in G$ we have $(a^{-1})^{-1} = a$.*

Proof. (i) We have $(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-} = a(e)a^{-1} = aa^{-1} = e$. Similarly, $(b^{-1}a^{-1})(ab) = e$. Thus the inverse of ab is $b^{-1}a^{-1}$.

(ii) We have $(a^{-1})a = a(a^{-1}) = e$, hence by uniqueness of inverse, the inverse of a^{-1} is equal to a. □

Proposition 3.3. *Let G be any group, and let $a, b \in G$. Then the equations $ax = b$ and $xa = b$ have unique solutions in G.*

Proof. Assume that $ax = b$. Multiplying on the left by a^{-1}, we obtain

$$a^{-1}(ax) = a^{-1}b.$$

Thus $x = ex = (a^{-1}a)x = a^{-1}(ax) = a^{-1}b$. Furthermore, $x = a^{-1}b$ satisfies the equation $ax = b$. Thus this equation has a unique solution. The converse is proves similarly. □

Proposition 3.4. (Left and Right Cancellation Laws) *Let G be a group and let $a, b, c \in G$.*

(i) *If $ab = ac$ then $b = c$.*

(ii) *If $ba = ca$, then $b = c$.*

Proof. (i) Assume that $ab = ac$. Multiplying both sides by a^{-1}, we have

$$a^{-1}(ab) = a^{-1}(ac).$$

By associativity of G, we can write the latter equality as $(a^{-1}a)b = (a^{-1}a)c$. Thus $eb = ec$, hence $b = c$.

(ii) The argument is similar and left as an exercise. □

By a group's defining properties we know that $g_1(g_2g_3) = g_1(g_2g_3)$ for all elements $g_1, g_2, g_3 \in G$. Is it also true that $g_1(g_2(g_3g_4)) = ((g_1g_2)g_3)g_4$? The next theorem answers this question.

Theorem 3.5. (General Associativity Theorem) *Let G be a group and let $g_1, \ldots, g_n \in G$. Then the value of the product $g_1 \star \cdots \star g_n$ is independent of the way brackets are placed.*

Proof. We prove the statement by strong induction on n.
Step 1. The statement is trivial for $n \leqslant 3$.
Step 2. Assume the statement holds for all $n \leqslant n_0$. Now assume $n = n_0 + 1$ and consider a way of parsing $g_1 \star \cdots \star g_n$ by brackets. There exists an outermost part of brackets, i.e., some $1 \leqslant k \leqslant n-1$ such that the expression looks like

$$(g_1 \star \cdots \star g_k) \star (g_{k+1} \star \cdots \star g_n).$$

By induction hypothesis we can assume that the left bracket is equal to

$$((g_1 \star g_2) \star \cdots) \star g_k)$$

and the right bracket is equal to

$$g_{k+1} \star (g_{k+1} \star \cdots \star (g_{n-1} \star g_n)).$$

Thus we have

$$\begin{aligned}
&(g_1 \star \cdots \star g_k) \star (g_{k+1} \star \cdots \star g_n) \\
&\qquad = ((\underbrace{g_1 \star \cdots \star g_{k-1}}_{a}) \star \underbrace{\quad g_k \quad}_{b}) \star \underbrace{(g_{k+1} \star (\underbrace{g_{k+1} \star \cdots \star g_n}_{c}))}_{c} \\
&\qquad = (\underbrace{g_1 \star \cdots \star g_{k-1}}_{a}) \star (\underbrace{\quad g_k \quad}_{b} \star (\underbrace{g_{k+1} \star \cdots \star g_n}_{c})),
\end{aligned}$$

where we are using $(ab)c = a(bc)$. Continuing in this fashion, we obtain

$$(g_1 \star \cdots \star g_k) \star (g_{k+1} \star \cdots \star g_n) = g_1 \star (g_2 \star (g_3 \star \cdots \star g_n)).$$

Thus the value of any bracketed expression $(g_1 \star \cdots \star g_k) \star (g_{k+1} \star \cdots \star g_n)$ is equal to the right-aligned bracketing $g_1 \star (g_2 \star (g_3 \star \cdots \star g_n))$. In particular, we have shown using the induction hypothesis that the value of bracketed expressions in any $g_1, \cdots, g_n \in G$ for $n = k+1$ is independent of the choice of the bracketing. This completes the proof by induciton. □

Powers of an element $g \in G$. Let G be a group. For every $g \in G$ and $n \in \mathbb{Z}$ we define g^n as follows:

$$g^n := \begin{cases} \underbrace{g \cdots g}_{n \text{ times}} & \text{if } n > 0, \\ e & \text{if } n = 0, \\ (g^{-1})^n & \text{if } n < 0. \end{cases}$$

Example 3.6. Let $G = (\mathbb{Z}, +)$; this notation means the binary operation of $\mathbb{Z}$ is the usual addition. Then in this notation $g^m = m \cdot g$ for $g \in G = \mathbb{Z}$ and $m \in \mathbb{Z}$. For instance, $3^5 = 15$ and $(-2)^4 = -8$. This notation should *not* be confused with the usual exponentiation of numbers.

Example 3.7. Let $G = \mathbb{Z}_{24}$. Then for $g \in G$ and $n \in \mathbb{Z}$ we have

$$g^n := ng \bmod 24.$$

Again, this notation should not be confused with the usual exponentiation on $\mathbb{Z}_{24}$, which pertains to the binary operator of multiplication on $\mathbb{Z}_{24}$. For example, $3^2 := 3 + 3 = 6$ and $2^3 := 2 + 2 + 2 = 6$.

Example 3.8. Let $G = U(24)$. This time the binary operation of G is the restriction of the multiplication of $\mathbb{Z}_{24}$. Therefore g^n is consistent with the usual notion of exponentiation that is obtained from the multiplication of $\mathbb{Z}_{24}$. Thus, for instance $5^2 = 25$ modulo $24 = 1$.

The concept of the power of an element can be a bit confusing when working with arithmetic modulo n:

- Suppose that $G = \mathbb{Z}_n = \{0, 1, \ldots, n-1\}$. Then the identity element is 0. In this group, the binary operation is "$+$". Therefore a^k for $a \in G$ and $k \in \mathbb{N}$ means

$$\underbrace{a \star \cdots \star a}_{k \text{ times}} = \underbrace{a + \cdots + a}_{k \text{ times}}.$$

  For example, if $n = 8$ and $a = 3$ and $k = 2$, then a^k means $a + a = 3 + 3 = 6$.
- Now suppose that $G = U(n)$. Note that $U(n) \subset \mathbb{Z}_n$, but the group operations of $\mathbb{Z}_n$ are $U(n)$ are <u>different</u>. The identity element of G is 1.

  The binary operation of G is induced by multiplication modulo n, which does <u>not</u> make $\mathbb{Z}_n$ a group. For $a \in G$ and $k \in \mathbb{N}$, the element a^k in G means

$$\underbrace{a \star \cdots \star a}_{k \text{ times}} = \underbrace{a \times \cdots \times a}_{k \text{ times}}.$$

  For example, if $n = 8$ and $a = 3$ and $k = 2$, then a^k means $a \times a = 3 \times 3 = 1$.

Theorem 3.9. *Let G be a group. Then the following statements hold:*

(i) $g^m g^n = g^{m+n}$ *for all* $g \in G$ *and* $m, n \in \mathbb{Z}$.
(ii) $(g^m)^n = g^{mn}$ *for all* $g \in G$ *and* $m, n \in \mathbb{Z}$.
(iii) $(gh)^{-n} = (h^{-1}g^{-1})^n$ *for all* , $h \in G$ *and* $n \in \mathbb{Z}$.

Proof. (i) We argue by considering various cases. Let us first assume that $m, n > 0$. Then

$$g^m g^n = (\underbrace{g \cdots g}_{m \text{ times}})(\underbrace{g \cdots g}_{n \text{ times}}) = \underbrace{g \cdots g}_{m+n \text{ times}} = g^{m+n}.$$

Next assume that $m > 0$ and $n = 0$. Then $g^{m+n} = g^m = g^m e = g^m g^n$. The argument for the case $m = 0$ and $n > 0$ is similar.

Next assume that $m > 0$ and $n < 0$. Then $n = -k$ for some $k > 0$. Thus

$$g^m g^n = (\underbrace{g \cdots g}_{m \text{ times}})(\underbrace{g^{-1} \cdots g^{-1}}_{k \text{ times}}) = \begin{cases} g^{m-k} = g^{m+n} \text{ if } m \geqslant k, \\ (g^{-1})^{k-m} = g^{m-k} \text{ if } m < k. \end{cases}$$

We leave the remaining cases as an exercise. □

Cartesian product of groups. Let G_1 and G_2 be two groups. We equip the Cartesisn product

$$G_1 \times G_2 := \{(g_1, g_2) \ : \ g_1 \in G_1, \ g_2 \in G_2\}$$

with the binary operation defined by

$$(g_1, g_2)(h_1, h_2) := (g_1 h_1, g_2 h_2) \ \text{ for } g_1, h_1 \in G_1, \ g_2, h_2 \in G_2.$$

Proposition 3.10. *$G_1 \times G_2$ is a group.*

Proof. We leave the details as an exercise. We only mention that the identity element of this group is (e_1, e_2), where e_i is the identity element of G_i for $i = 1, 2$, and the inverse of an element (g_1, g_2) is (g_1^{-1}, g_2^{-1}). □

Example 3.11. The set $\mathbb{Z}_6 \times \mathbb{Z}_4$ is a group. We have

$$(2, 3)^{-1} = (4, 1) \text{ and } (3, 3)(3, 3) = (3 + 3 \text{ modulo } 6, 3 + 3 \text{ modulo } 4) = (0, 2).$$

Isomorphisms of groups. Of course changing the labelings/names of elements of a group does not change its intrinsic structure. Therefore it is natural to only consider groups up to isomorphisms:

Definition 3.12. Let G_1 and G_2 be two groups. An *isomorphism* $f : G_1 \to G_2$ is a bijection such that $f(g_1 g_2) = f(g_1) f(g_2)$ for all $g_1, g_2 \in G_1$. If an isomorphism exists between G_1 and G_2, then we say G_1 and G_2 are *isomorphic*, and we write $G_1 \cong G_2$.

Remark 3.13. For an isomorphism, we have $f(e_1) = e_2$ where $e_i \in G_i$ denotes the identity element. This is because we ave

$$f(e_1) = f(e_1^2) = f(e_1)^2, \text{ or equivalently, } f(e_1)e_2 = f(e_1)f(e_1).$$

Now the left cancellation law in G_2 implies $e_2 = f(e_1)$.

Furthermore, $f(g^{-1}) = f(g)^{-1}$. This is because we have

$$f(g)f(g^{-1}) = f(gg^{-1}) = f(e_1) = e_2 \text{ and similarly } f(g^{-1})f(g) = e_1.$$

By uniqueness of inverses in G_2, we obtain $f(g)^{-1} = f(g^{-1})$.

Example 3.14. Consider the set $H : \{-1, +1\}$ with the usual multiplication, i.e.,

$$(1)(1) = 1, \ (1)(-1) = (-1)(1) = -1, \ (-1)(-1) = 1.$$

Then H is a group. Furthermore, the map

$$\phi : \mathbb{Z}_2 \to H,$$

given by $\phi(0) = 1$ and $\phi(1) = -1$ is an isomorphism.

Example 3.15. There is an isomorphism $\phi : \mathbb{Z}_2 \to U(3)$: Recall that

$$\mathbb{Z}_2 = \{0, 1\} \text{ and } U(3) = \{1, 2\}.$$

The isomorphism ϕ is given by

$$\phi(0) = 1 \text{ and } \phi(1) = 2.$$

Example 3.16. The map $\phi : \mathbb{Z}_2 \times \mathbb{Z}_3 \to \mathbb{Z}_6$ given by

$$\phi(0,0) = 0,\ \phi(0,1) = 4,\ \phi(0,2) = 2,\ \phi(1,0) = 3,\ \phi(1,1) = 1,\ \phi(1,2) = 5$$

is an isomorphism. To check this, you need to verify that

$$\phi(g_1 g_2) = \phi(g_1)\phi(g_2)$$

for every pair of elements g_1, g_2 chosen from $\mathbb{Z}_2 \times \mathbb{Z}_3$. Since this group has 6 elements, there are $6 \times 6 = 36$ pairs g_1, g_2 that one can choose, and therefore there are 36 relations to check. This is a tedious task. Later, we will study this isomorphism more conceptually.

Example 3.17. Let $k \in \mathbb{Z}_5$ such that $k \neq 0$. Then the map $\phi : \mathbb{Z}_5 \to \mathbb{Z}_5$, defined by

$$\phi(i) = ki,$$

is an isomorphism.

Exercises.

(1) Verify that the map in Examples 3.15, 3.16 and 3.17 are indeed isomorphisms.
(2) Is the map in Example 3.16 the only possible isomorphism?

GROUPS OF SMALL CARDINALITY

As shown below, it is not difficult to classify all groups of small cardinality up to isomorphism. Clearly up to isomorphism there is only one group with one element.

Proposition 4.1. *Let G be a group such that $|G| = 2$. Then $G \cong \mathbb{Z}_2$.*

Proof. Step 1. Since G has an identity element, it is of the form $G = \{e, a\}$ where $a \neq e$. Now if $a^2 = a$ then by cancellation laws we obtain $a = e$, which is impossible. Thus we should have $a^2 = e$. Step 2. Now consider the map

$$\phi : \mathbb{Z}_2 \to H, \;\; \phi(0) = e \text{ and } \phi(1) = a.$$

It is straightforward to verify that ϕ is an isomorphism. □

Proposition 4.2. *Let G be a group such that $|G| = 3$. Then $G \cong \mathbb{Z}_3$.*

Proof. As before we can express G as $G = \{e, a, b\}$ where e is the identity element.
Step 1. We show that $ab = e$. Assume the contrary. Then we should be in one of the following two cases:

(i) $ab = a$. Then cancellation implies $b = e$, which is a contradiction.
(ii) $ab = b$. Then cancellation implies $a = e$, which is a contradiction.

Step 2. We show that $a^2 = b$. Assume the contrary. Then we should be in one of the following two cases:

(i) $a^2 = a$. By cancellation this implies $a = e$ which is a contradiction.
(ii) $a^2 = e$. Then by Step 1 we have $a^2 = ab$, hence by cancellation we obtain $a = b$ which is a contradiction.

Step 3. Similar to Step 2, we can show that $b^2 = a$. Thus the distinct element of G are $G = \{e, a, a^2\}$, and we have $a^3 = a(a^2) = (a^2)a = e$ and $(a^2)(a^2) = a$.
Step 4. We can now easily verify that the map

$$\phi : \mathbb{Z}_3 \to G$$

defined by

$$\phi(0) = e, \; \phi(1) = a, \; \phi(2) = a^2$$

is an isomorphism. □

Before we describe all groups of order 4, we prove a general proposition:

Proposition 4.3. *Let G be any finite group. Then every row and every column in the Cayley table of G contains a permutation of the elements of G.*

Proof. The column of the Cayley table corresponding to $a \in G$ contains elements of the form xa for $x \in G$. Now from Proposition 3.3 it follows that every $g \in G$ appears precisely once in this column. The argument for the rows in analogous. □

Proposition 4.4. *Let G be a group such that $|G| = 4$. Then either $G \cong \mathbb{Z}_4$ or $G \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.*

Proof. We can write the elements of G as $G = \{e, a, b, c\}$ where e is the identity element.
Step 1. Similar to the proof of Proposition 4.2, we know that $ab \neq a$ and $ab \neq b$. Thus we have $ab \in \{e, c\}$. We consider two separate cases:

Case I: $ab = e$.
Step I.1. We show $ac = b$. Assume the contrary. Then we are in one of the following the following cases:

- $ac = e$. Then $ac = ab$, hence $c = b$ which is a contradiction.
- $ac = a$. Then by cancellation $c = e$ which is a contradiction.
- $ac = c$. Then by cancellation $a = e$ which is a contradiction.

Step I.2. We show $a^2 = c$. Assume the contrary. Then we are in one of the following cases:

- $a^2 = e$: Then $a^2 = ab$, hence $a = b$, which is a contradiction.
- $a^2 = a$: Then $a = e$ which is a contradiction.
- $a^2 = b$: Then $a^2 = ac$, hence $a = c$ which is a contradiction.

Step I.3. We show $bc = a$. Assume the contrary. Then we are in one of the following cases:

- $bc = b$. Then by cancellation $c = e$ which is a contradiction.
- $bc = e$. Then $bc = ba$ (since $ab = e$ implies $ba = e$), hence by cancellation $c = a$ which is a contradiction.
- $bc = c$. Then by cancellation $b = e$, which is a contradiction.

Step I.4. So far the Cayley table of G looks like

	e	a	b	c
e	e	a	b	c
a	a	c	e	b
b	b	e		a
c	c			

By Proposition 4.3 we can complete the rest of the Cayley table to the following:

	e	a	b	c
e	e	a	b	c
a	a	c	e	b
b	b	e	c	a
c	c	b	a	e

We can now check that that the map $\phi : \mathbb{Z}_4 \to G$ defined by

$$\phi(0) = e,\ \phi(1) = a,\ \phi(2) = c,\ \phi(3) = b$$

is an isomorphism.

Case II: $ab \neq e$.
Step II.1. By symmtry, we can assume that $ba \neq e$, $bc \neq e$, $cb \neq a$, $ca \neq e$, and $ac \neq e$ (otherwise after relabelling the variables we will be back in Case I).
Step II.2. It follows (since $ab \in \{e, c\}$, etc.) that

$$ab = c = ba,\ bc = a = ab,\ ca = b = ac.$$

Step II.3. So far the Cayley table of G looks like

	e	a	b	c
e	e	a	b	c
a	a		c	b
b	b	c		a
c	c	b	a	

By Proposition 4.3 we can complete the rest of the Cayley table to the following:

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	a
c	c	b	a	e

It is now a straightforward calculation that the map

$$\phi : \mathbb{Z}_2 \times \mathbb{Z}_2 \to G$$

defined by

$$\phi(0,0) = e,\ \phi(1,0) = a,\ \ \phi(0,1) = b,\ \ \phi(1,1) = c$$

is an isomorphism. □

Remark 4.5. The groups $\mathbb{Z}_4$ and $\mathbb{Z}_2 \times \mathbb{Z}_2$ are not isomorphic. In fact suppose that there exists an isomorphism

$$\phi : \mathbb{Z}_4 \to \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Set $(a, b) = \phi(1)$. Now on the one hand $1 + 1 \neq 0$, hence

$$\phi(1 + 1) \neq \phi(0),$$

since ϕ is a bijection. On the other hand for any $a \in \mathbb{Z}_2$ we have $a + a = 0$, hence

$$\phi(1 + 1) = \phi(1)\phi(1) = (a, b)(a, b) = (a + a, b + b) = (0, 0) = \phi(0),$$

which is a contradiction.

SUBGROUPS

Definition 5.1. Let G be any group. A subset $H \subseteq G$ is called a subgroup of G if the restriction of the binary operation of G to H makes H a group. If H is a subgroup of G, then we write $H \leqslant G$.

Example 5.2.
(i) $\mathbb{Z}$ is a subgroup of $\mathbb{Q}$ (the binary operation on both is the usual addition).
(ii) For any group G, the subset $\{e\}$ where $e = e_G$ is a subgroup. It is called the trivial subgroup of G.
(iii) $U(5)$ is not a subgroup of $\mathbb{Z}_5$ even though $U(5) \subseteq \mathbb{Z}_5$.
(iv) The set $\{\pm I, \pm \mathbf{I}\}$ is a subgroup of the quaternion group Q of Example 2.1.
(v) $\mathbb{Z} \times \{0\}$ is a subgroup of $\mathbb{Z} \times \mathbb{Z}$.
(vi) $\mathbb{Z} \times \{1\}$ is not a subgroup of $\mathbb{Z} \times \mathbb{Z}$.
(vii) Let $n \geqslant 2$. The set $\{\sigma \in S_n \ : \ \sigma(n) = n\}$ is a subgroup of S_n.

Remark 5.3.
(i) Note that if $H \subseteq G$ is a subgroup, then the identity elements e_H of H and e_G of G coincide: this is because

$$(e_H)^2 = e_H = e_H e_G,$$

and the cancellation law (in G) implies $e_H = e_G$.
(ii) It is also straightforward to verify that the inverse in H of an element $g \in H$ is the same as its inverse in G. We leave the verification of this statement as an exercise.

Theorem 5.4. (Subgroup Test) *Let G be any group and let $H \subseteq G$ be any subset of G. Then H is a subgroup of G if and only if the following are satisfied.*
(i) $e \in G$.
(ii) For all $h, h' \in H$ we have $hh' \in H$.
(iii) For all $h \in H$ we have $h^{-1} \in H$, where h^{-1} denotes the inverse of h in G.

Proof. Assume that conditions (i)–(iii) are satisfied. We show that H is a subgroup of G.
Step 1. From (ii) it follows that the binary operation of G induces a binary operation on H. Associativity of this operation follows from the fact that G is a group.

Step 2. From (i) it follows that H has an identity element (the identity of G is also an identity of H).

Step 3. From (iii) it follows that every $h \in H$ has an inverse in H.

Thus all of the conditions of Definition 2.15 are satisfied for H.
Conversely, if H is a subgroup of G, then it is straightforward to verify the conditions (i)–(iii). We leave this verification as an exercise.

□

Exercises.

(1) Using Theorem 5.4 show that the following sets of elements is a subgroup of the given G.
   (i) $\{X \in \mathrm{GL}_n(\mathbb{R}) \ : \ \det(X) = 1\}$, $G = \mathrm{GL}_n(\mathbb{R})$.
   (ii) $d\mathbb{Z} := \{kd \ : k \in \mathbb{Z}\} = \{0, \pm d, \pm 2d, \pm 3d, \ldots\}$ where d is a fixed integer, $G = \mathbb{Z}$.
(2) Determine all of the subgroups of $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Theorem 5.5. (Subgroup Test - Concise Version) (Subgroup Test - Finite Groups Version) *Let G be any group and let $H \subseteq G$ be a nonempty subset of G. Then H is a subgroup of G if and only if H satisfies the following property:*

$$\textit{for all } g, h \in H \textit{ we have } gh^{-1} \in H.$$

Proof. If H is a subgroup of G, then it satisfis the property stated in the theorem. We leave this straightforward argument as an exercise.

Conversely, assume that H is a nonempty subset that satisfies the given condition, i.e., for all $g, h \in H$ we have $gh^{-1} \in H$. To prove that H is a subgroup, we verify that the conditions (i)–(iii) of Theorem 5.4 are satisfied:

Step 1. Since H is nonempty, we can find $h_0 \in H$. Then $e = h_0 h_0^{-1} \in H$, thus assumption (i) of Theorem 5.4 holds.

Step 2. We show that for every $a \in H$ we have $a^{-1} \in H$. Since $e \in H$, setting $g := e$ and $h := a$, we obtain

$$gh^{-1} = ea^{-1} = a^{-1} \in H.$$

Note that this verifies assumption (iii) of Theorem 5.4.

Step 3. Finally, we show that for all $a_1, a_2 \in H$ we have $a_1 a_2 \in H$. This verifies assumption (ii) of Theorem 5.4.

Let $a_1, a_2 \in H$. Then by Step 2 we have $(a_2)^{-1} \in H$. Now setting $g = a_1$ and $a_2 = (a_2)^{-1}$ we obtain $gh^{-1} = a_1 a_2 \in H$. □

Theorem 5.6. (Subgroup Test - Finite Groups Version) *Let G be a finite group and let $H \subseteq G$ be a nonempty subset of G. Then H is a subgroup of G if and only if H is closed under the binary operation of G, i.e.,*

$$\textit{for all } h, h' \in H \textit{ we have } hh' \in H.$$

Proof. We show that assumptions (i) and (iii) of Theorem 5.4 are satisfied.
Step 1. Since $H \neq \varnothing$, we can choose some $h_0 \in H$. Now consider the set

$$S(h_0) := \{h_0, h_0^2, h_0^3, \ldots\} = \{h_0^k \ : \ k \in \mathbb{N}\}.$$

From the assumption of the theorem it follows that $S(h_0) \subseteq H$.
Step 2. Since H is finite, $S(h_0)$ should also be finite. Thus, there exist integers $k > \ell \geqslant 1$ such that $h_0^k = h_0^\ell$. Multiplying by $h_0^{k\ell}$, we obtain that

$$h_0^m = e \text{ where } m := k - \ell \in \mathbb{N}.$$

Thus $e \in S$. Consequently, $e \in H$.

Step 3. Finally, we show that for every $h \in H$, we have $h^{-1} \in H$. To this end, note that as in Step 2, we have h^m for some $m \in \mathbb{N}$. If $m = 1$, then $h = e$, and clearly $h^{-1} = e \in H$. If $m > 1$, then

$$h^{-1} = eh^{-1} = h^m h^{-1} = h^{m-1} \in S,$$

and therefore $h^{-1} \in H$. □

Example 5.7. Theorem 5.6 is not correct if G is not finite. For example, if $G = \mathbb{Z}$ (equipped with addition) and $H = \mathbb{N}$, then H satisfies the condition of Theorem 5.6, but H is not a subgroup of $\mathbb{Z}$ because $1 \in H$ but $-1 \notin H$.

Proposition 5.8. *Let G be a group and let $H \leqslant G$ and $K \leqslant G$. Then $H \cap K \leqslant G$.*

Proof. We verify the assumptions of Theorem 5.4 for $H \cap K$:

Step 1. Since $e \in H$ and $e \in K$, we have $e \in H \cap K$.

Step 2. For $a, b \in H \cap K$, we have $ab \in H$ because a and b are in H, and we have $ab \in K$ because a and b are in K. Thus $ab \in H \cap K$.

Step 3. By a similar argument as in Step 2, it follows that for $h \in H \cap K$ we have $h^{-1} \in H \cap K$.

□

Example 5.9. If H, K are subgroups of a group G, then $H \cup K$ is <u>not</u> necessarily a subgroup of G: Set $G = \mathbb{Z}_6$ and

$$H := \{0, 3\} \text{ and } K := \{0, 2, 4\}.$$

Then H and K are subgroups of $\mathbb{Z}_6$ but $H \cup K = \{0, 2, 3, 4\}$ is not a subgroup because $2 + 3 = 5 \notin H \cup K$.

The centre of a group. The centre and centralizer of an element of a group are some of the most ubiquitous examples subgroups of groups.

Definition 5.10. Let G be any group. The *centre* of G is the subset of G defined by

$$Z(G) := \{g \in G \ : \ gh = hg \text{ for every } h \in G.\}\,.$$

In other words, the centre of G is the set of elements of G which commute with every element of G.

Theorem 5.11. *Let G be any group. Then the centre of G is an abelian subgroup of G.*

Proof. We use the concise subgroup test (Theorem 5.5):

Step 1. Clearly $eg = ge = g$ for all $g \in G$, hence $e \in Z(G)$. In particular, $Z(G) \neq \varnothing$.

Step 2. Assume that $a, b \in Z(G)$. We want to show that $ab^{-1} \in G$. Then for every $g \in G$ we have

$$bg = gb.$$

Multiplying on both sides by b^{-1}, we obtain

$$gb^{-1} = b^{-1}g \text{ for all } g \in G.$$

It follows that for all $g \in G$ we hvve

$$(ab^{-1})g = a(b^{-1}g) = a(gb^{-1}) = (ag)b^{-1} = (ga)b^{-1} = g(ab^{-1}.$$

Thus $ab^{-1} \in Z(G)$.
Step 3. Finally, $Z(G)$ is abelian, because the definition of $Z(G)$ implies that the relation $ab = ba$ holds for all $a, b \in Z(G)$. □

Example 5.12. (i) IF G is an abelian group, then $Z(G) = G$.
(ii) Let $Q = \{\pm I, \pm\mathbf{I}, \pm\mathbf{J}, \pm\mathbf{K}\}$ be the quaternion group (see Proposition 2.27). Then $Z(Q) = \{\pm I\}$.
(iii) The centre of S_3 is the trivial subgroup of S_3. That is, $Z(S_3) = \{e\}$.

Exercises.

(1) Verify the claim of Example 5.12(iii).
(2) Let G and H be two groups, and let $f; G \to H$ be an isomorphism between G and H. Prove that for every subgroup G_1 of G, the set

$$f(G_1) := \{f(g) \ : \ g \in G_1\}$$

is a subgroup of H.

Definition 5.13. Let G be any group, and let $a \in G$ be any element. The *centralizer* of a, denoted by $C_G(a)$, is defined as

$$C_G(a) := \{g \in G \ : \ ag = ga\}.$$

Proposition 5.14. *Let G be any group. Then $C_G(a)$ is a subgroup of G.*

Proof. We use the Concise Subgroup Test. Note that $C_G(a) \neq \varnothing$ since $e \in C_G(a)$.

Step 1. Let $x, y \in C_G(a)$. We need to show that $xy^{-1} \in C_G(a)$. We have

$$\begin{aligned} ya = ay \ \Rightarrow \ y^{-1}(ya)y^{-1} = y^{-1}(ay)y^{-1} \ &\Rightarrow \ (y^{-1}y)ay^{-1} = y^{-1}a(yy^{-1}) \\ &\Rightarrow \ eay^{-1} = y^{-1}ae \ \Rightarrow \ ay^{-1} = y^{-1}a. \end{aligned}$$

Step 2. Now we have

$$(xy^{-1})a = x(y^{-1}a) = x(ay^{-1}) = (xa)y^{-1} = (ax)y^{-1} = a(xy^{-1}).$$

This implies that $xy^{-1} \in C_G(a)$. Thus the assumption of Theorem 5.5 holds. □

Example 5.15. (i) Let $Q = \{\pm I, \pm\mathbf{I}, \pm\mathbf{J}, \pm\mathbf{K}\}$ be the quaternion group (see Proposition 2.27). Then $C_Q(\mathbf{J}) = \{\pm I, \pm J\}$.
(ii) Let $G = \mathrm{GL}_2(\mathbb{R})$, and set $A := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Then

$$C_G(A) = \left\{ \begin{bmatrix} a & b \\ 0 & a \end{bmatrix} \ : \ a \in \mathbb{C}^* \text{ and } b \in \mathbb{C}. \right\}.$$

To prove the latter claim, take $B = \begin{bmatrix} x & y \\ z & t \end{bmatrix} \in C_G(A)$. Then

$$AB = BA \text{ implies } \begin{bmatrix} x+z & y+t \\ z & t \end{bmatrix} = \begin{bmatrix} x & x+y \\ z & z+t \end{bmatrix}.$$

It follows that $x + y = x$, hence $z = 0$, and $y + t = x + y$, hence $t = x$. Thus B is of the form

$$\begin{bmatrix} x & y \\ 0 & x \end{bmatrix}.$$

Since B is invertible, we have $x \neq 0$. Furthermore, by a direct calculation we can verify that all matrices of the latter form belong to $C_G(A)$.

CYCLIC GROUPS

Definition 6.1. (i) We say a group G is a *cyclic* group if there exists an element $g \in G$ such that
$$G = \{g^k \ : \ k \in \mathbb{Z}\}.$$
(ii) Let G be any group. A subgroup H of G which is of the form
$$H = \{h^k \ : \ k \in \mathbb{Z}\}$$
is called a *cyclic* subgroup of G generated by h. Such a cyclic subgroup is denoted by $\langle h \rangle$.

Example 6.2. (i) The group $\mathbb{Z}_n$ is cyclic because $\mathbb{Z}_n = \langle 1 \rangle$.
(ii) Note that the generator of a cyclic group is not necessarily unique. For instance, we have
$$\mathbb{Z}_6 = \langle 1 \rangle = \langle 5 \rangle.$$
(iii) The group $\mathbb{Z}$ is cyclic. Note that $\mathbb{Z} = \langle 1 \rangle = \langle -1 \rangle$.
(iv) The set $S := \{(a, 0) \ : \ a \in \mathbb{Z}_n\}$ is a cyclic subgroup of $\mathbb{Z}_n \times \mathbb{Z}_n$. Note that $S \cong \mathbb{Z}_n$, where the isomorphism is the natural map $(a, a) \mapsto a$.
(v) The group $U(9) = \{1, 2, 4, 5, 7, 8\}$ is cyclic. Indeed $U(9) = \langle 2 \rangle$. However, note that we have $\langle 4 \rangle = \{1, 4, 8\} \subsetneq U(9)$.

Our next goal is to classify all cyclic groups. We begin with the following definition.

Definition 6.3. Let G be any group, and let $g \in G$.
(i) If there exists some $k \in \mathbb{N}$ such that $g^k = e$, then we define the *order of* g to be $o(g) := \min\{k \in \mathbb{N} \ : \ g^k = e\}$.
(ii) If $g^k \neq e$ for all $k \in \mathbb{N}$, then we set $o(g) = \infty$.

Example 6.4. (i) The order of 3 in $\mathbb{Z}_{12}$ is 4. That is, $o(3) = 4$.
(ii) The order of 3 in $\mathbb{Z}$ is infinity. That is, $o(3) = \infty$.
(iii) $o(e) = 1$ for every group G.

Exercises.

(1) let G be any group. Show that $o(g) = o(g^{-1})$ for all $g \in G$.
(2) Assume that G is an abelian group and $g, h \in G$ such that $o(g) = 2$ and $o(h) = 3$. Prove that $o(gh) = 6$.

Lemma 6.5. *Let G be a finite group. Then $o(g) < \infty$ for every $g \in G$.*

Proof. Fix $g \in G$. The set $S(g) := \{g, g^2, g^3, \ldots\} = \{g^k \ : \ k \in \mathbb{N}\}$ is a subset of G, hence in particular it is finite. In particular, there exist integers $k > \ell \geqslant 1$ such that $g^k = g^\ell$. It follows that $g^{k-\ell} = g^k g^{-\ell} = e$. Note that $k - \ell \in \mathbb{N}$. Now from Definition 6.3(i) it follows that $o(g) < \infty$.

□

Theorem 6.6. *Let G be any group, and let $g \in G$ such that $o(g) < \infty$. Set $n := o(g)$.*

(i) *For $k \in \mathbb{Z}$, we have $g^k = e$ if and only if $n|k$.*
(ii) *For $k, \ell \in \mathbb{Z}$, we have $g^k = g^\ell$ if and only if $n|(k-\ell)$.*
(iii) $\langle g \rangle = \{e, g, \cdots, g^{n-1}\}$.

Proof. (i) For $k \in \mathbb{Z}$, by the division algorithm we can write $k = nq+r$, where $0 \leqslant r \leqslant n-1$. Note that

$$n|k \text{ if and only if } r = 0.$$

Now

$$g^k = g^{nq+r} = g^{nq}g^r = (g^n)^q g^r = e^q g^r = g^r.$$

If $r = 0$ then $g^k = e$, and if $1 \leqslant r < n$ then $g^k \neq e$ because $k < n = o(g)$. This completes the proof of (i).

(ii) Note that $g^k = g^\ell$ if and only if $g^{k-\ell} = e$. Now we write $(k-\ell) = nq + r$ by the division algorithm, where $0 \leqslant r \leqslant n-1$. From (i) it follows that $g^{k-\ell} = e$ if and only if $n|(k-\ell)$.

(iii) From (i) it follows that for any $k \in \mathbb{Z}$, $g^k = g^r$ for some $0 \leqslant r \leqslant n-1$. From (ii) it follows that the elements $e = g^0, g^1, \ldots, g^{n-1}$ are distinct. Thus $\langle g \rangle = \{e, g, \ldots, g^{n-1}\}$.

□

Theorem 6.7. *Let G be a cyclic group.*
(i) *If G is finite, then $G \cong \mathbb{Z}_n$ for some $n \in \mathbb{N}$.*
(ii) *If G is not finite, then $G \cong \mathbb{Z}$.*

Proof. By definition of a cyclic group, we know that $G = \langle g \rangle$ for some $g \in G$.

If G is finite, then by Lemma 6.5 we know that $o(g) \in \mathbb{N}$. Set $n := o(g)$. We define a map

$$\phi : \mathbb{Z}_n \to G \ : \ \phi(k) := g^k.$$

We show that this map is an isomorphism.

Step 1. Clearly $\phi(\mathbb{Z}_n) \subseteq \langle g \rangle$. From Theorem 6.6(iii) it follows that indeed ϕ is a bijection onto $\langle g \rangle$.

Step 2. We show that $\phi(a + b) = \phi(a)\phi(b)$. For further clarity, we represent the addition of $\mathbb{Z}_n$ by $\dot{+}$, and the usual addition of $\mathbb{Z}$ by $+$. We have

$$\phi(a \dot{+} b) = g^{a \dot{+} b} \text{ and } \phi(a)\phi(b) = g^a g^b = g^{a+b}.$$

From the definition of $\dot{+}$ it follows that $n|\,[(a+b) - (a \dot{+} b)]$, hence by Theorem 6.6(ii) it follows that $g^{a+b} = g^{a \dot{+} b}$.

Next assume that G is not finite. Since $G = \langle g \rangle$, we must have $o(g) = \infty$. Now we consider the map

$$\phi : \mathbb{Z} \to G \ , \ \phi(k) = g^k.$$

We will show that ϕ is an isomorphism.

Step 1. Clearly ϕ is a surjection onto $\langle g \rangle$. We show that it is also an injection. Assume, on the contrary, that $\phi(k) = \phi(\ell)$ for integers $k > \ell$. Then $g^{k-\ell} = g^k g^{-\ell} = e$. Since $k - \ell \in \mathbb{N}$, from Definition 6.3(i) it follows that we should have $o(g) < \infty$. This is a contradition.

Step 2. To complete the proof, it is enough to verify that $\phi(a+b) = \phi(a)\phi(b)$ for all $a, b \in \mathbb{Z}$. To this end, note that

$$\phi(a + b) = g^{a+b} = g^a g^b = \phi(a)\phi(b),$$

which verifies the latter claim. □

Computing $o(g^d)$ from $o(g)$ and d. Suppose we know $o(g)$ for some $g \in G$. How can we compute $o(g^d)$? This is answered by the next theorem.

Theorem 6.8. *Let G be any group and let $g \in G$ such that $o(g) < \infty$. Then for every $d \in \mathbb{Z}$ we have*

$$o(g^d) = \frac{o(g)}{\gcd(o(g), d)}.$$

Proof. Set $m := \frac{o(g)}{\gcd(o(g),d)}$ and $m' = \frac{d}{\gcd(o(g),d)}$. Note that $m, m' \in \mathbb{Z}$. Indeed $m \in \mathbb{N}$.

Step 1. We show that $(g^d)^m = 1$. We have

$$(g^d)^m = g^{dm},$$

and

$$dm = d\frac{o(g)}{\gcd(o(g), d)} = \frac{d}{\gcd(o(g), d)} o(g) = m' o(g).$$

From the latter equation it follows that $o(g)|dm$. Thus Theorem 6.6(i) implies that $g^{dm} = e$.

Step 2. Let $k \in \mathbb{N}$ be such that $(g^d)^k = e$. We claim that $m|k$. To prove this claim, note that From $g^{dk} = e$ and Theorem 6.6(i) it follows that $o(g)|dk$. Thus

$$dk = o(g)t \text{ for some } t \in \mathbb{Z}.$$

Now

$$dk = o(g)t \;\Rightarrow\; m'k = \frac{d}{\gcd(o(g), d)} k = \frac{o(g)}{\gcd(o(g), d)} t = mt \;\Rightarrow\; m|m'k.$$

From Corollary 1.11 it follows that

$$\gcd(m, m') = 1.$$

Therefore from $m|m'k$, $\gcd(m, m') = 1$, and Lemma 1.15(iii) it follows that $m|k$. In particular, $m \leqslant k$ (since $m, k \geqslant 1$).

Step 3. From Step 2 it follows that if $(g^d)^k = e$ for some $k \in \mathbb{N}$, then $m \leqslant k$, and therefore $m = \min\{k \in \mathbb{N} \,:\, (g^d)^k = e\}$, hence $m = o(g)$. □

Exercises.

Let G be a group and let $g, h \in G$ such that $o(g) = m$ and $o(h) = n$. Assume that $gh = hg$. Prove that $o(gh) = mn$.

Subgroups of cyclic groups. Our next goal is to classify subgroups of cyclic groups. We start with a general statement about the order of powers of an element.

Remark 6.9. For any $d, n \in \mathbb{N}$ such that $d|n$, set

$$d\mathbb{Z}_n := \{k \in \mathbb{Z}_n \ : \ d|k\} = \{0, d, 2d, \ldots, n-d\}.$$

Note that $d\mathbb{Z}_n$ is the set of elements of $\mathbb{Z}_n$ that are divisible by d. In particular,

$$n\mathbb{Z}_n = \{0\}.$$

It is straightforward to check using the subgroup test that $d\mathbb{Z}_n$ is a subgroup of $\mathbb{Z}_n$.

Example 6.10. Here are some examples of the groups $d\mathbb{Z}_n$:

(i) $4\mathbb{Z}_{12} = \{0, 4, 8\}$.
(ii) $3\mathbb{Z}_{18} = \{0, 3, 6, 9, 12, 15\}$.

Theorem 6.11. *Let $n \geqslant 1$. The subgroups of $\mathbb{Z}_n$ are precisely the sets $d\mathbb{Z}_n$ for integers $d \in \mathbb{N}$ such that $d|n$. Here*

$$d\mathbb{Z}_n := \{k \in \mathbb{Z}_n \ : \ d|k\} = \{0, d, 2d, \ldots, n-d\}.$$

Proof. Fix a subgroup H of $\mathbb{Z}_n$. If $H = \{0\}$ then we can set $d = n$. Next assume that $H \neq \{0\}$.

Step 1. Set $d := \min\{k \in H \ : k > 0\}$. Our goal is to prove that indeed $H = d\mathbb{Z}_n$.

Step 2. We prove that $d|n$. In order to prove this claim, write $n = qd + r$ for $0 \leqslant r < d$ by the division algorithm. (Note that $q \geqslant 0$ because $n \geqslant d \geqslant 1$.) To verify that $d|n$, it is enough to show that $r = 0$.

Now assume, on the contrary, that $r > 0$. Then from $r + qd = n$ it follows that r is the inverse of qd in $\mathbb{Z}_n$. However, $qd = \underbrace{d + \cdots + d}_{q \text{ times}} \in H$. It follows that the inverse of qd also belongs to H. Namely, $r \in H$. But this contradicts the fact that $d = \min\{k \in H \ : \ k > 0\}$.

Step 3. Since $d \in H$ and H is closed under the addition of $\mathbb{Z}_n$, it is clear that $d\mathbb{Z}_n \subseteq H$. Next we prove the reverse inclusion: $H \subseteq d\mathbb{Z}$.

Take any element $k \in H$. Then by the division algorithm we can write

$$k = qd + r \text{ for some } 0 \leqslant r < d.$$

As usual, let $-qd$ denote the inverse of qd in $\mathbb{Z}_n$. As in Step 2, we have $qd \in H$ and therefore $-qd \in H$. By adding $-qd$ to both sides of $k = qd + r$, we obtain

$$k + (-qd) = r, \text{ hence } r = k + (-qd) \in H.$$

If $r \neq 0$, then by the definition of d we should have $d \leqslant r$, but we know that $r < d$, and this is a contradiction. Thus we should have $r = 0$, and therefore $k = qd$. This implies that $k \in d\mathbb{Z}_n$. □

Example 6.12. The subgroups of $\mathbb{Z}_{12}$ are

$$\mathbb{Z}_{12}, \ 2\mathbb{Z}_{12}, \ 3\mathbb{Z}_{12}, \ 4\mathbb{Z}_{12}, \ 6\mathbb{Z}_{12}, \ 12\mathbb{Z}_{12}.$$

Exercises.

(1) Verify that for $n, d \in \mathbb{N}$ such that $d|n$, we have $(-d)\mathbb{Z}_n = d\mathbb{Z}_n$.
(2) Verify that the subgroup of $\mathbb{Z}_{18}$ generated by 10 is the same as the subgroup of $\mathbb{Z}_{18}$ generated by 2. Note that $2|18$ but $10 \nmid 18$. Why does this not contradict Theorem 6.11?
(3) Let $n, d \in \mathbb{N}$ such that $d|n$. Prove that the subgroup $d\mathbb{Z}_n$ of $\mathbb{Z}_n$ is isomorphic $\mathbb{Z}_{\frac{n}{d}}$. This implies that every subgroup of $\mathbb{Z}_n$ is a cyclic group.

Remark 6.13. For $d \in \mathbb{Z}$, set $d\mathbb{Z} := \{k \in \mathbb{Z} \ : \ d|k\}$. Note that $d\mathbb{Z}$ is precisely the set of integers that are divisible by d (indeed for $d = 0$ we have $0\mathbb{Z} = \{0\}$). Using the subgroup test, it is easy to verify that $d\mathbb{Z}$ is a subgroup of $\mathbb{Z}$.

Example 6.14. For example, $3\mathbb{Z} = \{0, \pm 3, \pm 6, \pm 9, \ldots\}$.

Theorem 6.15. *The subgroups of $\mathbb{Z}$ are precisely the sets $d\mathbb{Z} := \{k \in \mathbb{Z} \ : \ d|k\}$, where $d \in \mathbb{Z}$.*

Proof. The proof is similar to the one given for Theorem 6.11. Fix a subgroup H of $\mathbb{Z}$. If $H = \{0\}$ then $H = 0\mathbb{Z}$ and we can set $d = 0$. From now on, we assume that $H \neq \{0\}$.

Step 1. Set $d := \min\{x \in H \ : \ x > 0\}$. Our final goal is to show that $H = d\mathbb{Z}$. Let us verify that $d\mathbb{Z} \subseteq H$. Indeed from $d \in H$ it follows that

$$\forall_{k \in \mathbb{N}} \ : \ kd = \underbrace{d + \cdots + d}_{k \text{ times}} \in H$$

and $-kd$ is the inverse of kd in G, so $-kd \in H$ as well (since H is a group). This completes the proof of $d\mathbb{Z} \subseteq H$.

Step 2. We prove that $H \subseteq d\mathbb{Z}$. Assume, on the contrary, that there exists$m \in H$ such that $m \notin d\mathbb{Z}$. Now by the division algorithm we can write

$$m = qd + r \text{ where } 0 \leqslant r < d.$$

Indeed $r \neq 0$ because $m \notin d\mathbb{Z}$. Now $m, qd \in H$, and therefore

$$r = m - qd \in H.$$

Since $0 < r < d$, this contradicts the fact that d is the smallest positive element of H. $\square$

THE CIRCLE AND DIHEDRAL GROUPS

7.1 The circle group $\mathbb{T}$. Set $\mathbb{T} := \{z \in \mathbb{C} \ : \ |z| = 1\}$. Since we have

$$|z_1 z_2| = |z_1| \cdot |z_2|$$

for any two $z_1, z_2 \in \mathbb{C}$, the set of complex numbers $\mathbb{T}$ is closed under multiplication.

Proposition 7.1. *The set $\mathbb{T}$, equipped with multiplication, is an abelian group.*

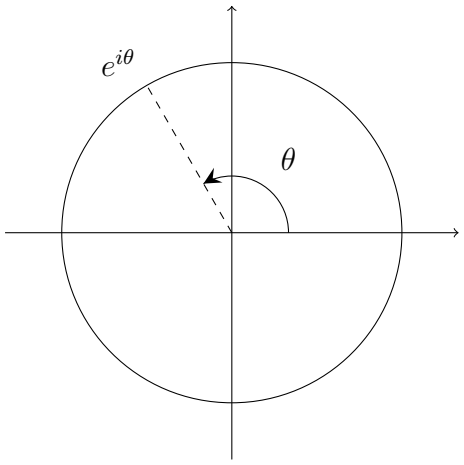
Proof. $\mathbb{T}$ is a subset of the group $\mathbb{C}^*$ of nonzero complex numbers. Using the Concise Subgroup Test it is straightforward to verify that $\mathbb{T}$ is indeed a subgroup of $\mathbb{C}^*$. We leave the details as an exercise. □

A geometric way to think about $\mathbb{T}$ is as follows. Every element of $\mathbb{T}$ is a point on the unit circle. This point forms an angle $0 \leqslant \theta < 2\pi$ with the x-axis (we measure the angle counterclockwise). We denote this point by $e^{i\theta}$. Recall that

$$i = \sqrt{-1}.$$

The standard Cartesian coordinates of $e^{i\theta}$ are $(\cos\theta, \sin\theta)$. Therefore we are defining

$$e^{i\theta} := \cos(\theta) + i\sin(\theta).$$



For example

$$e^{i\pi} = \cos(\pi) + i\sin(\pi) = -1, \text{ and } e^{\frac{2\pi i}{3}} = \cos(\tfrac{2\pi}{3}) + i\sin(\tfrac{2\pi}{3}) = \tfrac{1}{2} + \tfrac{\sqrt{3}}{2}i.$$

Using trigonometry, one can show that for any two numbers $\theta_1, \theta_2 \in \mathbb{R}$,

$$e^{i(\theta_1+\theta_2)} = e^{i\theta_1} e^{i\theta_2}. \tag{7.1}$$

Here the multiplication on the right hand side is taking place in complex numbers.

Exercises.

(1) Using the formulas for $\cos(a+b)$ and $\sin(a+b)$ in terms of the values of $\cos(a), \sin(a), \cos(b)$, and $\sin(b)$ from trigonometry, prove the relation (7.1).

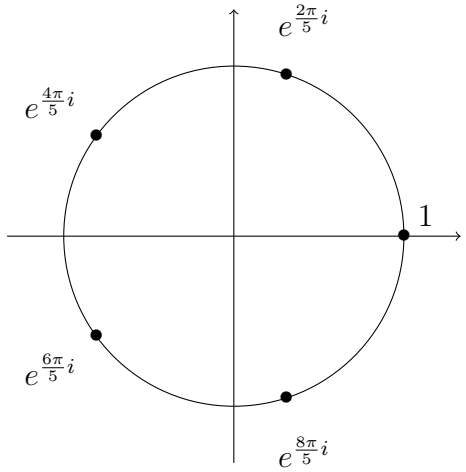
(2) Verify that $e^{\pi} = -1$ and $e^{2\pi i} = 1$ and $e^{\frac{\pi}{2}i} = i$ and $e^{\frac{\pi}{3}i} = \frac{1}{2} = \frac{\sqrt{3}}{2}i$.
(3) Prove that every nonzero complex number z can be written uniquely in the form $z = re^{i\theta}$, where $r > 0$ and $0 \leqslant \theta < 2\pi$. What happens for $z = 0$?
Hint: Set $r := |z|$.

Remark 7.2. You are most likely familiar with e^t for a real number t. You know that

$$e^t = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \cdots \tag{7.2}$$

In this section, we defined e^t for imaginary values of t. Interestingly enough, the formula (7.2) still remains valid. We will not need this formula. But it is good to know that everything remains consistent!

Let us now look at the points $e^{\frac{2\pi k}{n}i}$ for $k = 0, \ldots, n-1$. These points form the angles $\frac{2\pi k}{n}$ with the x-axis. Therefore they are the vertices of a regular n-gon. The following picture illustrates these points for $n = 5$:



Proposition 7.3. *Let $n \in \mathbb{N}$, and set*

$$R_n := \left\{1, e^{\frac{2\pi}{n}i}, \ldots, e^{\frac{2\pi k}{n}i}, \ldots, e^{\frac{2\pi(n-1)}{n}i}\right\} = \left\{e^{\frac{2\pi k}{n}i} \;:\; k \in \mathbb{Z} \text{ and } 0 \leqslant k \leqslant n-1\right\}.$$

Then R_n is the set of complex roots of the equation $z^n = 1$.

Proof. (i) First note that from (7.1) it follows that for every $k \in \mathbb{Z}$,

$$\left(e^{\frac{2\pi k}{n}i}\right)^n = e^{n\frac{2\pi k}{n}i} = e^{2\pi k i} = 1.$$

Thus every element of R_n is a root of $z^n = 1$.

Conversely, assume that $z \in \mathbb{C}$ satisfies $z^n = 1$. Then

$$1 = |1| = |z^n| = |z|^n,$$

and since $|z|$ is a non-negative real number, it follows that $|z| = 1$. This means that $z \in \mathbb{T}$. Therefore $z = e^{i\theta}$ for some $0 \leqslant \theta < 2\pi$. Now

$$z^n = (e^{i\theta})^n = e^{ni\theta} = \cos(n\theta) + i\sin(n\theta).$$

Since $z^n = 1$, we find that

$$\cos(n\theta) = 1 \text{ and } \sin(n\theta) = 0.$$

These relations (together with the well-known properties of $\sin x$ and $\cos x$) imply that

$$n\theta = 2\pi k \text{ for some } k \in \mathbb{Z}.$$

But we know that $0 \leqslant \theta < 2\pi$, and therefore $0 \leqslant n\theta < 2\pi n$. Consequently, we should have

$$n\theta = 2\pi k \text{ for some } k \in \{0, 1, \ldots, n-1\}.$$

Equivalently,

$$\theta = \frac{2\pi k}{n} \text{ for some } k \in \{0, 1, \ldots, n-1\}.$$

It follows that the complex number $z = e^{i\theta}$ belongs to R_n. □

Example 7.4. The set of complex roots of the equation $z^5 = 1$ is

$$R_5 = \left\{1, e^{\frac{2\pi}{5}i}, e^{\frac{4\pi}{5}i}, e^{\frac{6\pi}{5}i}, e^{\frac{8\pi}{5}i}\right\}.$$

The set of complex roots of the equation $z^4 = 1$ is

$$R_4 = \left\{1, e^{\frac{2\pi}{4}i}, e^{\frac{4\pi}{4}i}, e^{\frac{6\pi}{4}i}, \right\} = \{1, i, -1, -i\}.$$

Remark 7.5. Let $n \in \mathbb{N}$ Let $a \in \mathbb{Z}$, and let b be the remainder of a modulo n. (Thus $b \in \{0, \ldots, n-1\}$.) Then

$$e^{\frac{2\pi a}{n}i} = e^{\frac{2\pi b}{n}i}.$$

The reason is that by the division algorithm we can write $a = nq + b$ where $q \in \mathbb{Z}$. Then,

$$e^{\frac{2\pi a}{n}i} = e^{\frac{2\pi(nq+b)}{n}i} = e^{\frac{2\pi nq}{n}i} e^{\frac{2\pi b}{n}i} = e^{2\pi qi} e^{\frac{2\pi b}{n}i} = (e^{2\pi i})^q e^{\frac{2\pi b}{n}i} = (1)^q e^{\frac{2\pi b}{n}i} = e^{\frac{2\pi b}{n}i}.$$

Proposition 7.6. *Let $n \in \mathbb{N}$, and let R_n be the subset of $\mathbb{T}$ defined in Proposition 7.3. Then R_n is a subgroup of $\mathbb{T}$ and it is isomorphic to $\mathbb{Z}_n$.*

Proof. Step 1. We show that R_n is a subgroup of $\mathbb{T}$. This can be done by the Concise Subgroup Test (note that $1 \in R_n$, so that $R_n \neq \varnothing$). We need to show that for any two elements $z_1, z_2 \in R_n$, we have $z_1 z_2^{-1} \in R_n$. Recall that by Proposition 7.3, the set R_n is the set of all roots of $z^n = 1$ in $\mathbb{C}$. Thus it is sufficient to check that $z_1 z_2^{-1}$ is a root of the equation $z^n = 1$. This can be verified as follows:

$$(z_1 z_2^{-1})^n = z_1^n z_2^{-n} = (1)(1) = 1.$$

Step 2. We define an isomorphism $\phi : \mathbb{Z}_n \to R_n$. We set

$$\phi : \mathbb{Z}_n \to \mathbb{T}\ ,\ \phi(k) := e^{\frac{2\pi ik}{n}} \text{ for } k \in \{0, 1, \ldots, n-1\}.$$

This map is clearly a bijection. It is an isomorphism because

$$\phi(k_1 + k_2) = e^{\frac{2\pi(k_1+k_2 \bmod n)}{n}i} = e^{\frac{2\pi(k_1+k_2)}{n}i} = e^{\frac{2\pi k_1}{n}i} e^{\frac{2\pi k_2}{n}i} = \phi(k_1)\phi(k_2).$$

Note that in the last calculation, we are using Remark 7.5. □

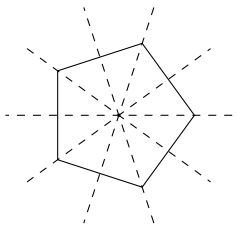
The dihedral groups D_n, $n \geqslant 1$.

Definition 7.7. Let $n \geqslant 3$ be an integer. A regular n-gon is a convex polygon with n sides of equal length such that the angles between its adjacent sides are all equal.

Example 7.8. An equilateral triangle is a regular 3-gon and a square is a regular 4-gon.

Remark 7.9. The points $e^{\frac{2\pi k}{n}i} \in \mathbb{C}$ with $k \in \{0, \ldots, n-1\}$ are the vertices of a regular n-gon.

Definition 7.10. Let $n \geqslant 3$ be an integer. The group of symmetries of a regular n-gon is called the *n-th dihedral group*.



The reflections of a regular pentagon.

Let us describe an explicit realization of the symmetries of the n-gon of Remark 7.9. Consider the following symmetries:

(i) n reflections about the lines passing through the origin and the point $e^{\frac{\pi k}{n}i}$ for $k = 0, \ldots, n-1$.
(ii) n counterclockwise rotations with angles $\frac{2k\pi}{n}$, for $k = 0, 1, \ldots, n-1$, around the origin.

Consider the following symmetries of the aforementioned n-gon:

(i) r : the counterclockwise rotation about the origin with angle $\frac{2\pi}{n}$.
(ii) s : the reflection about the x-axis.

Clearly $r, s \in D_n$. Furthermore, by geometric investigation we observe that

$$o(r) = n \text{ and } o(s) = 2 \text{ and } srs = srs^{-1} = r^{-1}.$$

Thus the elements of D_n are precisely

$$\left\{e, r, r^2, \ldots, r^{n-1}, s, rs, r^2s, \ldots, r^{n-1}s\right\}.$$

Remark 7.11. (i) $r^k s$ is reflection about the line passing throught the origin and the point $e^{\frac{\pi k}{n}i}$.
(ii) $sr^k s$ is a clockwise rotation with angle $\frac{2\pi k}{n}$, which is the same as a counterclockwise rotation with angle $\frac{2\pi(n-k)}{n}$.

Example 7.12. Consider the element $rsr^3sr^{-1}ssr \in D_5$. We can simplify this element as follows:

$$\begin{aligned} rsr^3sr^{-1}ssr &= rsr^3sr^4(ss)r = rsr^3s(r^4r) \\ &= rsr^3s = rsr^3s^{-1} = r(srs^{-1})^3 = r(r^{-1})^3 = r^{-2} = r^3. \end{aligned}$$

Exercises.

(1) Prove or disprove: $D_4 \cong Q$.
(2) Assume $n \geqslant 3$. Prove that $\{r^k \ : \ 0 \leqslant k \leqslant n-1\}$ is a cyclic subgroup of D_n.
(3) Determine all of the subgroups of D_6.

SYMMETRIC GROUPS

Recall that S_n denotes the group of permutations of $\{1, \ldots, n\}$. The number of elements of S_n is $n! = n(n-1)(n-2)\cdots 2 \cdot 1$.

Remark 8.1. From now on, we denote the identity permutation by ε. Thus

$$\varepsilon(k) = k \text{ for all integers } 1 \leqslant k \leqslant n.$$

We have already represented permutations as an array in (2.2). We are going to introduce a more effective representation of permutations. The idea is the following: given a permutation σ, we can start from some k and compute the values

$$\sigma(k)\ ,\ \sigma(\sigma(k)\ ,\ \sigma(\sigma(\sigma(k)))\ ,\ \ldots.$$

Since the values are always integers between 1 and n, some numbers should reoccur. Since σ is a bijection, we can "back up" from any repetition. This implies thatthere exists a $j \in \mathbb{N}$ such that $\sigma^{(j)}(k) = k$. Thus, the numbers

$$k, \sigma(k), \sigma(\sigma(k)), \ldots, \sigma^{(j-1)}(k)$$

form a "cycle".

Definition 8.2. Let $d, n \in \mathbb{N}$ such that $2 \leqslant d \leqslant n$. A *d-cycle* (or a cycle of length d) in S_n is a permutation $\sigma \in S_n$ with the following property: there exist d distinct numbers $a_1, \ldots, a_d$ such that

(i) $\sigma(a_i) = a_{i+1}$ for $1 \leqslant i \leqslant d-1$ and $\sigma(a_d) = a_1$,
(ii) $\sigma(k) = k$ for all $k \in \{1, \ldots, n\}$ such that $k \notin \{a_1, \ldots, a_d\}$.

We denote this d-cycle by

$$\sigma = (a_1\ \ a_2\ \ \cdots\ \ a_{d-1}\ \ a_d).$$

Example 8.3. The permutation

$$\sigma := \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 2 & 3 & 5 \end{pmatrix}$$

is a cycle of length 3 in S_5 because

$$\sigma(2) = 4\ ,\ \sigma(4) = 3\ ,\ \sigma(3) = 2$$

and $\sigma(k) = k$ for $k \notin \{1, 5\}$.

Proposition 8.4. *Let $n \geqslant 1$ and let $\sigma = (a_1\ \ a_2\ \ \cdots\ \ a_{d-1}\ \ a_d) \in S_n$ be a d-cycle. Then $\sigma^{-1} = (a_d\ \ a_{d-1}\ \ \cdots\ \ a_2\ \ a_1)$. In paricular, σ^{-1} is also a d-cycle.*

Proof. This is straightforward. □

Proposition 8.5. *Let $n \in \mathbb{N}$ and let $\sigma := (a_1\ \ a_2\ \ \cdots\ \ a_{d-1}\ \ a_d)$ be a d-cycle in S_n. Then*

$$\sigma = (a_i\ \ a_{i+1}\ \ \cdots\ \ a_{d-1}\ \ a_d\ \ a_1\ \ a_2\ \ \cdots\ \ a_{i-2}\ \ a_{i-1}) \text{ for every } 2 \leqslant i \leqslant d.$$

Proof. It is easy to verify that both cycles map a_i to a_{i+1} for $1 \leqslant i \leqslant d-1$, map a_r to a_1, and fix all $x \in \{1, \ldots, n\} \backslash \{a_1, \ldots, a_d\}$. □

Remark 8.6. Let $\sigma \in S_n$ and assume that σ is a d-cycle, i.e., $\sigma = (a_1 \ \cdots \ a_d)$. Obviously we have $n \geqslant \max\{a_1, \ldots, a_d\}$. Now, for any $m > n$, we can also think of σ as a d-cycle in S_m.

Products of cycles. Since cycles are permutations, products of cycles is simply the composition of their corresponding permutations. To compute the product of cycles in S_n, we should act ona letter $x \in \{1, \ldots, n\}$ from the rightmost cycle, and move towards the left.

Example 8.7. Let $\sigma_1 = (3 \ 1 \ 5)$ and $\sigma_2 = (5 \ 4 \ 1)$ and $\sigma_3 = (5 \ 2)$ be permutations in S_6. Then

$$(3 \ 1 \ 5)(5 \ 4 \ 1)(5 \ 2) = \sigma_1\sigma_2\sigma_3 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 1 & 5 & 2 & 6 \end{pmatrix}.$$

Disjoint cycles and cycle decomposition.

Definition 8.8. Two cycles

$$\sigma = (a_1 \ \cdots \ a_r) \text{ and } \tau = (b_1 \ \cdots \ b_s)$$

in S_n are called *disjoint* if the numbers $a_1, \ldots, a_r, b_1, \ldots, b_s$ are all distinct.

Example 8.9. The cycles $\sigma, \tau \in S_{10}$ given by

$$\sigma = (2 \ 4 \ 1 \ 6) \text{ and } \tau = (5 \ 8 \ 3)$$

are disjoint.

We now give a characterisation of disjoint cycles in terms of their fixed point sets.

Definition 8.10. Let $n \geqslant 1$ and let $\sigma \in S_n$. The *set of points moved by* σ, denoted by M_σ, is defined by

$$M_\sigma := \{x \in \mathbb{N} : \ 1 \leqslant x \leqslant n \text{ and } \sigma(x) \neq x\}.$$

Example 8.11. Let $\tau := (a_1 \ \cdots \ a_d)$ be a d-cycle in S_n. Then τ only moves $a_1, \ldots, a_d$. In other words, $M_\tau = \{a_1, \ldots, a_d\}$.

Remark 8.12. let $n \in \mathbb{N}$ and let σ, τ be two cycles in S_n. From the definition of disjoint cycles it follows that σ and τ are disjoint if and only if $M_\sigma \cap M_\tau = \varnothing$. In plain terms, σ and τ commute if and only if the set of elements of $\{1, \ldots, n\}$ that are moved by σ and by τ are disjoint sets.

An important property of disjoint cycles is that they commute.

Proposition 8.13. *Let $n \geqslant 1$ and let $\sigma, \tau \in S_n$ be disjoint cycles. Then $\sigma\tau = \tau\sigma$.*

Proof. Assume that

$$\sigma = (a_1 \ \cdots \ a_r) \text{ and } \tau = (b_1 \ \cdots \ b_s).$$

Fix $x \in \{1, \ldots, n\}$. We need to show that $\sigma(\tau(x)) = \tau(\sigma(x))$. One of the following 3 cases happens:

Case I. $x \in \{a_1, \ldots, a_r\}$. Then $\tau(x) = x$ because $x \notin \{b_1, \ldots, b_s\}$. Hence

$$\sigma(\tau(x)) = \sigma(x). \tag{8.1}$$

Also, $\tau(\sigma(x)) = \sigma(x)$ since $\sigma(x) \in \{a_1, \ldots, a_r\}$ and therefore $\sigma(x) \notin \{b_1, \ldots, b_s\}$. Consequently,

$$\tau(\sigma(x)) = \sigma(x). \tag{8.2}$$

From (8.1) and (8.2) it follows that $\sigma(\tau(x)) = \tau(\sigma(x))$.

Case II. $x \in \{b_1, \ldots, b_s\}$. This case is very similar to Case I. We leave the details of the proof as an exercise.

Case III. $x \notin \{a_1 \ldots, a_r\} \cup \{b_1, \ldots, b_s\}$. In this case $\sigma(x) = \tau(x) = x$ and therefore

$$\sigma(\tau(x)) = \sigma(x) = x \text{ and } \tau(\sigma(x)) = \tau(x) = x. \qquad \square$$

Definition 8.14. Let $\sigma \in S_n$ be a permutation. Starting from an element $x \in \{1, \ldots, n\}$, we consider the elements

$$x, \sigma(x), \sigma(\sigma(x)), \ldots, \sigma^{(k)}(x), \ldots$$

We call the set of these elements the *orbit* of x under σ and denote it by $O_\sigma(x)$. Thus

$$O_\sigma(x) = \left\{\sigma^{(i)}(x) \ : \ i \in \mathbb{N} \cup \{0\}\right\}.$$

The number of distinct elements of $O_\sigma(x)$ is called the *length* of the orbit of x.

Example 8.15. For the permutation $\sigma \in S_{10}$ given by

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 4 & 10 & 1 & 7 & 6 & 5 & 3 & 8 & 2 & 9 \end{pmatrix},$$

the orbits are as follows:

$$\begin{aligned} &O_\sigma(1) = \{1, 4, 7, 3\} = O_\sigma(3) = O_\sigma(4) = O_\sigma(7). \\ &O_\sigma(2) = \{2, 10, 9\} = O_\sigma(9) = O_\sigma(10). \\ &O_\sigma(5) = \{5, 6\} = O_\sigma(6) \\ &O_\sigma(8) = \{8\}. \end{aligned}$$

Note that several orbits are identical. Also, non-identical orbits are disjoint sets. These properties indeed hold in general. Our next goal is to prove these two properties in Proposition 8.16 and Proposition 8.17.

Proposition 8.16. *Let $\sigma \in S_n$.*

(i) *Let $x \in \{1, \ldots, n\}$. Then there exists an integer $k \in \mathbb{N}$ such that $\sigma^{(k)}(x) = x$. If*

$$\ell := \min\{k \in \mathbb{N} \ : \ \sigma^{(k)}(x) = x\},$$

then the elements $x, \sigma(x), \ldots, \sigma^{(\ell-1)}(x)$ are all distinct and

$$O_\sigma(x) = \{x, \sigma(x), \ldots, \sigma^{(\ell-1)}(x)\} = \left\{\sigma^{(i)}(x) \ : \ 0 \leqslant i \leqslant \ell - 1\right\}.$$

(ii) *For any $y \in O_\sigma(x)$, we have $O_\sigma(y) = O_\sigma(x)$.*

Proof. (i) Assume, on the contrary, that $\sigma^{(j)}(x) = \sigma^{(i)}(x)$ for some i, j such that $0 \leqslant i < j \leqslant \ell - 1$. Since σ is a bijection, we have

$$\sigma^{(j-1)}(x) = \sigma^{(i-1)}(x) \text{ and } \sigma^{(j-2)}(x) = \sigma^{(i-2)}(x) \text{ and } \ldots$$

Continuing in this fashion, we obtain $\sigma^{(j-i)}(x) = x$. But then note that $1 \leqslant j - i \leqslant \ell - 1$, hence

$$\ell = \min\{k \in \mathbb{N} \,:\, \sigma^{(k)}(x) = x\} \leqslant j - i \leqslant \ell - 1,$$

which is a contradiction.

Clearly, $\{x, \sigma(x), \ldots, \sigma^{(\ell-1)}(x)\} \subseteq O_\sigma(x)$. Furthermore,

$$\sigma(\sigma^{(\ell-1)}(x)) = \sigma^{(\ell)}(x) = x, \text{ hence } \sigma^{(k)}(x) = \sigma^{(k \text{ modulo } \ell)}(x).$$

Therefore there are no other elements in $O_\sigma(x)$.

(ii) Observe that $y = \sigma^{(i)}(x)$ for some $1 \leqslant i \leqslant \ell - 1$. Thus clearly,

$$O_\sigma(y) \subseteq O_\sigma(x). \tag{8.3}$$

Furthermore,

$$x = \sigma^{(\ell)}(x) = \sigma^{(\ell-i)}(\sigma^{(i)}(x)) = \sigma^{(\ell-i)}(y) \in O_\sigma(y).$$

That is, $x \in O_\sigma(y)$, hence

$$O_\sigma(x) \subseteq O_\sigma(y). \tag{8.4}$$

From (8.3) and (8.4) it follows that $O_\sigma(x) = O_\sigma(y)$. □

Proposition 8.17. *Let $\sigma \in S_n$, and let $x, y \in \{1, \ldots, n\}$. Then either*

$$O_\sigma(x) = O_\sigma(y),$$

or $O_\sigma(x) \cap O_\sigma(y) = \varnothing$.

Proof. Assume that $O_\sigma(x) \cap O_\sigma(y) \neq \varnothing$. We prove that $O_\sigma(x) = O_\sigma(y)$.

Let ℓ_x and ℓ_y denote the orbit lengths of x and y, respectively. Step 1. Take $a \in O_\sigma(x) \cap O_\sigma(y)$. Then there exist $0 \leqslant i \leqslant \ell_x - 1$ and $0 \leqslant j \leqslant \ell_y - 1$ such that

$$a = \sigma^{(i)}(x) = \sigma^{(j)}(y).$$

Step 2. First assume that $i \geqslant j$. (Because of the symmetry between x and y, the argument for the case $i < j$ is similar.) Since σ is a bijection, by precomposing with σ^{-1} we have

$$\sigma^{(i-1)}(x) = \sigma^{(j-1)}(y) \text{ and } \sigma^{(i-2)}(x) = \sigma^{(j-2)}(y) \text{ and } \ldots$$

By continuing in this fashion, we obtain $\sigma^{(i-j)}(x) = y$. This means that $y \in O_\sigma(x)$, hence Proposition 8.16(ii) implies that $O_\sigma(x) = O_\sigma(y)$. □

In the next proposition, we explain how to obtain the cycle decomposition of a permutation from its orbits.

Proposition 8.18. *Let $\sigma \in S_n$, where $n \in \mathbb{N}$. Suppose that $x_1, \ldots, x_r$ are chosen such that the collection of subsets*

$$\{O_\sigma(x_1), \ldots, O_\sigma(x_r)\}$$

contains each of the orbits of σ of length $\geqslant 2$ precisely once. For each $1 \leqslant i \leqslant r$, let ℓ_i be the orbit length of x_i, and let τ_i be the ℓ_i-cycle defined by

$$\tau_i = \big(x_i, \sigma(x_i), \ldots, \sigma^{(\ell_i - 1)}(x_i)\big).$$

Then the cycles $\tau_1, \ldots, \tau_r$ are mutually disjoint and $\sigma = \tau_1 \cdots \tau_r$.

Proof. Step 1. We prove that the given cycles are disjoint. For each i such that $1 \leqslant i \leqslant r$, the cycle τ_i only moves the elements of $O_\sigma(x_i)$. In other words, $M_{\tau_i} = O_\sigma(x_i)$. Now by Proposition 8.17, for $i \neq j$ we have

$$O_\sigma(x_i) \cap O_\sigma(x_j) = \varnothing.$$

Therefore τ_i and τ_j are disjoint cycles (see Remark 8.12).

Step 2. We prove that $\sigma(x) = \tau_1 \cdots \tau_r(x)$ for every $x \in \{1, \ldots, n\}$. Fix $x \in \{1, \ldots, n\}$. One of the following two cases can occur:

Case I: $x \in O_\sigma(x_i)$ for some $1 \leqslant i \leqslant r$. Since the cycles $\tau_1, \ldots, \tau_r$ are disjoint, τ_i is the only cycle among $\tau_1, \ldots, \tau_r$ that moves the elements of $O_\sigma(x_i)$. But we know that $x, \sigma(x) \in O_\sigma(x_i)$. Therefore

$$\tau_j(x) = x \text{ and } \tau_j(\sigma(x)) = \sigma(x) \text{ for all } j \neq i.$$

It follows that

$$\tau_1 \cdots \tau_r(x) = \tau_1 \cdots \tau_i(x) = \tau_1 \cdots \tau_{i-1}(\sigma(x)) = \sigma(x).$$

Case II. x does not belong to any of $O_\sigma(x_1), \ldots, O_\sigma(x_r)$. Then the orbit of x should have length 1, and therefore $\sigma(x) = x$. In addition, each τ_i only moves the elements of $O_\sigma(x_i)$, and therefore $\tau_i(x) = x$ for all $1 \leqslant i \leqslant r$. It follows that

$$\tau_1 \cdots \tau_r(x) = x = \sigma(x). \qquad \square$$

Example 8.19. Let us apply 8.18 to the permutation of Example 8.15. The collection of distinct orbits of length $\geqslant 2$ is:

$$\{O_\sigma(1), O_\sigma(2), O_\sigma(5)\}.$$

Therefore

$$\sigma = (1\ \ 4\ \ 7\ \ 3)(2\ \ 10\ \ 9)(5\ \ 6).$$

Theorem 8.20. (Cycle Decomposition Theorem) *Every permutation $\sigma \in S_n$ can be written in a unique way as a product of mutually disjoint cycles. In other words:*

(i) There exist mutually disjoint cycles $\tau_1, \ldots, \tau_r$ such that $\sigma = \tau_1 \cdots \tau_r$.

(ii) If $\tau'_1, \ldots, \tau'_s$ is another list of mutually disjoint cycles such that

$$\sigma = \tau'_1 \ldots \tau'_s,$$

then $r = s$ and the τ'_j's are identical to the τ_i's up to a rearragement.

Proof. Existence follows immediately from Proposition 8.18.

Uniqueness is proved as follows. Suppose that $\sigma = \tau'_1 \cdots \tau'_s$, for disjoint cycles $\tau'_1, \ldots, \tau'_s$. Then the elements moved by each cycle (i.e., the sets $M_{\tau'_i}$) must be the orbits of σ with length $\geqslant 2$. Therefore the decomposition $\sigma = \tau'_1 \cdots \tau'_s$ is uniquely determined by the orbits of σ. $\square$

Exercises.

(1) Using the method of proof of Proposition 8.13, prove that the following more general statement is true: for any two permutations $\sigma, \tau \in S_n$ such that $M_\sigma \cap M_\tau = \varnothing$, we have $\sigma\tau = \tau\sigma$.
(2) Compute the cycle decompostion of
$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 7 & 3 & 4 & 2 & 6 & 8 & 5 & 1 \end{pmatrix}$$
(3) Let σ be a permutation such that $\sigma = \sigma^{-1}$. Prove that in the cycle decomposition of σ, all of the cycles are 2-cycles.
(4) Let σ be a permutation such that the cycles in its cycles decomposition are either 2-cycles or 3-cycles. Prove that $\sigma^6 = \varepsilon$.

Transpositions and parity. Our next goal is to describe another useful way of expressing permutations as products of (not necessarily disjoint) cycles.

Definition 8.21. A 2-cycle is called a *transposition*.

Lemma 8.22. *Let $\sigma := (a_1 \ \cdots \ a_d)$ be a d-cycle in S_n, where $n \geqslant d \geqslant 2$. Then*
$$\begin{aligned} \sigma &= (a_1 \ a_2)(a_2 \ a_3) \cdots (a_i \ a_{i+1}) \cdots (a_{d-1} \ a_d) \\ &= (a_{d-1} \ a_d)(a_{d-2} \ a_d) \cdots (a_i \ a_d) \cdots (a_1 \ a_d). \end{aligned}$$

Proof. It is not difficult to trace the "journey" of an a_i under the product. For the first product, we have
$$a_i \to \cdots \to a_i \underbrace{\to}_{(a_i \ a_{i+1})} a_{i+1} \to a_{i+1} \to \cdots \to a_{i+1} \quad \text{for } i < d,$$
and
$$a_d \underbrace{\to}_{(a_{d-1} \ a_d)} a_{d-1} \underbrace{\to}_{(a_{d-2} \ a_{d-2})} a_{d-2} \to \cdots \to a_1.$$
For the second product, we leave the argument as an exercise. □

Exercises.

Prove the remaining part of Lemma 8.22.

Theorem 8.23. (Parity Theorem) *Let $n \in \mathbb{N}$. Then every permutation $\sigma \in S_n$ can be written as a product of transpositions. Furthermore, if*
$$\sigma = \tau_1 \cdots t\tau_r = \tau_1' \cdots \tau_s'$$
where the τ_i's and τ_j''s are transpositions, then r and s have the same parity (that is, they are either both even or both odd).

Proof. From the Cycle Decomposition Theorem and Lemma 8.22 it follows that every permutation can be expressed as a product of transpositions. Next we prove the statement about parity of r and s. Note that $(\tau_j')^{-1} = \tau_j'$, and therefore from the equality $\tau_1 \cdots \tau_r = \tau_1' \cdots \tau_s'$ it follows that
$$\tau_1 \cdots \tau_r \tau_s' \cdots \tau_1' = \varepsilon.$$

Thus, we have reduced the statement to the following claim:

$$\text{if } \varepsilon = \eta_1 \cdots \eta_k \text{ where the } \eta_i\text{'s are transpositions, then } k \text{ is even.}$$

We give a tricky proof for this claim, which is simple and beautiful, but it looks like its idea came out of the blue!

Let I denote the $n \times n$ identity matrix. Clearly $\det(I) = 1$. We know from linear algebra that if we swap two columns of any matrix, the determinant changes sign. Now consider the effect of the column swaps by the η_i's on the columns of a matrix: if $\eta_i = (a \;\; b)$, then we swap the a-th column and the b-th column of the matrix.

By applying $\eta_k, \eta_{r-1}, \ldots, \eta_1$ successively on the columns of I, each time the sign of the determinant of the result matrix changes. Therefore the final matrix should have determinant $(-1)^k$. However, the final matrix is the identity matrix (because $\eta_1 \cdots \eta_k = \varepsilon$). Therefore its determinant should be 1. It follows that

$$(-1)^k = 1, \text{ hence } k \text{ is even.} \qquad \square$$

Theorem 8.23 allows us to make the following definition.

Definition 8.24. Let $n \in \mathbb{N}$. A permutation S_n is called *even* if it can be expressed as a product of an even number of transpositions. A permutation in S_n is called *odd* if it can be expressed as a product of an odd number of transpositions.

Example 8.25. (i) Let $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 4 & 1 & 5 \end{pmatrix}$. Then σ is even because

$$\sigma = (1 \;\; 3)(3 \;\; 4).$$

(ii) For every $n \in \mathbb{N}$, the identity permutation $\varepsilon \in S_n$ is even.

Proposition 8.26. *Let $n \in \mathbb{N}$, and let $\sigma, \tau \in S_n$ be arbitrary permutations.*

(i) If σ and τ are both even, then $\sigma\tau$ is even.
(ii) If σ is even and τ is odd, then $\sigma\tau$ is odd.
(iii) If σ is odd and τ is even, then $\sigma\tau$ is odd.
(iv) If σ and τ are both odd, then $\sigma\tau$ is even.

Proof. These statements are consequences of obvious properties of even and odd numbers. For example, we can prove (i) as follows: assume that σ and τ are both even. Then

$$\sigma = \eta_1 \cdots \eta_{2r} \text{ and } \tau_1 = \eta_1' \cdots \eta_{2s}' \text{ for } r, s \in \mathbb{N} \cup \{0\},$$

where the η_i's and the η_j''s are transpositions. Now

$$\sigma\tau = \eta_1 \cdots \eta_{2r} \eta_1' \cdots \eta_{2s}',$$

which is a product of $2(r+s)$ transposition. Hence $\sigma\tau$ is even. The proofs of (ii)–(iv) are similar. $\square$

Lemma 8.27. *Let $n \in \mathbb{N}$ and let $\sigma \in S_n$.*

(i) If σ is even, then σ^{-1} is also even.
(ii) If σ is odd, then σ^{-1} is also odd.

Proof. (i) Assume that σ is even. Then we can express σ as $\sigma = \eta_1 \cdots \eta_{2r}$, where the η_i's are transpositions. Since $(\eta_i)^{-1} = \eta_i$ for every $1 \leqslant i \leqslant 2r$, we have

$$\sigma^{-1} = \eta_{2r} \cdots \eta_1.$$

Therefore σ^{-1} is an even permutation.
(ii) The argument is similar. □

Proposition 8.28. *Let $n \in \mathbb{N}$. Then the subset A_n of S_n, consisting of the even permutations in S_n, is a subgroup of S_n.*

Proof. From 8.26(i) and Lemma 8.27(i) it follows that for all $\sigma, \tau \in A_n$ we have $\sigma\tau^{-1} \in A_n$. Thus by the Concise Subgroup Test, A_n is a subgroup of S_n. □

Definition 8.29. The subgroup A_n of the symmetric group S_n is called the *alternating group on n letters*.

Exercises.

(1) Write down all of the elements of A_3.
(2) Prove that for every integer $n \geqslant 2$, the group A_n has precisely $\dfrac{n!}{2}$ elements.

Conjugate permutations. So far we have not determined the centre of the group S_n. If σ is in the center of S_n, then for any $\tau \in S_n$ we have $\sigma\tau = \tau\sigma$, or equivalently, $\tau^{-1}\sigma\tau = \sigma$. Therefore understanding the centre of S_n naturally leads to investigating permutations of the form $\tau^{-1}\sigma\tau$.

Definition 8.30. Two permutations $\sigma, \sigma' \in S_n$ are called *conjugate* if there exists a permutation $\tau \in S_n$ such that $\sigma' = \tau^{-1}\sigma\tau$.

Remark 8.31. Note that the relation $\sigma' = \tau^{-1}\sigma\tau$ can be written as

$$\sigma = \tau\sigma'\tau^{-1} = \eta^{-1}\sigma'\eta \text{ where } \eta = \tau^{-1}.$$

This implies that being conjugate is a symmetric relation on permutations (i.e, if σ is conjugate to σ', the σ' is also conjugate to σ).

Example 8.32. The permutations

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix} \text{ and } \tau = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix}$$

are conjugate. Indeed $\sigma = \eta^{-1}\tau\eta$ where $\eta = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix}$. But how do we find such an η? This is explained in the rest of this section.

Let $n \in \mathbb{N}$, and let $\sigma \in S_n$ be an arbitrary permutation. Suppose that the cycle decomposition of σ is as follows:

$$\sigma = (a_1 \ \cdots \ a_{d_1})(b_1 \ \cdots \ b_{d_2}) \cdots$$

Now let $\tau \in S_n$ be another permutation. What is the cycle decomposition of $\tau^{-1}\sigma\tau$? We will answer this question in the next proposition.

Proposition 8.33. *Let σ and τ be as above. Then the cycle decomposition of $\tau^{-1}\sigma\tau$ is as follows:*

$$\tau^{-1}\sigma\tau = \big(\tau^{-1}(a_1) \ \cdots \ \tau^{-1}(a_{d_1})\big)\big(\tau^{-1}(b_1) \ \cdots \ \tau^{-1}(b_{d_2})\big)\cdots$$

Proof. We can trace the "journey" of an element $x \in \{1, \ldots, n\}$ under the composition $\tau^{-1}\sigma\tau$. For example, if $x = \tau^{-1}(a_i)$ for some $1 \leqslant i < d_1$ then it is moved as follows:

$$\tau^{-1}(a_i) \underbrace{\to}_{\text{by } \tau} a_i \underbrace{\to}_{\text{by } \sigma} a_{i+1} \underbrace{\to}_{\text{by } \tau^{-1}} \tau^{-1}(a_{i+1}),$$

and if $x = \tau^{-1}(a_{d_1})$ then a similar reasoning yields that under $\tau^{-1}\sigma\tau$ the element x is mapped to $\tau^{-1}(a_1)$. For the other cycles the argument is the same. □

Example 8.34. Let $\sigma = (2 \ 4 \ 6 \ 3 \ 5)(7 \ 1)$, and let $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 3 & 1 & 2 & 8 & 4 & 6 & 7 \end{pmatrix}$. Then the cycle decomposition of $\tau^{-1}\sigma\tau$ is

$$(4 \ 6 \ 7 \ 2 \ 1)(8 \ 3).$$

Note that the cycles that appear in the decompositions of σ and $\tau^{-1}\sigma\tau$ are different, but the lengths of the cycles that appear in the two decompositions are the same.

Example 8.35. Are the permutations $\sigma_1, \sigma_2 \in S_8$ defined by

$$\sigma_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 7 & 1 & 5 & 2 & 3 & 6 & 4 & 8 \end{pmatrix} \text{ and } \sigma_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 3 & 6 & 5 & 1 & 2 & 7 & 8 \end{pmatrix}$$

conjugate? To answer this question, we write the cycle decompositions of σ_1 and σ_2:

$$\sigma_1 = (1 \ 7 \ 4 \ 2)(3 \ 5) \text{ and } \sigma_2 = (1 \ 4 \ 5)(2 \ 6 \ 3).$$

Now one consequence of Proposition 8.33 is the following: if two permutations σ_1 and σ_2 in S_n are conjugate, then for any $1 \leqslant \ell \leqslant n$, the number of ℓ-cycles that appear in the cycle decompositions of σ_1 and σ_2 should be the same. However, there is one 2-cycle in the cycle decomposition of σ_1, and there are no 2-cycles in the cycle decomposition of σ_2. Therefore σ_1 and σ_2 are not conjugate.

Theorem 8.36. *Let $n \geqslant 3$. Then $Z(S_n)$ is equal to the trivial subgroup $\{\varepsilon\}$.*

Proof. Let $\sigma \neq \varepsilon$. We will prove that $\sigma \notin Z(S_n)$. We can express σ as a product of disjoint cycles. Then one of the following cases occurs:

Case I: There is at least one cycle in the cycle decomposition of σ of length at least 3. Thus, we can write the cycle decomposition as:

$$\sigma = (a_1 \ \cdots \ a_{d_1})(b_1 \ \cdots \ b_{d_2})\cdots,$$

where $d_1 \geqslant 3$. Now let consider the permutation $\tau \in S_n$ defined as follows:

$$\tau(a_1) = a_2, \ \tau(a_2) = a_1,$$

and $\tau(x) = x$ for all $x \notin \{a_1, a_2\}$. Then

$$\tau^{-1}\sigma\tau = (a_2 \ a_1 \ a_3 \ a_4 \ a_5 \ \cdots \ a_{d_1})(b_1 \ \cdots \ b_{d_2})\cdots$$

Clearly $\tau^{-1}\sigma\tau(a_1) = a_3$ whereas $\sigma(a_1) = a_2$. Thus $\tau^{-1}\sigma\tau \neq \sigma$, and therefore $\sigma \notin Z(G)$.

Case II: The cycle decomposition of σ has at least two 2-cycles. Thus we can express this cycle decomposition as

$$\sigma = (a\ \ b)(c\ \ d)\tau_3 \cdots \tau_r,$$

where a, b, c, d are distinct. Now consider the permutation $\tau \in S_n$ defined by

$$\tau(a) = c, \tau(c) = b, \tau(b) = a, \tau(d) = d,$$

and $\tau(x) = x$ for all $x \notin \{a, b, c, d\}$. Then the cycle decomposition of $\tau^{-1}\sigma\tau$ is

$$(b\ \ c)(a\ \ d)\tau_1 \cdots \tau_r.$$

In particular, $\sigma(a) = b$, whereas $\tau^{-1}\sigma\tau(a) = d$. Therefore $\sigma \neq \tau^{-1}\sigma\tau$, and hence $\sigma \notin Z(S_n)$.

Case III: The cycle decomposition of τ consists of a single 2-cycle. Thus

$$\sigma = (a\ \ b)$$

for distinct a, b. Since $n \geqslant 3$, we can choose an element $c \in \{1, \ldots, n\}$ such that $c \neq a$ and $c \neq b$. Now let $\tau = (a\ \ b\ \ c)$. Then

$$\sigma(a) = b \text{ and } \tau^{-1}\sigma\tau(a) = c,$$

therefore $\sigma \neq \tau^{-1}\sigma\tau$, and in particular $\sigma \notin Z(S_n)$. □

COSETS AND LAGRANGE'S THEOREM

Definition 9.1. Let G be any group and let H be a subgroup of G. For an element $a \in G$, the *left coset of H generated by a* is the set $aH := \{ah \ : \ h \in H\}$. The *right coset of H generated by a* is the set $Ha := \{ha \ : \ h \in H\}$.

Example 9.2. (i) Let $G = \mathbb{Z}$ and let

$$H = 3\mathbb{Z} = \{x \in \mathbb{Z} \ : \ x = 3k \text{ for some } k \in \mathbb{Z}\} = \{0, \pm 3, \pm 6, \pm 9, \ldots\}.$$

Then $0 + H = H = 3\mathbb{Z}$ and

$$1 + H = \{x \in \mathbb{Z} \ : \ x = 3k + 1 \text{ for some } k \in \mathbb{Z}\}$$

and

$$2 + H = \{x \in \mathbb{Z} \ : \ x = 3k + 2 \text{ for some } k \in \mathbb{Z}\}.$$

Note that

$$0 + H = H + 0 \ , \ 1 + H = H + 1 \ , \ 2 + H = H + 2.$$

(ii) Let $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ and let $H = \mathbb{Z}_2 \times \{0\} = \{(0,0), (1,0)\}$. Then the distinct left cosets of H are

$$(0,0) + H = H \text{ and } (0,1) + H = \{(0,1), (1,1)\}.$$

(iii) Let $G = S_3$ and let $H = \{\varepsilon, (1\ 2)\}$. Then we have

$$(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\} \quad \text{but} \quad H(1\ 3) = \{(1\ 3), (1\ 3\ 2)\}.$$

In particular, the left coset $(1\ 3)H$ and the right coset $H(1\ 3)$ are *not* identical.

Theorem 9.3. *Let G be a group, let $H \subseteq G$ be a subgroup of G, and let $a, b \in G$.*

(i) $H = He = eH$ where $e \in G$ denotes the identity of G.
(ii) $Ha = H$ if and only if $a \in H$.
(iii) $Ha = Hb$ if and only if $ab^{-1} \in H$.
(iv) $a \in Hb$ if and only if $Ha = Hb$.
(v) Either $Ha = Hb$ or $Ha \cap Hb = \varnothing$.
(vi) The right cosets of H form a partition of G.

Proof. (i) is obvious.

(ii) If $Ha = H$ then $a = ea \in Ha$, hence $a \in H$.

Conversely, if $a \in H$, then for any $x \in Ha$ we have $x = ha$ for some $x \in H$, hence $ha \in H$. This implies $Ha \subseteq H$. Also, for any $h \in H$ we have

$$h = ha^{-1}a,$$

and since $a^{-1} \in H$, we have $h \in Ha$. This implies $H \subseteq Ha$.

(iii) Assume that $Ha = Hb$. Then $a = ea \in Ha$, hence $a = hb$ for some $h \in H$. It follows that $ab^{-1} = hbb^{-1} = h \in H$.

Conversely, assume that $ab^{-1} \in H$. Then $ab^{-1} = h$ for some $h \in H$, that is, $a = hb$. Now:

- For every $x \in Ha$ we have $x = h_1 a$ for some $h_1 \in H$. Thus
$$x = h_1 a = h_1(hb) = (h_1)hb \text{ and } h_1 h \in H \;\Rightarrow\; x \in Hb.$$
This proves that $Ha \subseteq Hb$.
- Similarly, for every $x \in Hb$ we have $x = h_2 b$ for some $h_2 \in H$. Thus
$$x = h_2 b = h_2(h^{-1}a) = (h_2 h^{-1})a \text{ and } h_2 h^{-1} \in H \;\Rightarrow\; x \in Ha.$$
This proves that $Hb \subseteq Ha$.

(iv) If $a \in Hb$ then $a = hb$ for some $h \in H$, hence $ab^{-1} = hbb^{-1} = h \in H$. Now (iii) implies that $Ha = Hb$. Conversely, if $Ha = Hb$, then $a = ea \in Ha = Hb$, so $a \in Hb$.

(v) Assume that $Ha \cap Hb \neq \varnothing$. Choose $x \in Ha \cap Hb$. Then $x = h_1 a = h_2 b$ for $h_1, h_2 \in H$. Thus
$$a = (h_1)^{-1} h_1 a = (h_1)^{-1} h_2 b \in Hb.$$
Now (iv) implies that $Ha = Hb$.

(vi) Note that $a \in Ha$ for any $a \in G$, hence $G = \bigcup_{a \in G} Ha$. From (v) we know that the right cosets are either identical or disjoint. This implies that the right cosets induce a partition of the elements of G. □

Remark 9.4. There is a result analogous to Theorem 9.3 for left cosets. The only difference is that Theorem 9.3(iii) should be formulated as
$$aH = bH \text{ if and only if } a^{-1}b \in H.$$

Example 9.5. (i) Let $G = \mathbb{Z}_6$ and let $H = 2\mathbb{Z}_6 = \{0, 2, 4\}$. Then the distinct left cosets of H in G are
$$0 + H = \{0, 2, 4\} = 2 + H = 4 + H \text{ and } 1 + H = \{1, 3, 5\} = 3 + H = 5 + H.$$

(ii) Let $G = \mathbb{R}^* = \mathbb{R}\backslash\{0\}$ be the group of nonzero real numbers (where the binary operation is the usual multiplication). Let
$$H = \mathbb{R}^+ := \{x \in \mathbb{R} \,:\, x > 0\}.$$
Then the distinct left cosets of H in G are
$$1 \cdot H = \mathbb{R}^+ \text{ and } (-1)H = \mathbb{R}^- = \{x \in \mathbb{R} \,:\, x < 0\}.$$

(iii) Let $G = \mathbb{R}^2$, the 2-dimensional real vector space equipped with the usual vector addition. Also let $H = \mathbb{R} \times \{0\} = \{(x, 0) \,:\, x \in \mathbb{R}\}$. Then H is a subspace of G, and therefore also a subgroup of G. The distinct left cosets of H are the sets
$$(0, b) + H = \{(a, b) \,:\, a \in \mathbb{R}\} \quad \text{for every } b \in \mathbb{R}.$$

Example 9.6. More generally, let $G = \mathbb{Z}_n$ for some $n \in \mathbb{N}$, and let $H = d\mathbb{Z}_n$ for some $d \in \mathbb{N}$ which satisfies $d|n$. We want to determine the distinct left cosets of H.
Step 1. Observe that $d\mathbb{Z} = \{0, d, 2d, \ldots, n - d\}$ is the set of elements of $\{1, \ldots, n\}$ that are divisible by d, i.e., they have remainder 0 module d.
Step 2. There are d possible remainders modulo d:
$$0, 1, \ldots, d - 1.$$

For any $r \in \{0, 1, \ldots, d-1\}$, the elements of the left coset $r + H = r + d\mathbb{Z}$ are precisely those elements in $\mathbb{Z}_n$ which have remainder r modulo d. Thus, every element of $\mathbb{Z}_n$ belongs to *precisely one* of the sets $r + H$. Therefore there are in total d distinct left cosets of H, namely:

$$0 + H,\ 1 + H,\ \ldots, (d-1) + H.$$

Example 9.7. Let $G = \mathbb{Z} \times \mathbb{Z}$ and let $H = 2\mathbb{Z} \times 3\mathbb{Z} = \{(2k, 3s) \ : \ k, s \in \mathbb{Z}\}$. It is straightforward to verify that H is a subgroup of G. We would like to determine the distinct left cosets of H. First note that for any $(a, b) \in \mathbb{Z} \times \mathbb{Z}$, we have

$$\begin{aligned}(a, b) + H &= \{(2k + a, 3s + b) \ : \ k, s \in \mathbb{Z}\} \\ &= \{(m, n) \in \mathbb{Z} \times \mathbb{Z} \ : \ m \equiv a \bmod 2 \text{ and } n \equiv b \bmod 3\}.\end{aligned}$$

Thefore the left coset $(a, b) + H$ is determined by the remainders of the components modulo 2 and 3.

The possible remainders modulo 2 are 0,1 and the possible remainders modulo 3 are 0,1,2. Therefore we have a total of $2 \times 3 = 6$ pairs of possible remainders:

$$(0, 0),\ (0, 1),\ (0, 2),\ (1, 0),\ (1, 1),\ (1, 2).$$

For each of the above six pairs (a, b) we can form the corresponding left coset $(a, b) + H$, and every element $(m, n) \in \mathbb{Z} \times \mathbb{Z}$ appears in *precisely one* of these cosets (according to the remainder of m modulo 2, and the remainder of n modulo 3). Therefore in total there are six distinct left cosets of H:

$$(0, 0) + H,\ (0, 1) + H,\ (0, 2) + H,\ (1, 0) + H,\ (1, 1) + H,\ (1, 2) + H.$$

Example 9.8. Let $G = S_5$ and let $H = \{\sigma \in G \ : \ \sigma(5) = 5\}$. Note that H is a subgroup of G, and indeed $H \cong S_4$. Let us determine the distinct left cosets of H in G.

For any permutation $\sigma \in S_5$ we have $\sigma(5) \in \{1, 2, 3, 4, 5\}$. Now consider the transpositions

$$\tau_i = (5\ \ i).$$

Then the following statement holds for any $\sigma \in S_5$:

$$\text{Let } 1 \leqslant i \leqslant 5. \text{ If } \sigma(5) = i, \text{ then } \tau_i\sigma \in H. \text{ Conversely, if } \tau_i\sigma \in H, \text{ then } \sigma(5) = i.$$

For the "conversely" part, note that $\tau_i^2 = \varepsilon$, so that

$$\sigma(5) = \tau_i^2\sigma(5) = \tau_i\tau_i(\sigma(5)) = \tau_i(5) = i.$$

It follows that $\sigma_i H = \{\sigma \in S_5 \ : \ \sigma(5) = i\}$. Thus, the left cosets

$$\sigma_1 H, \sigma_2 H, \sigma_3 H, \sigma_4 H, \sigma_5 H$$

are distinct and their union is equal to $G = S_5$. Consequently, the left cosets

$$\sigma_1 H, \sigma_2 H, \sigma_3 H, \sigma_4 H, \sigma_5 H$$

are all of the distinct left cosets of H.

Remark 9.9. If G is an abelian group, then for any subgroup H of G and any $a \in G$ we have $aH = Ha$.

Exercises.

(1) Determine all of the left and right cosets of $H = \{\varepsilon, (1\ 2)\}$ in S_3.
(2) Let $G = \mathbb{R} \times \mathbb{R}$. Let H be the set of elements of G of the form (x, x), i.e.,

$$H = \{(x, y) \in G \ : \ x = y\}.$$

Prove that H is a subgroup of G. Also, show that the sets

$$(x, 0) + H \text{ for all } x \in \mathbb{R}.$$

are all of the distinct cosets of H in G.

Lemma 9.10. *Let G be a group, and let H be a subgroup of G. Then the following statements are true:*

(i) *For every $a \in G$, the maps*

$$\ell_a : H \to aH \ , \ h \mapsto ah$$

and

$$r_a : H \to Ha \ , \ h \mapsto ha$$

are bijections.
(ii) *If G is finite, then for every $a \in H$ we have $|H| = |Ha| = |aH|$.*

Proof. (i) We prove the statement for the first map. (For the second map the argument is similar.)

To show that the map ℓ_a is an injection:

$$\ell_a(x) = \ell_a(y) \ \Rightarrow \ ax = ay \ \Rightarrow \ x = y,$$

where the last step follows from the Left Cancellation Law.

To show ℓ_a is surjective: given any $x \in aH$ we have $x = ah$ for some $h \in H$. Therefore $\ell_a(h) = ah = x$. Hence ℓ_a is a surjection onto aH.

(ii) Follows from (i). □

Theorem 9.11. (Lagrange's Theorem) *Let G be a finite group, and let H be a subgroup of G. Then*

(i) *$|H|$ divides $|G|$.*
(ii) *The number of distinct right cosets of H in G is equal to $\frac{|G|}{|H|}$.*
(iii) *The number of distinct left cosets of H in G is equal to $\frac{|G|}{|H|}$.*

Proof. By Theorem 9.3(vi), G is partitioned into the right cosets of H. By Lemma 9.10(ii), all of the right cosets of H have $|H|$ elements. It follows that

$$|G| = |H| \times \text{ number of distinct right cosets of } H.$$

This completes the proof of (i) and (ii). The proof of (iii) is similar. □

Definition 9.12. Let G be a group and let H be a subgroup of G. The *index* of H in G, denoted by $[G : H]$, is defined as the number of distinct left (or right) cosets of H in G.

Example 9.13. Let $G = \mathbb{Z}_9$ and let $H = 3\mathbb{Z}_9 = \{0, 3, 6\}$. Then the distinct left cosets of H in G are $0 + H = \{0, 3, 6\}$, $1 + H = \{1, 4, 7\}$, and $2 + H = \{2, 5, 8\}$. Thus $[G : H] = 3$.

Corollary 9.14. *Let G be a finite group and let $g \in G$. Then $o(g)$ divides $|G|$.*

Proof. Note that $o(g) < \infty$ because G is finite (see Lemma 6.5). Set $H = \langle g \rangle$, the cyclic subgroup generated by G. Then Theorem 6.6(iii) implies that $|H| = o(g)$. Furthermore, Theorem 9.11(i) implies that $|H|$ divides $|G|$. Therefore $o(G)$ divides $|G|$. □

Example 9.15. Does S_5 have an element of order 48? The answer is no, because $|S_5| = 5! = 120$ and $48 \nmid 120$.

Corollary 9.16. *Let G be a finite group and let $g \in G$. Set $n := |G|$. Then $g^n = e$.*

Proof. From Corollary 9.14 it follows that $o(g)|n$. Thus

$$g^n = (g^{o(g)})^{\frac{n}{o(g)}} = e^{\frac{n}{o(g)}} = e.$$

□

Let $n \in \mathbb{N}$. Recall that $U(n)$ is the (multiplicative) group of invertible elements of $\mathbb{Z}_n$. We define

$$\phi(n) := |U(n)|.$$

For example $\phi(1) = \phi(2) = 1$, $\phi(3) = \phi(4) = 2$, etc. The function $\phi(n)$ is usually called the *Euler function*.

Corollary 9.17. (Fermat's Little Theorem/Euler's Theorem) *Let $n \in \mathbb{N}$. For every $a \in \mathbb{Z}$ such that $\gcd(a, n) = 1$, the residue of $a^{\phi(n)}$ modulo n is equal to 1.*

Proof. The statement is equivalent to showing that for any $k \in U(n)$, we have $k^{\phi(n)} = 1$ in $U(n)$. In this form, the statement follows from Corollary 9.16 for $G = U(n)$. □

Example 9.18. Suppose we want to compute the remainder of the division of 2^{1000} modulo 37. Then we note that 37 is a prime number, hence $\phi(37) = 36$. Note that Corollary 9.17 implies that the remainder of 2^{36} modulo 37 is 1. Now

$$1000 = 36 \times 27 + 28,$$

and therefore

$$2^{1000} = (2^{36})^{27} \times 2^{28} = 2^{28} \text{ modulo } 37.$$

To further simplify this calculation, note that $2^7 = 128 = 3 \times 37 + 17$, hence

$$2^{28} = (2^7)^4 = 17^4 = 83521 = 2257 \times 37 + 12.$$

Consequently, 2^{1000} modulo 37 is equal to 12.

Example 9.19. Let $G = D_n$, the n-th dihedral group (see Definition 7.10). Then $|D_n| = 2n$ and therefore by Lagrange's theorem for every $g \in D_n$ we have $o(g)|2n$. In fact there are *no* elements of order $2n$. Below we determine the orders of elements of D_6. We have

$$D_6 = \{e, r, r^2, r^3, r^4, r^5, s, sr, sr^2, sr^3, sr^4, sr^5\},$$

where $s^2 = e = r^6$ and $srs^{-1} = r^{-1}$. The orders of the elements of D_6 are given as follows:

g	e	r	r^2	r^3	r^4	r^5	s	sr	sr^2	sr^3	sr^4	sr^5
$o(g)$	1	6	3	2	3	6	2	2	2	2	2	2

All of these numbers of divisors of 12, and this is consistent with Lagrange's theorem.

Theorem 9.20. *Let G be a finite group such that $p := |G|$ is a prime number. Then G is isomorphic to $\mathbb{Z}_p$.*

Proof. Let $g \in G$ such that $g \neq e$. Then $o(g) \neq 1$, and by Corollary 9.14 we know that $o(g)$ divides p. However, p is prime and its only positive divisors are 1 and p. It follows that $o(g) = p$.

Now consider the cyclic subgroup of G generated by g, i.e., $H = \langle g \rangle$. By Theorem 6.6(iii), H has $o(g) = p$ elements. Thus $H = G$. It follows that G is a cyclic group with p elements. From the classification of cyclic groups, it follows that $G \cong \mathbb{Z}_p$. □

Exercises.

(1) Show that if n is a prime number, then $\phi(n) = n - 1$.
(2) Show that if $n = pq$ where p and q are prime numbers, then
$$\phi(n) = (p-1)(q-1).$$
(3) Calculate the remainder of 3^{1000} modulo 55.

CAYLEY'S THEOREM

The symmetric groups S_n have so far been the most interesting family of examples of finite non-abelian groups. These groups have the following remarkable property:

Theorem 10.1. (Cayley's Theorem) *Let G be a finite group, and set $n := |G|$. Then the group S_n has a subgroup isomorphic to G.*

Proof. Since $|G| = n$, we can think of S_n as the group of permutations of the elements of G. We denote the group of permutations of elements of G by S_G. Thus, $S_G \cong S_n$. Every element of S_G is a bijection from G onto G.

Step 1. For any $g \in G$, define the map $\sigma_g : G \to G$ by $\sigma_g(x) := gx$. By the Left Cancellation Law, the maps σ_g are injective. By Proposition 3.3, these maps are also surjective. Therefore the maps σ_g are permutations of the elements of G.

Step 2. Let $g_1, g_2 \in G$. We show that $\sigma_{g_1 g_2} = \sigma_{g_1} \circ \sigma_{g_2}$. To verify this statement, observe that for every $x \in G$,

$$\sigma_{g_1} \circ \sigma_{g_2}(x) = \sigma_{g_1}(g_2 x) = g_1(g_2 x) = (g_1 g_2)x = \sigma_{g_1 g_2}(x).$$

Step 3. From Step 2 it follows in particular that

$$\sigma_g \circ \sigma_{g^{-1}} = \sigma_e = \varepsilon \text{ and } \sigma_{g^{-1}} \circ \sigma_g = \varepsilon.$$

Therefore $(\sigma_g)^{-1} = \sigma_{g^{-1}}$ for every $g \in G$. (Here we interpret the left hand side of the last equation as the inverse of σ_g in S_G.)

Step 4. Set $H := \{\sigma_g \ : g \in G\}$. Thus H is a subset of S_G. We prove that H is a subgroup of G. To this end, we use the Concise Subgroup test (clearly $H \neq \varnothing$). For any two elements $\sigma_g, \sigma_h \in H$, where $g, h \in G$, by Steps 2 and 3 we have

$$\sigma_g \sigma_h^{-1} = \sigma_g \sigma_{h^{-1}} = \sigma_{gh^{-1}} \in H.$$

Thus by the Concise Subgroup Test, H is a subgroup of S_G.

Step 5. We now construct an isomorphism $\phi : G \to H$. Set

$$\phi(g) := \sigma_g \text{ for all } g \in G.$$

Clearly ϕ is a surjection. Furthermore, ϕ is an injection because

$$\phi(g_1) = \phi(g_2) \ \Rightarrow \ \sigma_{g_1} = \sigma_{g_2} \ \Rightarrow \ \sigma_{g_1}(e) = \sigma_{g_2}(e) \ \Rightarrow \ g_1 = g_2.$$

Finally, for every $g_1, g_2 \in G$,

$$\phi(g_1 g_2) = \sigma_{g_1 g_2} = \sigma_{g_1} \circ \sigma_{g_2} = \phi(g_1)\phi(g_2).$$

Thus $\phi : G \to H$ is an isomorphism.

□

Example 10.2. Let $G = \mathbb{Z}_3$. Then Theorem 10.1 states that G is isomorphic to a subgroup of S_3. The embedding is as follows: we have

$$G = \mathbb{Z}_3 = \{0, 1, 2\},$$

and therefore

$$\sigma_0(x) = x,\ \sigma_1(x) = x + 1,\ \sigma_2(x) = x + 2 \text{ for } x = 0, 1, 2.$$

Thus as permutations of $\{0, 1, 2\}$ we have

$$\sigma_0 = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix},\ \sigma_1 = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \end{pmatrix},\ \sigma_2 = \begin{pmatrix} 0 & 1 & 2 \\ 2 & 0 & 1 \end{pmatrix}.$$

Remark 10.3. Theorem 10.1 is a "no-go" theorem: it basically says that a complete understanding of all of the subgroups of S_n will include a complete understanding of all finite groups, and therefore it is an extremely difficult problem.

However, it is still useful to know that every finite group appears as a subgroup of a permutation group. Indeed in the first half of 20th century, qualitative results about subgroups of permutation groups played a major role in the development of group theory.

In the rest of this section we give an application of Cayley's theorem. We begin by two lemmas.

Lemma 10.4. *Let G be a group such that $|G|$ is even. Then there exists an element $g \in G$ such that $o(g) = 2$.*

Proof. We need to show that there is element $g \in G$ such that $g = g^{-1}$ and $g \neq e$.

Step 1. Suppose $G = \{g_1, \ldots, g_{2n}\}$ where $n \in \mathbb{N}$. Connect each $g \in G$ to g^{-1} by an arc. If $g = g^{-1}$, then this arc will become a loop at g.

Step 2. There is at least one loop (it is the arc connecting e to itself). Observe that the claim of the lemma is equivalent to showing that there is at least one more loop.

Step 3. Now suppose that, on the contrary, there are no loops other than the one connecting e to e. If there are k other arcs (which are not loops), then the end points of each of these arcs identify two distinct elements of G (which are also different from e). Therefore the total number of elements of G sohuld be $(2 \times k) + 1$, which is an odd number. This contradicts the assumption of the lemma. □

Remark 10.5. The following generalization of Lemma 10.4 is also true: if G is a finite group, then for every prime number p which divides $|G|$ there exists an element $g \in G$ such that $o(g) = p$. However, the proof of this statement requires techniques that are beyond this course.

Lemma 10.6. *Let G be a subgroup of S_n, where $n \in \mathbb{N}$. Assume that $G \not\subset A_n$, where $A_n \subseteq S_n$ is the alternating subgroup of S_n. Then $G \cap A_n$ is a subgroup of G that satisfies $[G : G \cap A_n] = 2$. In particular, $|G|$ is an even number.*

Proof. Since $G \cap A_n$ is the intersection of two subgroups of S_n, it is also a subgroup of S_n (or G).

Next we show that $[G : G \cap A_n] = 2$. Since $G \not\subset A_n$, at least one element of G is an odd permutation. Moreover, $\varepsilon \in G$ is an even permutation. Now let $m = |G|$, and write the elements of G as

$$\sigma_1, \ldots, \sigma_m,$$

such that $\sigma_1, \ldots, \sigma_k$ are odd permutations, whereas $\sigma_{k+1}, \ldots, \sigma_m$ are even permutations. By the above discussion, we have $1 < k < m$.

Step 1. Consider the following elements of G

$$\sigma_1\sigma_1, \ldots, \sigma_1\sigma_i, \ldots, \sigma_1\sigma_m.$$

By the Left Cancellation Law, these m elements are distinct. Since G has m elements, every element of G occurs in the above list. Thus, the above list contains every element of G precisely once.

Step 2. By Proposition 8.26, the permutations $\sigma_1\sigma_i$ are even for $1 \leqslant i \leqslant k$, and odd for $k + 1 \leqslant i \leqslant m$. It follows that the elements of G should consist of k even permutations and $m - k$ odd permutations.

Step 3. Comparing the conclusion of Step 2 with the definition of k prior to Step 1, we obtain that $k = m - k$. Thus the number of elements of G is $m = 2k$ (hence it is an even number), and the number of elements of $G \cap A_n$ is k. It follows that

$$[G : G \cap A_n] = \frac{|G|}{|G \cap A_n|} = 2. \qquad \square$$

Theorem 10.7. *Let G be a group such that $|G| = 2n$, where n is an odd integer. Then G contains a subgroup of order n.*

Proof. Step 1. By Cayley's Theorem, we can assume that G is a subgroup of S_G. Recall from the proof of Cayley's Theorem that the element of S_G corresponding to any $g \in G$ is the permutation

$$\sigma_g : G \to G \ , \ \sigma_g(x) = gx.$$

Step 2. By Lemma 10.4, there exists an element $g_\circ \in G$ such that $o(g_\circ) = 2$. Note that in particular $g_\circ \neq e$. Then for any $x \in G$ we have

$$\sigma_{g_\circ}(x) = g_\circ x \neq x \text{ and } \sigma_{g_\circ}(\sigma_{g_\circ}(x)) = g_\circ^2 x = x.$$

It follows that $\sigma_{g_\circ}$ is a product of n disjoint transpositions of the form $(x, \sigma_{g_\circ}(x))$.

Step 3. Since n is odd, Step 2 implies that the permutation $\sigma_{g_\circ}$ is an odd permutation. Now let A_G denote the alternating subgroup of S_G. (In other words A_G is the subgroup of S_G consisting of even permutations.) Since $\sigma_{g_\circ} \in G$, it follows that $G \not\subset A_G$. Therefore Lemma 10.6 implies that $[G : G \cap A_G] = 2$. Consequently, $G \cap A_G$ is a subgroup of G that satisfies $|G \cap A_G| = n$. $\square$

NORMAL SUBGROUPS AND QUOTIENTS

This section introduces an important construction in algebra: the quotient construction.

Definition 11.1. Let G be any group and let H be a subgroup of G. We call H a *normal* subgroup of G if $gH = Hg$ for every $g \in H$.

Example 11.2. (i) Let $G = \mathbb{Z}$ and let $H = 2\mathbb{Z} = \{0, \pm 2, \pm 4, \pm 6, \ldots\}$. The left cosets of H are of the form $k + H$ for $k \in \mathbb{Z}$, and the distincts right cosets of H are of the form $H + k$ for $k \in \mathbb{Z}$. Since $\mathbb{Z}$ is abelian, we have

$$k + H = H + k \text{ for all } k \in \mathbb{Z}.$$

Therefore H is a normal subgroup of G.

(ii) Let $G = S_3$, and let $H = \{\varepsilon, (1\ 2)\}$. In Example 9.2(iii) we noted that

$$(1\ 3)H \neq H(1\ 3).$$

Therefore H is not a normal subgroup of G.

(iii) Let G be an abelian group. Then every subgroup of G is normal.

Let G be a group and let H be a subgroup of G. Then for every $g \in G$ we define

$$gHg^{-1} = \{ghg^{-1} \ : \ g \in G\}.$$

Theorem 11.3. *Let G be any group and let H be a subgroup of G. Then the following statements are equivalent:*

(i) H is a normal subgroup of G.
(ii) $gHg^{-1} = H$ for every $g \in G$.
(iii) $gHg^{-1} \subseteq H$ for every $g \in G$.

Proof. (i)⇒(ii): Let $g \in G$. Then $gH = Hg$, and therefore

$$gHg^{-1} = (gH)g^{-1} = (Hg)g^{-1} = H(gg^{-1}) = He = H.$$

(ii)⇒(iii): This is trivial.
(iii)⇒(i): Let $g \in G$. Then $gHg^{-1} \subseteq H$, and replacing g by g^{-1} we have $g^{-1}Hg \subseteq H$. Multiplying the latter relation on the left by g and on the right by g^{-1}, we obtain :

$$g(g^{-1}Hg)g^{-1} \subseteq gHg^{-1} \text{ and consequently, } H \subseteq gHg^{-1}.$$

Now we have

$$gHg^{-1} \subseteq H \text{ and } H \subseteq gHg^{-1},$$

hence $H = gHg^{-1}$. Multiplying the latter relation on the right by g, and using associativity of G, we obtain

$$Hg = (gHg^{-1})g = (gH)(g^{-1}g) = gH(e) = gH. \qquad \square$$

Proposition 11.4. *The alternating group A_n is a normal subgroup of the symmetric group S_n for every $n \in \mathbb{N}$.*

Proof. Step 1. If σ is an even permutation then $\sigma A_n = A_n$, because $\sigma \in A_n$ and therefore the map $\eta \mapsto \sigma\eta$ is a permutation of the elements of A_n. (This follows from Proposition 3.3.)
Step 2. A similar argument also implies that If σ is an even permutation then $A_n\sigma = A_n$.
Step 3. Now if σ is an odd permutation, then the elements of σA_n are a product of an even permutation and an odd permutation. By Proposition 8.26, such products are always odd permutations. Therefore $\sigma A_n \subseteq S_n \backslash A_n$. In addition, every odd permutation $\tau \in S_n$ belongs to σA_n. Indeed we have

$$\tau = \varepsilon\tau = (\sigma\sigma^{-1})\tau = \sigma(\sigma^{-1}\tau),$$

and $\sigma^{-1}\tau \in A_n$ because σ^{-1} and τ are both odd permutations (see Proposition 8.26 and Lemma 8.27).
Step 4. By Step 3, for every odd permutation we have $\sigma A_n = S_n \backslash A_n$. A small modification of the argument of Step 3 shows that we also have $A_n\sigma = S_n \backslash A_n$.
Step 5. From Steps 1, 2, and 4 it follows that

$$\begin{cases} \sigma A_n = A_n\sigma = A_n & \text{if } \sigma \text{ is even,} \\ \sigma A_n = A_n\sigma = S_n \backslash A_n & \text{if } \sigma \text{ is odd.} \end{cases}$$

Thus in particular A_n is a normal subgroup of S_n. □

Proposition 11.5. *Let G be any group. Then the centre of G is a normal subgroup of G.*

Proof. Let $g \in G$. Then for any $h \in Z(G)$ we have $ghg^{-1} = h$. Therefore,

$$gZ(G)g^{-1} = \{ghg^{-1} \ : \ h \in Z(G)\} = Z(G).$$

Consequently, $Z(G)$ is a normal subgroup of G. □

The following statement extends Proposition 11.4.

Proposition 11.6. *Let G be a group and let H be a subgroup of G such that $[G : H] = 2$. Then H is a normal subgroup of G.*

Proof. Step 1. Since $[G : H] = 2$, the subgroup H has only two distinct right cosets. These cosets can be represented as $H = He$ and Hg, where $g \notin H$. Since $H \cup Hg = G$ and $H \cap Hg = \varnothing$, it follows that $Hg = G \backslash H$. Consequently, we have

$$Hg = \begin{cases} H & \text{if } h \in H, \\ G\backslash H & \text{if } h \notin H. \end{cases}$$

Step 2. The argument of Step 1 also works for left cosets, and therefore we obtain that

$$gH = \begin{cases} H & \text{if } h \in H, \\ G\backslash H & \text{if } h \notin H. \end{cases}$$

Step 3. From Step 1 and Step 2 it follows that $Hg = gH$ for every $g \in G$. Hence H is a normal subgroup of G. □

Exercises.

(1) Let G be any group. Prove that every subgroup of $Z(G)$ is a normal subgroup of G.
(2) Let D_5 be the 5-th dihedral group (which is the group of symmetries of a regular pentagon). Does D_5 have a normal subgroup with 5 elements?
(3)

Lemma 11.7. *Let G be any group, and let H and K be two normal subgroups of G. Then $H \cap K$ is also a normal subgroup of G.*

Proof. In Proposition 5.8 we proved that the intersection of any two subgroups is a subgroup. Hence $H \cap K$ is a subgroup of G. Next we show that it is normal.

Since H and K are normal, for every $g \in G$ we have

$$gHg^{-1} = H \text{ and } gKg^{-1} = K.$$

Thus

$$g(H \cap K)g^{-1} = \{gxg^{-1} \in G \ : \ x \in H \cap K\} \subseteq gHg^{-1} \cap gKg^{-1} = H \cap K.$$

Now Theorem 11.3 implies that $H \cap K$ is a normal subgroup of G. □

Remark 11.8. Let G be a group and let H, K be subgroups of G. We define

$$HK := \big\{x \in G \ : \ \exists_{h\in H}\exists_{k\in K}(x = hk)\big\}.$$

Note that HK is not necessarily a subgroup of G. For example, $G = S_3$, $H = \{\varepsilon, (1\ 2)\}$, and $K = \{\varepsilon, (1\ 3)\}$. Then

$$HK = \{\varepsilon, (1\ 2), (1\ 3), (1\ 3\ 2)\}$$

and it is easy to verify that HK is not a subgroup of S_3.

In the next proposition, we provide some conditions that determine when HK is a subgroup of G.

Proposition 11.9. *Let G be a group and let H, K be subgroups of G. The following statements are equivalent:*

(i) HK is a subgroup of G.
(ii) $HK = KH$.
(iii) KH is a subgroup of G.

Proof. (i) ⇒ (ii): We need to show that $HK \subseteq KH$ and $KH \subseteq HK$.

$\underline{KH \subseteq HK}$: Let $x \in KH$. Then $x = hk$ for some elements $h \in H$ and $k \in K$. We need to show that $kh \in HK$. We have

$$h^{-1} \in H \text{ and } k^{-1} \in K,$$

since H and K are subgroups. Thus $h^{-1}k^{-1} \in HK$. Set $g = h^{-1}k^{-1}$. Since HK is a subgroup, we have $g^{-1} \in HK$. But $g^{-1} = (h^{-1}k^{-1})^{-1} = kh$. Thus $kh \in HK$, as desired.

$\underline{HK \subseteq KH}$: Let $x \in HK$. Then $x^{-1} \in HK$ because HK is a subgroup of G. It follows that

$$x^{-1} = hk \text{ for some elements } h \in H \text{ and } k \in K.$$

Now

$$x = (x^{-1})^{-1}(hk)^{-1} = k^{-1}h^{-1} \in KH,$$

because $k^{-1} \in K$ and $h^{-1} \in H$ (since H and K are subgroups of G).

(ii) $\Rightarrow$ (iii): As usual let e denote the identity element of G. Since $e \in H$ and $e \in K$, we have $e \in KH$. We now use the Concise Subgroup Test. To this end, we need to verify that if $k_1h_1 \in KH$ and $k_2h_2 \in KH$, then

$$(k_1h_1)(k_2h_2)^{-1} \in KH.$$

We have $(k_1h_1)(k_2h_2)^{-1} = k_1h_1h_2^{-1}k_2^{-1}$. Now

$$k_1h_1h_2^{-1} \in KH = HK,$$

hence there there exist elements $h_3 \in H$ and $k_3 \in K$ such that $k_1h_1h_2^{-1} = h_3k_3$. It follows that

$$(k_1h_1)(k_2h_2)^{-1} = (k_1h_1h_2^{-1})k_2^{-1} = (h_3k_3)k_2^{-1} = h_3(k_2k_3)^{-1} \in HK,$$

as desired.

The proof of (iii)$\Rightarrow$(ii) is similar to (i)$\Rightarrow$(ii), and the proof of (ii)$\Rightarrow$(iii) is similar to (ii)$\Rightarrow$(i). $\square$

Proposition 11.10. *Let G be a group and let H, K be subgroups of G.*

(i) If at least one of H and K is a normal subgroup of G, then HK is a subgroup of G.
(ii) If both H and K are normal subgroups of G, then HK is also a normal subgroup of G.

Proof. (i) First assume that H is normal.

Step 1. By Proposition 11.9, in order to show that HK is a subgroup of G it is enough to verify that $HK = KH$. To this end, we will verify that $HK \subseteq KH$ and $KH \subseteq HK$.

Step 2. We verify that $HK \subseteq KH$. Let $g \in HK$. Then there exist elements $h \in H$ and $k \in K$ such that $g = hk$. Now

$$g = hk = ehk = (kk^{-1})hk = k(k^{-1}hk).$$

Since H is normal, we have $k^{-1}Hk \subseteq H$ and therefore $k^{-1}hk \in H$. Therefore $g \in KH$.

Step 3. The proof of the reverse inclusion $KH \subseteq HK$ is similar to to Step 2:

$$\begin{aligned} g \in KH &\Rightarrow g = k_1h_1 \text{ where } k_1 \in K, h_1 \in H \\ &\Rightarrow g = k_1h_1 = (k_1hk_1^{-1})k_1 \in HK \Rightarrow g \in HK, \end{aligned}$$

since $k_1Hk_1^{-1} \subseteq H$.

Now assume that K is normal. Then by the argument given above, it follows that KH is a subgroup of G. In particular, Proposition 11.9 implies that $KH = HK$. Thus HK is a subgroup of G.

(ii) By part (i), we know that HK is a subgroup of G. Thus we only need to show that HK is normal. For any $g \in G$,

$$gHKg^{-1} = gHeKg^{-1} = gHg^{-1}gKg^{-1} = (gHg^{-1})(kHk^{-1}) = (H)(K) = HK.$$

Therefore for every $g \in G$ we have $gHKg^{-1} = HK$. Consequently, HK is a normal subgroup of G. $\square$

Quotient groups. Let G be a group and let N be a normal subgroup of G. Our next goal is to define a group structure on the set of left cosets of G. We define

$$G/N = \{gN \ : \ g \in G\}.$$

Example 11.11. Let $G = \mathbb{Z}_4 = \{0, 1, 2, 3\}$ and let $N = 2\mathbb{Z}_4 = \{0, 2\}$. Then N is a normal subgroup of G, and it has two distinct left cosets: $0 + N$ and $1 + N$. Therefore G/N is the two-element set

$$G/N = \{0 + N, 1 + N\}.$$

We now define the following binary operation on G/N:

$$(g_1N) \star (g_2N) := g_1g_2N. \tag{11.1}$$

Lemma 11.12. *Let G be a group and let N be a normal subgroup of G. Then the binary operation defined in* (11.1) *is well-defined, i.e., if $g_1N = h_1N$ and $g_2N = h_2N$ for $g_1, g_2, h_1, h_2 \in G$, then*

$$g_1N \star g_2N = h_1N \star h_2N.$$

Proof. From the definition of the binary operation $\star$ on the cosets, given in Equation (11.1) above, it follows that we need to verify that

$$g_1g_2N = h_1h_2N.$$

By Theorem 9.3, this is equivalent to the relation

$$(h_1h_2)^{-1}(g_1g_2) \in N.$$

In the following steps we prove this relation.

Step 1. From $g_1N = h_1N$ and Theorem 9.3 it follows that $h_1^{-1}g_1 \in N$. Similarly, from $g_2N = h_2N$ it follows that $h_2^{-1}g_2 \in N$.

Step 2. We can now write

$$(h_1h_2)^{-1}g_1g_2 = h_2^{-1}h_1^{-1}g_1g_2 = h_2^{-1}(h_1^{-1}g_1)g_2. \tag{11.2}$$

Since $h_1^{-1}g_1 \in N$, we have $(h_1^{-1}g_1)g_2 \in Ng_2$. Since N is normal, we have $Ng_2 = g_2N$. Therefore $(h_1^{-1}g_1)g_2 \in g_2N$. In particular, there exists an element $\overline{h} \in N$ such that

$$(h_1^{-1}g_1)g_2 = g_2\overline{h}.$$

Step 3. Now using Equation (11.2) we have

$$(h_1h_2)^{-1}g_1g_2 = h_2^{-1}(g_2\overline{h}) = (h_2^{-1}g_2)\overline{h} \in N,$$

because $h_2^{-1}g_2 \in N$ by Step 1 and $\overline{h} \in N$. □

Example 11.13. Let G and N be as in Example 11.11. Then the binary operation on G/N has the following Cayley table:

$\star$	$0+N$	$1+N$
$0+N$	$0+N$	$1+N$
$1+N$	$1+N$	$0+N$

Example 11.14. Let $G = \mathbb{Z}$ and let $N = 3\mathbb{Z}$. Then

$$G/N = \{0 + \mathbb{Z}, 1 + \mathbb{Z}, 2 + \mathbb{Z}\},$$

and the binary operation on G/N has the following Cayley table:

$\star$	$0+N$	$1+N$	$2+N$
$0+N$	$0+N$	$1+N$	$2+N$
$1+N$	$1+N$	$2+N$	$0+N$
$2+N$	$2+N$	$0+N$	$1+N$

Theorem 11.15. *Let G be a group and let H be a normal subgroup of G. Then the set G/N equipped with the binary operation* (11.1) *is a group.*

Proof. We verify the three properties of a group:

Associativity of $\star$: we have

$$(g_1N \star g_2N) \star g_3N = (g_1g_2)N \star g_3N = ((g_1g_2)g_3)N$$

and similarly,

$$g_1 \star (g_2N \star g_3N) = (g_1(g_2g_3))N.$$

Now associativity of G/N follows from $(g_1g_2)g_3 = g_1(g_2g_3)$, i.e., from associativity of G.

Identity element: as usual let $e \in G$ denote the identity element. We have

$$eN \star gN = egN = gN \text{ and } gN \star eN = geN = gN.$$

Thus eN is the identity element of G/N.

Inverse: for every $gN \in G/N$, we have

$$gN \star g^{-1}N = gg^{-1}N = eN \text{ and } g^{-1} \star gN = g^{-1}gN = eN$$

□

Definition 11.16. Let G be a group and let N be a normal subgroup of G. Then the group G/N whose binary operation is defined in (11.1) is called the *quotient* of G by N.

Example 11.17. Let $G = \mathbb{Z}$ and let $N = 3\mathbb{Z}$. Then

$$G/N = \{0+N, 1+N, 2+N\}.$$

By Theorem 11.15, G/N is a group. Since $|G/N| = 3$, it follows from Proposition 4.2 that G/N is isomorphic to the group $\mathbb{Z}_3$.

Example 11.18. More generally, let $G = \mathbb{Z}$ and let $N = d\mathbb{Z}$ where $d \in \mathbb{N}$. Then

$$G/N = \{0+N, 1+N, \ldots, (d-1)+N.$$

The identity element of G/N is $0+N$. Furthermore if we set $x = 1+N$, then for every $1 \leqslant k \leqslant d-1$ we have:

$$\underbrace{x \star \cdots \star x}_{k \text{ times}} = \underbrace{(1+N) \star \cdots \star (1+N)}_{k \text{ times}} = (\underbrace{1+\cdots+1}_{k \text{ times}}) + N = k+N,$$

and

$$\underbrace{x \star \cdots \star x}_{d \text{ times}} = 0+N.$$

It follows that G/N is generated by $x = 1+N$. Therefore G/N is a cyclic group of order d. Hence $G/N \cong \mathbb{Z}_d$.

Example 11.19. Let $n \geqslant 1$ and let $G = \mathbb{Z}_n$. Also let $N = d\mathbb{Z}_n$ where $d|n$. Then N is a normal subgroup of G (because G is abelian). As we showed in Example 9.6, the elements of G/N are

$$\{0 + N, 1 + N, \ldots, (d-1) + N\}.$$

The identity element of G/N is $0 + N$, and furthermore

$$k + N = \underbrace{(1 + N) + \cdots + (1 + N)}_{k \text{ times}} \text{ for every } 1 \leqslant k \leqslant d - 1.$$

Therefore G/N is a cyclic group of order d. It follows that $G/N \cong \mathbb{Z}_d$.

Example 11.20. Let $G = \mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ and let $H = \mathbb{R} \times \{0\}$. In Example 9.5(iii) we showed that the distinct left cosets of H are of the form

$$(0, y) + H \text{ for all } y \in \mathbb{R}.$$

Thus there is a natural bijective correspondence between elements of $\mathbb{R}$ and the distinct left cosets of H. This bijection is given by the map

$$\phi : \mathbb{R} \to G/H \ , \ \phi(y) = (0, y) + H.$$

Moreover, from the definition of the operation $\star$ on G/H it follows that for every $y_1, y_2 \in \mathbb{R}$, we have

$$((0, y_1) + H) \star ((0, y_2) + H) = ((0, y_1) + (0, y_2)) + H = (0, y_1 + y_2) + H.$$

The left hand side of the above relation is $\phi(y_1) \star \phi(y_2)$, and its right hand side is $\phi(y_1 + y_2)$. It follows that $\phi(y_1 + y_2) = \phi(y_1) \star \phi(y_2)$. Therefore the bijection ϕ is indeed an isomorphism of groups. Thus $G/H \cong \mathbb{R}$.

Example 11.21. Let $G = \mathbb{R} \times \mathbb{R}$ and let $H = \{(x, x) \ : \ x \in \mathbb{R}\}$. In one of the exercises of Section 9 you have shown that H is a subgroup of G, and the sets

$$(x, 0) + H$$

are all of the distinct left cosets of G. Thus there is a natural bijection ϕ between elements of $\mathbb{R}$ and G/H:

$$\phi : \mathbb{R} \to G/H \ , \ \phi(t) = (t, 0) + H.$$

Moreover, from the definition of the operation $\star$ on G/H we have

$$((x_1, 0) + H) \star (H + (x_2, 0) + H) = ((x_1, 0) + (x_2, 0)) + H = (x_1 + x_2) + H.$$

The left hand side of the above relation is $\phi(x_1) \star \phi(x_2)$, and its right hand side is $\phi(x_1 + x_2)$. It follows that $\phi(x_1 + x_2) = \phi(x_1) \star \phi(x_2)$. Consequently, ϕ is an isomorphism. This implies that $G/H \cong \mathbb{R}$.

Exercises.

(1) Let G be a group and let N be a subgroup of G. Prove that if the binary operation defined on G/N in (11.1) is well-defined, then N should be normal.

(2) Let G be any group and let $H \subseteq G \times G$ be defined by:

$$H = \{(x, x) \ : \ x \in G\}.$$

Prove that H is a subgroup of $G \times G$. Further, prove that H is normal if and only if G is abelian.

(3) Let G be a group and let H be the subset of the Cartesian product $G \times G$ defined as follows:

$$H = \{(x, y) \in G \times G \ : \ y = x^{-1}\}.$$

Prove that H is a subgroup of $G \times G$ if and only if G is abelian.

GROUP HOMOMORPHISMS

We will now generalize the notion of group isomorphisms.

Definition 12.1. Let G and H be two groups. A map $\phi : G \to H$ which satisfies

$$\phi(g_1 g_2) = \phi(g_1)\phi(g_2) \text{ for all } g_1, g_2 \in G$$

is called a *group homomorphism* from G to H.

Lemma 12.2. *Let G and H be two groups and let $\phi : G \to H$ be a group homomorphism. Then the following statements hold:*

(i) $\phi(e_G) = e_H$, where e_G and e_H denote the identity elements of G and G.
(ii) $\phi(g^{-1}) = \phi(g)^{-1}$ for every $g \in G$.

Proof. The proof of this lemma is exactly the same as the argument given for Remark 3.13. We leave the adaptation of the proof as an exercise. □

Example 12.3. (i) Let $G = H = \mathbb{Z}$, and let $\phi : G \to H$ be defined by $\phi(k) = 2k$ for every $k \in G = \mathbb{Z}$. Then ϕ is a group homomorphism. However, ϕ is not surjective and therefore it is not an isomorphism.

(ii) Let $G = \mathbb{Z}$ and let $H = \mathbb{Z}_3$. Consider the map

$$\phi : \mathbb{Z} \to \mathbb{Z}_3 \ , \ \phi(k) = \text{ the remainder of } k \text{ modulo } 3.$$

Then ϕ is a group homomorphism because for every $k_1, k_2 \in \mathbb{Z}$,

$$\begin{aligned}\phi(k_1 + k_2) &= \text{ the remainder of } k_1 + k_2 \text{ modulo } 3 \\ &= (\text{ the remainder of } k_1 \text{ modulo } 3) + (\text{ the remainder of } k_2 \text{ modulo } 3)\,.\end{aligned}$$

However, ϕ is not an isomorphism because it is not a bijection. Indeed

$$\phi^{-1}(0) = \{0, \pm 3, \pm 6, \pm 9, \pm 12, \ldots\}.$$

(iii) Let $n \in \mathbb{N}$ and let $G = S_n$. Also, let $H = \mathbb{Z}_2$. Then consider the map

$$\phi : S_n \to \mathbb{Z}_2 \ , \ \phi(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ is an even permutation,} \\ 1 & \text{if } \sigma \text{ is an odd permutation.} \end{cases}$$

From Proposition 8.26 it follows that ϕ is a homomorphism of groups. Furthermore, $\phi^{-1}(0) = A_n$, where A_n is as usual the subgroup of even permutations of S_n.

Definition 12.4. Let $\phi : G \to H$ be a group homomorphism. The *image* of ϕ, denoted by $\operatorname{im} \phi$, is defined as

$$\operatorname{im} \phi = \{\phi(g) \ : \ g \in G\}.$$

The *kernel* of ϕ, denoted by $\ker \phi$, is defined by

$$\ker \phi = \{g \in G \ : \ \phi(g) = e_H\}.$$

Example 12.5. Here are some examples of image and kernel of homomorphisms.

(i) Let $G = \mathbb{Z}$, $H = \mathbb{Z}$, and $\phi : G \to H$ be the map defined by $\phi(k) := 2k$ for every $k \in \mathbb{Z}$. Then $\operatorname{im}\phi = 2\mathbb{Z} \subset \mathbb{Z}$, and $\ker\phi = \{0\}$.

(ii) Let $G = \mathbb{Z}$, $H = \mathbb{Z}_3$, and $\phi : \mathbb{Z} \to \mathbb{Z}_3$ be the homomorphism defined in Example 12.3(ii). Then

$$\operatorname{im}\phi = \mathbb{Z}_3 \text{ and } \ker\phi = 3\mathbb{Z} = \{0, \pm 3, \pm 6, \pm 9, \ldots\}.$$

(iii) More generally, let $d \in \mathbb{N}$ and let $\phi : \mathbb{Z} \to \mathbb{Z}_d$ be the map defined as follows: for every $k \in \mathbb{Z}$ we set

$$\phi(k) = \text{ the remainder of } k \text{ modulo } d.$$

Then ϕ is a group homomorphism: to prove this claim, take any $k_1, k_2 \in \mathbb{Z}$ and set $r_1 = \phi(k_1)$ and $r_2 = \phi(k_2)$. Then by the division algorithm we have

$$k_1 = dq_1 + r_1 \text{ and } k_2 = dq_2 + r_2 \text{ where } q_1, q_2 \in \mathbb{Z}.$$

Thus

$$(k_1 + k_2) - (r_1 + r_2) = d(q_1 + d_2),$$

and it follows that the remainder of $k_1 + k_2$ modulo d is equal to the remainder of $r_1 + r_2 = \phi(k_1) + \phi(k_2)$ modulo d. Consequently,

$$\phi(k_1 + k_2) = \phi(k_1) + \phi(k_2) \text{ in } \mathbb{Z}_d.$$

The kernel of ϕ is the set of all elements of $\mathbb{Z}$ which have remainder 0 modulo d. Thus

$$\ker\phi = d\mathbb{Z}.$$

(iv) Let $G = S_n$, let $H = \mathbb{Z}_2$, and let $\phi : G \to H$ be the homomorphism defined in Example 12.3(iii). Then

$$\operatorname{im}\phi = \mathbb{Z}_2 \text{ and } \ker\phi = A_n.$$

(v) Let G be any group and let $H = \{e\}$ be the one-element group. Then the map $\phi : G \to H$ which maps every element of g to $e \in H$ is a group homomorphism. Moreover, $\ker\phi = G$.

Example 12.6. Let $G = H = \mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$, and let $\phi : G \to H$ be the map defined by

$$\phi\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 1 & -3 \\ 2 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Here we are representing the elements of $G = \mathbb{R} \times \mathbb{R}$ as a vertical vector. Thus we have

$$\phi(x, y) = (x - 3y, 2x - 6y).$$

Then $\ker\phi$ is equal to $\ker(A)$, i.e., the solution set of the homogeneous linear system

$$\begin{cases} x - 3y = 0 \\ 2x - 6y = 0 \end{cases}$$

In this linear system x is a basis variable and y is a free variable. Therefore:

$$\ker\phi = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \ : \ x = 3y \right\} = \left\{ \begin{bmatrix} 3t \\ t \end{bmatrix} \ : \ t \in \mathbb{R} \right\}.$$

Proposition 12.7. *Let G and H be two groups and let $\phi : G \to H$ be a group homomorphism. Then the following statements hold:*

(i) $\operatorname{im}\phi$ *is a subgroup of* H.
(ii) $\ker\phi$ *is a normal subgroup of* G.
(iii) ϕ *is an injection if and only if* $\ker\phi = \{e\}$, *where* e *denotes the identity element of* G.

Proof. (i) Clearly $\operatorname{im}\phi$ is nonempty. We now use the Concise Subgroup Test.

For all $x, y \in \operatorname{im}\phi$,

$$x = \phi(a) \text{ and } y = \phi(b) \text{ for some } a, b \in G.$$

Since $\phi(b)^{-1} = \phi(b^{-1})$, we have

$$xy^{-1} = \phi(a)\phi(b)^{-1} = \phi(a)\phi(b^{-1}) = \phi(ab^{-1}),$$

hence $ab^{-1} \in \operatorname{im}(\phi)$.

(ii) First we show that $\ker\phi$ is a subgroup of G. Note that $e \in \ker\phi$ so $\ker\phi \neq \varnothing$. Now we use the Concise Subgroup Test. Let $a, b \in \ker\phi$. Then

$$\phi(a) = e \text{ and } \phi(b) = e \;\Rightarrow\; \phi(ab^{-1}) = \phi(a)\phi(b^{-1}) = \phi(a)\phi(b)^{-1} = e,$$

and it follows that $ab^{-1} \in \ker\phi$.

Next we show that $\ker\phi$ is a normal subgroup of G. To this end, it is enough to show that $g(\ker\phi)g^{-1} \subseteq \ker\phi$ for every $g \in G$. To prove the latter statement, note that for any $x \in \ker\phi$ we have

$$\begin{aligned}\phi(gxg^{-1}) = \phi(g)\phi(x)\phi(g^{-1}) &= \phi(g)e\phi(g^{-1}) \\ &= \phi(g)\phi(g^{-1}) = \phi(gg^{-1}) = \phi(e) = e,\end{aligned}$$

hence $gxg^{-1} \in \ker\phi$.

(iii) Assume that ϕ is an injection. If $\phi(x) = e_H$, then $\phi(x) = e_H = \phi(e_G)$, and injectivity of ϕ implies that $x = e_G$.

Conversely, assume that $\phi(x) = e_H$ implies that $x = e_G$. Now assume $\phi(a) = \phi(b)$ for some $a, b \in G$. Then

$$\phi(ab^{-1}) = \phi(a)\phi(b^{-1}) = \phi(a)\phi(b)^{-1} = e_H,$$

hence $ab^{-1} = e_G$. It follows immediately that $a = b$. □

Example 12.8. Let $G = \mathbb{Z} \times \mathbb{Z}$, let $H = \mathbb{Z}$, and let $\phi = G \to H$ be given by $\phi(x, y) = 3x - 2y$. Then

$$\ker(\phi) = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \;:\; 3x - 2y = 0\}.$$

We have

$$3x = 2y \;\Rightarrow\; 3|2y, \text{ but } \gcd(2, 3) = 1, \text{ therefore by we have } 3|y.$$

A similar reasoning implies that $2|x$. Thus

$$x = 2x_1 \text{ and } y = 3y_1 \text{ for } x_1, y_1 \in \mathbb{Z}.$$

Now

$$3x = 2y \;\Rightarrow\; 3(2x_1) = 2(3y_1) \;\Rightarrow\; 6x_1 = 6y_1 \;\Rightarrow\; x_1 = y_1.$$

Denoting $x_1 = y_1$ by k, we obtain that

$$H = \{(2k, 3k) \;:\; k \in \mathbb{Z}\}.$$

Example 12.9. Let $m > n \geqslant 1$ be integers. Now let $\phi : \mathbb{R}^m \to \mathbb{R}^n$ be the map defined by

$$\phi(x_1, \ldots, x_m) = (x_1, \ldots, x_n) \text{ for every } (x_1, \ldots, x_m) \in \mathbb{R}^m.$$

It is straightforward to verify that this map is a homomorphism of groups. We have

$$(x_1, \ldots, x_n) \in \ker \phi \text{ if and only if } x_{n+1} = \cdots = x_m = 0.$$

Thus $\ker \phi = \{(x_1, \ldots, x_m) \in \mathbb{R}^m \ : \ x_1 = \cdots = x_n = 0\}$.

Example 12.10. Let A be an $n \times m$ matrix with entries in $\mathbb{R}$, where $m, n \in \mathbb{N}$. We denote the elements of the groups/vectorspaces $\mathbb{R}^m$ and $\mathbb{R}^n$ by vertical vectors. Now let $\phi : \mathbb{R}^m \to \mathbb{R}^n$ be the map defined by $\phi(v) = Av$ for every $v \in \mathbb{R}^m$. Then ϕ is a group homomorphism because

$$\phi(v + w) = A(v + w) = Av + Aw = \phi(v) + \phi(w).$$

The image of A is the set of all vectors in $w \in \mathbb{R}^n$ such that $w = phi(v)$ for some $v \in \mathbb{R}^m$. Equivalently, it is the set of all vectors $w \in \mathbb{R}^n$ of the form

$$w = Av \text{ for some } v \in \mathbb{R}^m.$$

From linear algebra, we know that:

$$w \in \operatorname{im} \phi \text{ if and only if } w \in \operatorname{Col} A.$$

Thus

$$\operatorname{im} \phi = \operatorname{Col} A.$$

Now let us determine $\ker \phi$. We have $v \in \ker \phi$ if and only if $\phi(v) = 0$, or equivalently if and only if $Av = 0$. Thus

$$\ker \phi = \operatorname{Nul} A.$$

Example 12.11. Let $\mathbb{T}$ be the circle group, and let $n \in \mathbb{N}$. Now let $\phi_n : \mathbb{T} \to \mathbb{T}$ be the map defined by $\phi_n(z) = z^n$ for every $z \in \mathbb{T}$. Then ϕ is a group homomorphism because

$$\phi_n(z_1 z_2) = (z_1 z_2)^n = z_1^n z_2^n = \phi_n(z_1)\phi_n(z_2) \text{ for every } z1, z_2 \in \mathbb{T}.$$

This homomorphism is surjective because every $z_\circ \in \mathbb{T}$ is of the form

$$z_\circ = e^{i\theta} \text{ for some } 0 \leqslant \theta < 2\pi,$$

and we have $\phi_n(e^{\frac{\theta}{n}i}) = z_\circ$. The kernel of ϕ_n is the set of complex numbers $z \in \mathbb{T}$ such that $z^n = 1$. Thus

$$\ker \phi_n = R_n = \left\{ e^{\frac{2\pi k}{n}i} \ : \ 0 \leqslant k \leqslant n - 1 \right\}.$$

For example, if $n = 4$ then $\ker \phi_n = \{1, -1, i, -i\}$.

Recall that if $\phi : G \to H$ is an isomorphism of groups, then for every $g \in G$ we have $o(g) = o(\phi(g))$. This property might not hold for group homomorphisms, but a weaker version of that is true:

Lemma 12.12. *Let $\phi : G \to H$ be a group homomorphism, and let $g \in G$. Then $o(\phi(g)) \big| o(g)$.*

Proof. Set $m = o(g)$. Then $\phi(g)^m = \phi(g^m) = e_H$ Therefore from Theorem 6.6(i) it follows that $o(\phi(g))|m$.

□

Image and inverse image of subgroups. Let $\phi : G \to H$ be a group homomorphism. For any subset $S \subseteq G$, we define

$$f(S) := \{f(g) \ : \ g \in S\}.$$

Thus $f(S) \subseteq H$.

Similarly, for any subset $T \subseteq H$, we define

$$f^{-1}(T) := \{x \in G \ : \ f(g) \in T\}.$$

Thus $f^{-1}(T) \subseteq G$.

Proposition 12.13. *Let $\phi : G \to H$ be a group homomorphism. Then:*

(i) If $K \subseteq G$ is a subgroup of G, then $\phi(K)$ is a subgroup of H.

(ii) If $L \subseteq H$ is a subgroup of H, then $\phi^{-1}(L)$ is a subgroup of G.

Proof. (i) Recall that $\phi(K) = \{\phi(k) \ : \ k \in K\}$. Obviously $\phi(K) \neq \varnothing$. To prove that $\phi(K)$ is a subgroup of H we use the Concise Subgroup Test: take any $x, y \in \phi(K)$. Then $x = \phi(a)$ and $y = \phi(b)$ where $a, b \in K$. Now

$$xy^{-1} = \phi(a)\phi(b)^{-1} = \phi(a)\phi(b^{-1}) = \phi(ab^{-1}) \in \phi(K),$$

because $ab^{-1} \in K$.

(ii) Recall that $\phi^{-1}(L) = \{g \in G \ : \ \phi(g) \in L\}$. Obviously $\phi^{-1}(L) \neq \varnothing$. To prove that $\phi^{-1}(L)$ is a subgroup of G we use the Concise Subgroup Test: take any $x, y \in \phi^{-1}(L)$ and set $a = \phi(x)$ and $b = \phi(y)$. Then $a, b \in L$. Now

$$\phi(xy^{-1}) = \phi(x)\phi(y^{-1}) = \phi(x)\phi(y)^{-1} = ab^{-1} \in L,$$

because L is a subgroup of H. □

Proposition 12.14. *Let $\phi : G \to H$ be a surjective group homomorphism. Then:*

(i) If $K \subseteq G$ is a normal subgroup of G, then $\phi(K)$ is a normal subgroup of H.

(ii) If $L \subseteq H$ is a normal subgroup of H, then $\phi^{-1}(L)$ is a normal subgroup of G.

Proof. Proposition 12.13 implies that $\phi(K)$ and $\phi^{-1}(L)$ are subgroups of the corresponding groups. Next we show that they are normal:

(i) Take any $h \in H$. Since ϕ is surjective, we have $h = \phi(g)$ for some $g \in G$. Now

$$h\phi(K)h^{-1} = \phi(g)\phi(K)\phi(g)^{-1} = \phi(gKg^{-1}) \subseteq \phi(K),$$

since K is normal. Now Proposition 11.3 implies that $\phi(K)$ is a normal subgroup of H.

(ii) Take any $g \in G$. We need to prove that $g\phi^{-1}(L)g^{-1} \subseteq \phi^{-1}(L)$. Now take any $x \in \phi^{-1}(L)$. We have $\phi(x) \in L$, and therefore

$$\phi(gxg^{-1}) = \phi(g)\phi(x)\phi(g^{-1}) \in \phi(g)L\phi(g)^{-1} \subseteq L.$$

It follows that

$$gxg^{-1} \in \phi^{-1}(L).$$

This completes the proof of the inclusion $g\phi^{-1}(L)g^{-1} \subseteq \phi^{-1}(L)$. □

Proposition 12.15. *Let $f : G \to H$ be a surjective group homomorphism. Then:*

(i) For every subgroup $K \subseteq G$ such that $\ker f \subseteq K$, we have $f^{-1}(f(K)) = K$.

(ii) For every subgroup $L \subseteq H$, we have $f(f^{-1}(L)) = L$.

Proof. (i) We need to prove two inclusions:

$$S \subseteq f^{-1}(f(K)) \text{ and } f^{-1}(f(K)) \subseteq K.$$

$\underline{K \subseteq f^{-1}(f(K))}$: Let $x \in K$ and set $A = f(K)$. Then $f(x) \in f(K) = A$, and therefore by definition of $f^{-1}(A)$ we have $x \in f^{-1}(A) = f^{-1}(f(K))$. This argument does not need surjectivity of f or the assumption $\ker f \subseteq K$.

$\underline{f^{-1}(f(K)) \subseteq K}$: Let $x \in f^{-1}(f(K))$. Then $f(x) \in f(K)$, and hence

$$f(x) = f(k) \text{ for some } k \in K.$$

Since f is a homomorphism, it follows that

$$f(xk^{-}) = f(x)f(k)^{-1} = e_H, \text{ that is, } xk^{-1} \in \ker f.$$

but $\ker f \subseteq K$ and therefore $x = (xk^{-1})k$ is a product of two elements of K. It follows that $x \in K$. This argument uses $\ker f \subseteq K$ but does not need surjectivity of f.

(ii) Again we need to show that

$$f(f^{-1}(L)) \subseteq L \text{ and } L \subseteq f(f^{-1}(L)).$$

$\underline{f(f^{-1}(L)) \subseteq L}$: Let $x \in f(f^{-1}(L))$. Then $x = f(g)$ for some $g \in f^{-1}(L)$. Thus,

$$x = f(x) \text{ and } f(g) \in L.$$

It follows that $x \in L$. This argument does not need surjectivity of f.

$\underline{L \subseteq f(f^{-1}(L))}$: Let $x \in L$. By surjectivity of L, we have

$$x = f(g) \text{ for some } g \in G.$$

Now

$$f(g) = x \text{ and } x \in L \text{ imply that } g \in f^{-1}(L).$$

Consequently,

$$x = f(g) \in f(f^{-1}(L)).$$

□

Exercises.

(1) Let $\alpha : G \to H$ and $\beta : H \to K$ be group homomorphisms. Prove that $\beta \circ \alpha : G \to K$ is also a group homomorphism.

(2) Let $\phi_1 : G_1 \to H_1$ and $\phi_2 : G_2 \to H_2$ be group homomorphisms. Prove that the map $\phi : G_1 \times G_2 \to H_1 \times H_2$ defined by

$$\phi(g_1, g_2) = (\phi_1(g_1), \phi_2(g_2)) \text{ for all } g_1 \in G_1, g_2 \in G_2$$

is also a group homomorphism. (Typically, the map ϕ is denoted by $\phi_1 \times \phi_2$.)

(3) Let G be a group and let K be a subgroup of G such that $K \subseteq Z(G)$. Assume that G/K is a cyclic group. Prove that G is abelian.

(4) Let $G = \mathbb{R}$ and let $H = \mathbb{C}^*$, where $\mathbb{R}$ is considered as a group with respect to addition, whereas H is a group with respect to multiplication. As usual we set $i = \sqrt{-1}$. Prove that the map $\phi : G \to H$, $\phi(t) = e^{it}$ is a group homomorphism. Determine and $\ker \phi$ explicitly.

ISOMORPHISM THEOREMS

We now come back to the quotient group G/N, and we explore the relation between its structure and the group structure of G more thoroughly.

Theorem 13.1. *Let G be a group and let N be a normal subgroup of G. Then:*

(i) The map

$$\phi : G \to G/N \ , \ \phi(g) := gN$$

is a group homomorphism.

(ii) $\mathrm{im}\phi = G/N$ *and* $\ker\phi = N$.

(iii) Set

$$\mathcal{A} = \{H \subseteq G \ : \ H \text{ is a subgroup of } G \text{ and } N \subseteq H\}$$

and

$$\mathcal{B} = \{K \subseteq G/N \ : \ K \text{ is a subgroup of } G/N\}.$$

Then the assignments

$$\mathcal{A} \to \mathcal{B} \ , \ H \mapsto \phi(H)$$

and

$$\mathcal{B} \to \mathcal{A} \ , \ K \mapsto \phi^{-1}(K)$$

are bijections between $\mathcal{A}$ and $\mathcal{B}$. These bijections are inverse to each other.

Proof. (i) We have $\phi(g_1g_2) = g_1g_2N = g_1N \star g_2N = \phi(g_1) \star \phi(g_2)$.

(ii) Every element of G/N is of the form gN for some $g \in G$, and we have $\phi(g) = gN$. Thus ϕ is surjective.

Next we determine the kernel of ϕ. If $g \in \ker\phi$, then $\phi(g) = eN$, hence $gN = eN$, and therefore $e^{-1}g \in N$, which means $g \in N$. This proves that $\ker\phi \subseteq N$. Furthermore, for $g \in N$ we have $\phi(g) = gN = N = eN$, hence $g \in \ker\phi$. Thus $N \subseteq \ker\phi$.

(iii) Let H be a subgroup of G such that $N \subseteq H$. Then by Proposition 12.14, $\phi(H)$ is a subgroup of G/N. Similarly, if K is a subgroup of G/N, then by Proposition 12.14, $\phi^{-1}(K)$ is a subgroup of G, and indeed $N \subseteq \phi^{-1}(K)$ because for any $x \in N$, from $N = \ker\phi$ (see part (ii) above) it follows that

$$x \in \ker\phi, \text{ hence } \phi(x) = e_H \in K, \text{ and consequently } x \in \phi^{-1}(K).$$

Thus the assignments $H \mapsto \phi(H)$ and $K \mapsto \phi^{-1}(K)$ are maps from $\mathcal{A}$ to $\mathcal{B}$ and from $\mathcal{B}$ to $\mathcal{A}$ respectively.

Finally, Proposition 12.15 implies that these assignments are inverse to each other (and in particular, they are bijections). □

Proposition 13.2. *In the bijective correspondence of Theorem 13.1(iii), normal subgroups of G/N correspond to normal subgroups of G that contain N.*

Proof. This follows from Proposition 12.14. □

Example 13.3. Let $G = \mathbb{Z}_4 \times \mathbb{Z}_4$ and let

$$N = \mathbb{Z}_4 \times \{0\} = \{(x, 0) \ : \ x \in \mathbb{Z}_4\}.$$

Then N is a normal subgroup of G because G is abelian. The distinct left cosets of N in G are:

$$(0,0) + N, \ (1,0) + N, \ (2,0) + N, \ (3,0) + N.$$

By writing down the Cayley table of G/N we can see that $G/N \cong \mathbb{Z}_4$. Indeed there is a natural isomorphism $f : \mathbb{Z}_4 \to G/N$, given by

$$f(k) = k + \mathbb{Z}_4 \text{ for every } k \in \mathbb{Z}_4.$$

Now Theorem 13.1(iii) states that there is a bijection between subgroups of G/N and subgroups of $G \cong$ which contain N. This correspondence can be described explicitly as follows. Subgroups of $G/N \cong \mathbb{Z}_4$ are:

- $K_1 = \{(0,0) + N\}$.
- $K_2 = \{(0,0) + N, (2,0) + N\}$.
- $K_3 = \{(0,0) + N, (1,0) + N, (2,0) + N, (3,0) + N\}$.

Subgroups of $G = \mathbb{Z}_4 \times \mathbb{Z}_4$ that contain N are:

- $H_1 = N = \{(x, 0) \ : \ x \in \mathbb{Z}_4\}$.
- $H_2 = \{(x, y) \ : \ x \in \mathbb{Z}_4 \text{ and } y \in 2\mathbb{Z}_4\}$.
- $H_3 = G = \mathbb{Z}_4 \times \mathbb{Z}_4$.

In the bijective correspondence of Theorem 13.1, the subgroup H_i corresponds to the subgroup K_i for $i = 1, 2, 3$. More precisely,

$$\phi(H_i) = K_i \text{ and } \phi^{-1}(K_i) = H_i \text{ for } i = 1, 2, 3.$$

Example 13.4. Let $G = \mathbb{Z}$ and let $N = d\mathbb{Z}$ where $d \in \mathbb{N}$. Then N is a normal subgroup of G because G is abelian. In Example 11.18 we saw that $G/N \cong \mathbb{Z}_d$, and the map yielding this isomoprhism is

$$f : \mathbb{Z}_d \to \mathbb{Z}/d\mathbb{Z} \ , \ f(k) = k + d\mathbb{Z} \text{ for all } k \in \{0, 1, \ldots, d-1\}.$$

Theorem 13.1(iii) states that there is a bijective correspondence between the following:

- subgroups of $\mathbb{Z}_d$,
- subgroups of $\mathbb{Z}$ which contain $d\mathbb{Z}$.

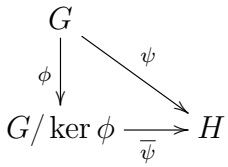
This correspondence can be described explicitly as follows: subgroups of $\mathbb{Z}_d$ are of the form $m\mathbb{Z}_d$ where $m \in \mathbb{N}$ and $m|d$. The subgroup of $\mathbb{Z}$ that corresponds to $m\mathbb{Z}_d$ via Theorem 13.1 is $m\mathbb{Z}$. More precisely, we have

$$\phi(m\mathbb{Z}) = m\mathbb{Z}_d \text{ and } \phi^{-1}(m\mathbb{Z}_d) = m\mathbb{Z}.$$

Note that from $m|d$ it follows that $d\mathbb{Z} \subseteq m\mathbb{Z}$.

Theorem 13.5. (First Isomorphism Theorem) *Let $\psi : G \to H$ be a group homomorphism.*

(i) Set $N = \ker\psi$. Then there exists a unique group homomorphism $\overline{\psi} : G/N \to H$ such that $\overline{\psi} \circ \phi = \psi$, where $\phi : G \to G/N$ is the canonical group homomorphism defined by $\phi(g) := gN$ (see Theorem 13.1).



(ii) The map $\overline{\psi}$ of part (i) provides an isomorphism $G/\ker\psi \cong \operatorname{im}\psi$.

Proof. (i) The relation $\overline{\psi} \circ \phi = \psi$ uniquely defines $\overline{\psi}$: we must define

$$\overline{\psi}(gN) = \psi(g) \text{ for all } g \in G.$$

Step 1. We show that $\overline{\psi}$ is well defined. Let $g_1, g_2 \in N$ such that $g_1N = g_2N$. Then $g_1^{-1}g_2 \in N = \ker\psi$, hence $\psi(g_1^{-1}g_2) = e_H$. It follows that

$$\psi(g_1)^{-1}\phi(g_2) = \psi(g_1^{-1})\psi(g_2) = \psi(g_1^{-1}g_2) = e_H,$$

and consequently $\psi(g_1) = \psi(g_2)$. Thus $\overline{\psi}(g_1N) = \overline{\psi}(g_2N)$.
Step 2. We prove that $\overline{\psi}$ is a group homomorphism. Let $g_1, g_2 \in G$. Then

$$\overline{\psi}(g_1N \star g_2N) = \overline{\psi}(g_1g_2N) = \psi(g_1g_2) = \psi(g_1)\psi(g_2) = \overline{\psi}(g_1N)\overline{\psi}(g_2N).$$

Consequently, $\overline{\psi}$ is a group homomorphism.

(ii) The image of $\overline{\psi}$ is equal to $\psi(G)$, and therefore we can consider $\overline{\psi}$ as a map from $G/\ker\psi$ onto $\operatorname{im}\psi$. Thus the only thing left to show is that $\overline{\psi}$ is an injection. By Proposition 12.7(iii), it suffices to show that $\ker\overline{\psi} = \{e_{G/N}\}$, where $e_{G/N} = e_GN$ is the identity element of $G/N = G/\ker\psi$.

Take any element $gN \in \ker\overline{\psi}$. Then from the definition of $\overline{\psi}$ we have

$$\psi(g) = \overline{\psi}(gN) = e_H,$$

hence $g \in \ker\psi$, that is, $g \in N$. This implies that

$$gN = eN = e_{G/N}.$$

Consequently, $\ker\overline{\psi} = \{e_{G/N}\}$. □

Example 13.6. Let $d \in \mathbb{N}$, and let $\psi : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}_d$ be the map defined by

$$\psi(x, y) = (2x, y \text{ modulo } d) \text{ for all } (x, y) \in \mathbb{Z} \times \mathbb{Z}.$$

$\underline{\ker\psi}$: Assume that $\psi(x, y) = (0, 0)$. Then $(2x, y \text{ modulo } d) = (0, 0)$, hence $x = 0$ and $d|y$. Therefore $\ker\psi = \{0\} \times d\mathbb{Z}$.

$\underline{\operatorname{im}\psi}$: Clearly $(2x, y \text{ modulo } d) \in 2\mathbb{Z} \times \mathbb{Z}_d$, and ψ is a surjection onto $2\mathbb{Z} \times \mathbb{Z}_d$. Therefore

$$\operatorname{im}\psi = 2\mathbb{Z} \times \mathbb{Z}_d.$$

Now by the First Isomorphism Theorem we have

$$(\mathbb{Z} \times \mathbb{Z})/(\{0\} \times d\mathbb{Z}) \cong 2\mathbb{Z} \times d\mathbb{Z}.$$

Example 13.7. Let $\psi : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ be the map defined by

$$\psi(x, y) = 5x + 3y \text{ for all } (x, y) \in \mathbb{Z} \times \mathbb{Z}.$$

$\underline{\text{im}\,\psi}$: We need to determine which integers $n \in \mathbb{Z}$ can be written in the form $n = 5x + 3y$ for $x, y \in \mathbb{Z}$. By Bezout's Theorem (see Theorem 1.8), since $\gcd(5, 3) = 1$, every $n \in \mathbb{Z}$ can be written of this form. Hence $\text{im}, \psi = \mathbb{Z}$.

$\underline{\ker \psi}$: Assume that $\psi(x, y) = 0$. Then we have $5x + 3y = 0$, hence $5x = -3y$. We have:

$$5|5x \;\Rightarrow\; 5| - 3y \text{ but } \gcd(5, 3) = 1, \text{ hence } 5|y \text{ because of Lemma 1.15(iii).}$$

For a similar reason, $3|x$. Now we can write

$$x = 3x_1 \text{ and } y = 5y_1 \text{ for some } x_1, y_1 \in \mathbb{Z},$$

and thus

$$5x + 3y = 0 \text{ implies that } 15x_1 + 15y_1 = 0, \text{ hence } x_1 = -y_1.$$

Denoting $x_1 = -y_1$ by $k \in \mathbb{Z}$, it follows that

$$(x, y) = (3k, -5k).$$

Conversely, any element of $\mathbb{Z} \times \mathbb{Z}$ of the form $(3k, -5k)$ is in $\ker \phi$ because:

$$\phi(3k, -5k) = 5(3k) + 3(-5k) = 15k - 15k = 0.$$

Therefore $\ker \psi = \{(3k, -5k) \;:\; k \in \mathbb{Z}\}$.

Now the First Isomorphism Theorem implies that $(\mathbb{Z} \times \mathbb{Z})/\ker \psi \cong \mathbb{Z}$.

FINITELY GENERATED ABELIAN GROUPS

The problem of classifying all finite groups up to isomorphisms is an extremely difficult one, and there is no solution to it at the moment. However, the problem of classifying all finite *abelian* groups (and even more generally, all finitely generated abelian groups) is much simpler, and its solution is a fundamental theorem in group theory.The goal of this section is describe this classification.

Definition 14.1. Let G be a group. We call G *finitely generated* if there exists a finite subset $S = \{s_1, \dots, s_r\} \subseteq G$ such that every element of $g \in G$ can be written as

$$g = s_{i_1}^{\pm 1} \cdots s_{i_N}^{\pm 1},$$

where $1 \leqslant i_1, \dots, i_N \leqslant r$. If this condition holds, then we say G is *generated by* S, and we call S a generating set for G.

Example 14.2. (i) The group $\mathbb{Z}$ is finitely generated because we can set $S = \{1\}$ or $S = \{-1\}$ or $S = \{1, 3\}$ (there are many other choices for S).

(ii) The group $\mathbb{Z} \times \mathbb{Z}$ is finitely generated because we can set

$$S = \{(1, 0), (0, 1)\}.$$

Then for every element $(x, y) \in \mathbb{Z}$ we have:

– If $x, y \geqslant 0$ then

$$(x, y) = \underbrace{(1, 0) + \cdots + (1, 0)}_{x \text{ times}} + \underbrace{(0, 1) + \cdots + (0, 1)}_{y \text{ times}}.$$

– If $x \geqslant 0$ and $y < 0$ then

$$(x, y) = \underbrace{(1, 0) + \cdots + (1, 0)}_{x \text{ times}} + \underbrace{(0, -1) + \cdots + (0, -1)}_{-y \text{ times}}.$$

– If $x < 0$ and $y \geqslant 0$ then

$$(x, y) = \underbrace{(-1, 0) + \cdots + (-1, 0)}_{-x \text{ times}} + \underbrace{(0, 1) + \cdots + (0, 1)}_{y \text{ times}}.$$

– If $x, y < 0$ then

$$(x, y) = \underbrace{(-1, 0) + \cdots + (-1, 0)}_{-x \text{ times}} + \underbrace{(0, -1) + \cdots + (0, -1)}_{-y \text{ times}}.$$

Note that $\mathbb{Z} \times \mathbb{Z}$ is also generated by any subset S' satisfying $S' \supseteq S$.

(iii) Every cyclic group $G = \langle a \rangle$ is finitely generated because we cans set $S = \{a\}$.

(iv) Every finite group G is finitely generated because we can set $S = G$.

(v) For every $n \in \mathbb{N}$, the group $\mathbb{Z}^n$ is finitely generated because we can take $S = \{e_1, \ldots, e_n\}$, where

$$e_i = (0, \ldots, 0, 1, 0, \ldots, 0), \text{ for every } 1 \leqslant i \leqslant n,$$

is the vector containing only one 1 in the i-th coordinate and 0's elsewhere. Indeed every $g = (g_1, \ldots, g_n) \in \mathbb{Z}^n$ can be expressed as

$$g = g_1 e_1 + \cdots + g_n e_n.$$

Remark 14.3. From now on, we focus on abelian groups, and we write the operations of abelian groups additively. That is, instead of gh for two elements g, h in an abelian group G, we write $g + h$. As a result, instead of g^n for an integer n, we will write ng. This is consistent with the explicit description of the group $\mathbb{Z}^n$. The identity element of an abelian group G will also be denoted by "0".

With this notation, the definition of a finitely generated abelian group can be expressed as follows:

An abelian group G is finitely generated if and only if there exists a finite set $S = \{s_1, \ldots, s_N\} \subseteq G$ such that every element $g \in G$ can be written as

$$g = m_1 g_1 + \cdots + m_N g_N.$$

Generating subsets of $\mathbb{Z}^n$. Clearly the set of vectors $S = \{e_1, \ldots, e_n\}$ that was defined in Example 14.2(v) is a generating set of $\mathbb{Z}^n$. Now starting from any generating subset $S = \{s_1, \ldots, s_n\}$ of $\mathbb{Z}^n$ we can construct new generating subsets with the following operations:

I) Replacing one s_i by $-s_i$ (the inverse of s_i). If $g \in \mathbb{Z}^n$ can be expressed as

$$g = k_1 s_1 + \cdots + k_n s_n,$$

then we can write g as

$$g = k_1 s_1 + \cdots + k_{i-1} s_{i-1} + (-k_i)(-s_i) + k_{i+1} s_{i+1} + \cdots + k_n s_n.$$

Note that the coefficients of the above linear combination are still in $\mathbb{Z}$.

II) Replacing some s_i with $s_i + s_j$ with $j \neq i$. if $g \in \mathbb{Z}^n$ can be expressed as

$$g = k_1 s_1 + \cdots + k_n s_n,$$

then we can write g as

$$g = k_1 s_1 + \cdots + k_i(s_i + s_j) + \cdots + (k_j - k_i) s_j + \cdots + k_n s_n.$$

Note that the coefficients of the above linear combination are still in $\mathbb{Z}$.

Example 14.4. Let $S = \{e_1, e_2, e_3\} = \{(1,0,0), (0,1,0), (0,0,1)\}$ be the standard generating set of $\mathbb{Z}^3$. Then using the operations (I) and (II) above, we can form new generating sets of S, as follows:

$$\begin{aligned}\{e_1, e_2, e_3\} &\xrightarrow{\text{(I)}} \{e_1, -e_2, e_3\} \xrightarrow{\text{(II)}} \{e_1 - e_2, -e_2, e_3\} \xrightarrow{\text{(II)}} \{e_1 - 2e_2, -e_2, e_3\} \\ &\xrightarrow{\text{(I)}} \{e_1 - 2e_2, e_2, e_3\} \xrightarrow{\text{(I)}} \{e_1 - 2e_2, e_2, -e_3\} \\ &\xrightarrow{\text{(II)}} \{e_1 - 2e_2, e_2, e_2 - e_3\} \to \cdots\end{aligned}$$

Definition 14.5. A generating set $S = \{v_1, \ldots, v_n\} \subseteq \mathbb{Z}^n$ is called a *basis* if it satisfies the following property: if $m_1v_1 + \cdots + m_nv_n = 0$, for some integers $m_1, \ldots, m_n$, then $m_1 = \ldots = m_n = 0$.

Proposition 14.6. *Let $S := \{v_1, \ldots, v_n\}$ be a basis of $\mathbb{Z}^n$. Then the generating set obtained from S by each of the operations (I) and (II) is also a basis of $\mathbb{Z}^n$.*

Proof. We leave the proof as an exercise. □

Remark 14.7. The operations (I) and (II) above can also be applied to generating sets of *any* finitely generated abelian group. Thus, if $\{s_1, \ldots, s_n\}$ is a generating set of a finitely generated abelian group G, then the operations (I) and (II) above will produce new generating sets of G.

Proposition 14.8. *Let G be an abelian group. Then G is finitely generated if and only if there exists an integer $n \in \mathbb{N}$ and a surjective homomorphism*

$$\phi : \mathbb{Z}^n \to G.$$

In particular, every finite abelian group G is isomorphic to a quotient group of the form $\mathbb{Z}^n/K$, where K is a subgroup of $\mathbb{Z}^n$.

Proof. Assume that G is generated by a subset $S = \{s_1, \ldots, s_n\} \subseteq G$. Then we define the map $\mathbb{Z}^n \to G$ as follows:

$$\phi : \mathbb{Z}^n \to G \ , \ \phi(k_1, \ldots, k_n) := k_1s_1 + \cdots + k_ns_n.$$

Since S generates G, the map ϕ is a surjection. Furthermore, ϕ is a group homomorphism because

$$\begin{aligned}
\phi(k_1 + k_1', \ldots, k_n + k_n') &= (k_1 + k_1')s_1 + \cdots + (k_n + k_n')s_n \\
&= (k_1s_1 + k_1's_1) + \cdots + (k_ns_n + k_n's_n) \\
&= (k_1s_1 + \cdots + k_ns_n) + (k_1's_1 + \cdots + k_n's_n) \\
&= \phi(k_1, \ldots, k_n) + \phi(k_1', \ldots, k_n').
\end{aligned}$$

Next assume that a surjective group homomorphism $\phi : \mathbb{Z}^n \to G$ exists. Set

$$s_i = \phi(e_i) \text{ where } e_i = (0, \ldots, 0, 1, 0, \ldots, 0),$$

that is, e_i is the unit vector in $\mathbb{Z}^n$ with a 1 in the i-th position and 0's elsewhere. We show that G is generated by S. Indeed for every $g \in G$ we have $g = \phi(k_1, \ldots, k_n)$ for some element $(k_1, \ldots, k_n) \in \mathbb{Z}^n$. Then

$$g = \phi(k_1, \ldots, k_n) = k_1s_1 + \cdots + k_ns_n.$$ □

Corollary 14.9. *Every finite abelian group is isomorphic to a quotient $\mathbb{Z}^n/K$ where K is a subgroup of $\mathbb{Z}^n$.*

Proposition 14.10. *Let $n \in \mathbb{N}$. Then every subgroup of $\mathbb{Z}^n$ is a finitely generated abelian group that is generated by a subset $S \subseteq G$ such that $|S| \leqslant n$.*

Proof. We prove the statement by induction on n. For $n = 1$, the statement follows from the classification of subgroups of $\mathbb{Z}$ (see Theorem 6.15).

Next assume that the statement holds for subgroups of $\mathbb{Z}^{n-1}$. We prove that the statement also holds for all subgroups of $\mathbb{Z}^n$. From now on, we consider a subgroup G of $\mathbb{Z}^n$.

Step 1. Write every element $g \in G$ as $g = (g_1, \ldots, g_n)$, and let F be the subset of $\mathbb{Z}$ consisting of the integers g_1 for all $g \in G$. We show that F is a subgroup of $\mathbb{Z}$. Clearly $0 \in F$ because $(0, \ldots, 0) \in G$. Next we use the Concise Subgroup Test: take $a, b \in \mathbb{Z}$; then there exist elements $g, h \in G$ such that

$$g = (g_1, \ldots, g_n) \text{ with } g_1 = a, \text{ and } h = (h_1, \ldots, h_n) \text{ with } h_1 = b.$$

We need to verify that $a - b \in F$. But $a - b$ is the first component of $g - h \in G$, because

$$g - h = (g_1 - h_1, \ldots, g_n, h_n).$$

Step 2. Set $G_0 := G \cap \{0\} \times \mathbb{Z}^{n-1}$. This is the subset of G consisting of elements whose first coordinate is equal to 0. Then G_0 is a subgroup of $\mathbb{Z}^n$ because it is the intersection of two subgroups of $\mathbb{Z}^n$.

Step 3. Note that $G_0 \subseteq \{0\} \times \mathbb{Z}^{n-1} \cong \mathbb{Z}^{n-1}$, and therefore by the induction hypothesis G_0 is generated by a subset $S_0 \subseteq G_0$ such that $|S_0| \leqslant n - 1$.

Step 4. By the classification of subgroups of $\mathbb{Z}$ (see Theorem 6.15), we know that $F = d\mathbb{Z}$ for some $d \in \mathbb{Z}$. If $d = 0$, then it follows that

$$G = G_0,$$

and the proof is complete by Step 3. Next, we consider the case $d \neq 0$.

Step 5. Recall that by Step 4 we can assume that $F = d\mathbb{Z}$ for some $d \in \mathbb{Z}\backslash\{0\}$. Now take an element $x \in G$ such that

$$x = (x_1, \ldots, x_n) \text{ with } x_1 = d.$$

We claim that G is generated by $S = S_0 \cup \{x\}$. Indeed for any $g \in G$ we have

$$g = (g_1, \ldots, g_n) \text{ where } g_1 = dk \text{ for some } k \in \mathbb{Z}.$$

Thus we can write

$$g = kx + (g - kx),$$

and

$$g - kx = (g_1 - kx_1, \ldots, g_n - kx_n) = (0, g_2 - kx_2, \ldots, g_n - kx_n) \in G_0.$$

Now Step 3 implies that $g - kx$ can be expressed as a linear combination with integer coefficients of the elements of S_0. Consequently, g can be expressed as a linear combination of the elements of $S_0 \cup \{x\}$ with integer coefficients.

□

The Smith normal form of an integral matrix. Let A be an $m \times n$ matrix with integer entries. We are interested in the following six row and column operations:

RI: Multiplying the i-th row by (-1), where $1 \leqslant i \leqslant m$.

RII: Adding the i-th row to the j-th row, where $1 \leqslant i, j \leqslant m$ and $i \neq j$.

RIII: Switching the i-th row with the j-th row, where $1 \leqslant i, j \leqslant m$ and $i \neq j$.

CI: Multiplying the i-th column by (-1), where $1 \leqslant i \leqslant n$.

CII: Adding the i-th column to the j-th column, where $1 \leqslant i, j \leqslant n$ and $i \neq j$.

CIII: Switching the i-th column with the j-th column, where $1 \leqslant i, j \leqslant n$ and $i \neq j$.

Lemma 14.11. *Let $Y = [y_{ij}]$ be an $m \times n$ matrix with integer entries. Assume that $y_{11} \in \mathbb{N}$. Then by operations RI, RII, CI, and CII, we can replace the entries in the first row and first column of Y, other than the top-left corner of Y, by their remainders modulo y_{11}.*

Proof. This can be done as follows. Say for some $1 < a \leqslant m$ we have

$$y_{1a} = qy_{11} + r,$$

with $q, r \in \mathbb{Z}$ and $0 \leqslant r < y_{11}$. If $q < 0$, then

$$r = y_{1a} - qy_{11},$$

and by adding the first column of Y to the a-th column q times (this is CII), the entry y_{1a} will be substituted by r. No other entry in the first row and first column of Y is affected.

If $q > 0$, then we first multiply the first column of Y by (-1); this is CI. Then the top-left corner of the resulting matrix contains $-y_{11}$. Now note that

$$r = y_{1a} + q(-y_{11}),$$

and again by adding the first column to the a-th column q-times, the entry y_{1a} will be replaced by r. Finally we multiply the first column of the resulting matrix by (-1) for a second time, so that the entries in the first column are not affected by the whole process.

Finally, if $q = 0$ then $y_{1a} = r$ and there is nothing to be done. □

Proposition 14.12. (Smith Normal Form) *Let A be an $m \times n$ matrix with entries in Z. Then using the operations RI–RIII and CI–CIII we can obtain a matrix in the form*

$$\begin{bmatrix} D_k & 0_{k\times(n-k)} \\ 0_{(m-k)\times k} & 0_{k\times k} \end{bmatrix} \quad \textit{where} \quad D = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & \cdots & 0 & d_k \end{bmatrix} \tag{14.1}$$

where $d_1, \ldots, d_k \in \mathbb{N}$ and $d_i | d_{i+1}$ for $1 \leqslant i \leqslant k-1$.

Proof. If $A = 0_{m\times n}$ then there is nothing to prove. Thus from now on we assume that $A \neq 0_{m\times n}$. We claim that by the above six operations we can transform A to a matrix of the form

$$\begin{bmatrix} d & 0_{1\times(n-1)} \\ 0_{(m-1)\times 1} & B \end{bmatrix} \tag{14.2}$$

where $d \in \mathbb{N}$ and $B = [b_{ij}]$ is an $(m-1) \times (n-1)$ matrix with entries in $\mathbb{Z}$ such that $d | b_{i,j}$ for every entry $b_{i,j}$ of B. The claim of the proposition follows from this statement, by continuing the same process in the submatrix B, and noting that if all of the entries of a matrix are divisible by d, then after any of the operations RI–RIII and CI–CIII all of the entries of the resulting matrix will also be divisible by d.

We now prove the above claim. For every nonzero $m \times n$ matrix X, we set

$$m(X) = \min\{|x_{ij}| \ : \ x_{ij} \neq 0\}.$$

We repeatedly execute the following process on A:

- **Input**: an $m \times n$ matrix $X = [x_{ij}]$.

Step 1. Choose i_0, j_0 such that $x_{i_0 j_0} = m(X)$. If either $x_{11} = 0$ or $|x_{i_0 j_0}| < |x_{11}|$, then using the operations RIII and CIII, move the entry $x_{i_0 j_0}$ to the top-left corner position.

Step 2. Using RI if necessary, we can assume that the topleft corner entry is non-negative. Call the resulting matrix $X = [x'_{ij}]$. Thus, $x'_{11} = m(X) = |x_{i_0 j_0}|$.

Step 3. Using RI, RII, CI, and CII, obtain a matrix $Y = [y_{ij}]$ from X' such that

$$y_{1a} = x'_{1a} \text{ modulo } x'_{11}$$

and

$$y_{b1} = x'_{b1} \text{ modulo } x'_{11}$$

for all $1 < a \leqslant n$ and $1 < b \leqslant m$. This is possible by Lemma 14.11.

- **Output**: The $m \times n$ matrix Y.

If some of the entries in the first row or first column of X' other than the top-left corner are nonzero, then $0 \leqslant m(X') < m(X)$. However, by executing the above process repeatedly, the value of $m(X)$ cannot decrease indefinitely (because it is in $\mathbb{N}$). This means that after finitely many iterations of the above process, we obtain a a matrix A' from A in the block-diagonal form (14.2) such that $m(A') \leqslant m(A)$.

If the top-left corner entry d of A' divides all of the entries of B, then we are done. However, if there exists an entry $b_{i_0 j_0}$ of B such that $d \nmid b_{i_0 j_0}$, then we do the following:

Step 4. Add the first row to the i_0-th row (this is RII), and then by a technique similar to Lemma 14.11, use the operations RI, RII, CI, and CII to obtain a a matrix A'' in which $b_{i_0 j_0}$ is replaced by $b_{i_0 j_0}$ modulo d. Clearly $m(A'') < d = m(A')$. We now execute Steps 1–4 as many times as possible. Since the value of $m(A)$ keeps decreasing throughout the process, the process should come to an end. This happens precisely when the resulting matrix is of the form (14.2) where d divides every entry of B.

□

Using the Smith normal form, we can classify subgroups of $\mathbb{Z}^n$ explicitly. Informally, the next theorem states that every subgroup of $\mathbb{Z}^n$ is naturally embeds as a product of cyclic groups, after a suitable change of coordinates.

Theorem 14.13. *Let G be a subgroup of $\mathbb{Z}^n$. Then there exist $d_1, \ldots, d_k \in \mathbb{N}$, where $k \leqslant n$, and an isomorphism*

$$\phi : \mathbb{Z}^n \to \mathbb{Z}^n$$

such that

$$\begin{aligned} \phi(G) &= d_1\mathbb{Z} \times \cdots \times d_k\mathbb{Z} \times \{0\} \times \cdots \times \{0\} \\ &= \big\{(d_1 m_1, \ldots, d_k m_k, 0, \ldots, 0) \ : \ m_1, \ldots, m_k \in \mathbb{Z}\big\}, \end{aligned}$$

and $d_i | d_{i+1}$ for $1 \leqslant i \leqslant k-1$.

Proof. Recall that $E = \{e_1, \ldots, e_n\}$ is the standard generating set of $\mathbb{Z}^n$. By Proposition 14.10 we can also choose a generating set $S = \{s_1, \ldots, s_m\}$ for G (which satisfies $m \leqslant n$).

Step 1. Each $s_i \in S$ can be expressed as

$$s_i = \sum_{j=1}^{n} c_{ij} e_j.$$

We can now form the $m \times n$ matrix

$$A = \begin{bmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & & \vdots \\ c_{m1} & \cdots & c_{mn} \end{bmatrix}$$

Thus, each row of A represents the "coordinates" of one of the s_i's when it is expressed in terms of $e_1, \ldots, e_n$.

Step 2. We now interpret the operations RI–RIII and CI–CIII in terms of operations on the generating sets E and S.

Operations RI–RIII mean that we are changing the generating set of G. For example, if we multiply the i-th row of A by (-1), this means that we are replacing the set S by the set

$$\{s_1, \ldots, s_{i-1}, -s_i, s_{i+1} \ldots, s_m\}.$$

Similarly, if we add the i-th row of A to the j-th row, this means that we are replacing the set S by the set

$$\{s_1, \ldots, s_{j-1}, s_j + s_i, s_{j+1} \ldots, s_m\}.$$

Operations CI—CIII on A mean that we are changing the geneating set E of $\mathbb{Z}^n$, and therefore the rows of the "new" A give the coordinates of $s_1, \ldots, s_m$ when they are expressed in terms of this new generating set of $\mathbb{Z}^n$.

Step 3. Consider a sequence of the operations RI–RIII and CI—CIII on a matrix A. Each "R" operation is equivalent to multiplication on the left by an invertible $m \times m$ matrix P. Each "C" operation is equivalent to multiplication on the right by an invertible $n \times n$ matrix Q. Thus, the sequence of operations of "R" and "C" type turns A into $B = P_1 \cdots P_r A Q_1 \cdots Q_s$. Since matrix multiplication is associative, the "R" operations commute with the "C" operations. Consequently, in the process of obtaining B from A, we can assume that first all the "R" operations are implemented, then all the "C" operations are implemented.

Step 4. By Proposition 14.12 we can find a sequence of "R" and "C" operations such that implementing them on A yields a matrix

$$B = P_1 \ldots P_r A Q_1 \ldots Q_s$$

in Smith normal form. We now need the following observations:

- The rows of $M := P_1 \ldots P_r A$ give coordinates of elements of a generating set $S' = \{s'_1, \ldots, s'_m\}$ of G in terms of the standard generating set E of $\mathbb{Z}^n$.
- The rows of $B = MQ_1 \ldots Q_s$ give coordinates of the elements of S' in terms of a generating set $E' = \{e'_1, \ldots, e'_n\}$ of $\mathbb{Z}^n$.

Now consider the map $\phi : \mathbb{Z}^n \to \mathbb{Z}^n$ defined by

$$\phi(\textstyle\sum_{i=1}^n x_i e_i') := \sum_{i=1}^n x_i e_i \text{ for all } x_1, \ldots, x_n \in \mathbb{Z}.$$

Proposition 14.6 implies that this map is well-defined and injective. Since $E = \{e_1, \ldots, e_n\}$ is a generating set for $\mathbb{Z}6n$, this map is also surjective. Therefore ϕ is indeed an isomorphism $\mathbb{Z}^n \to \mathbb{Z}^n$.

Step 5. Finally, we investigate the structure of $\phi(G)$ under the map ϕ of Step 4. Note that

$$s_i' = d_i e_i' \text{ for } 1 \leqslant i \leqslant k \text{ and } s_i' = 0 \text{ for } i > k.$$

Therefore we have

$$\phi(s_i') = \phi(d_i e_i') = d_i e_i \text{ for } 1 \leqslant i \leqslant k \text{ and } \phi(s_i') = 0 \text{ for } i > k.$$

Since the group $\phi(G)$ is generated by $\phi(s_1'), \ldots, \phi(s_k')$, it follows that

$$\phi(G) = \big\{(d_1 m_1, \ldots, d_k m_k, 0, \ldots, 0) \ : \ m_1, \ldots, m_k \in \mathbb{Z}\big\}. \qquad \square$$